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Informatica® Multidomain MDM
10.5 HotFix 5

Provisioning Tool Guide

Informatica Multidomain MDM Provisioning Tool Guide
10.5 HotFix 5
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Table of Contents

Preface 10

Informatica Resources 10

- Informatica Network 10
- Informatica Knowledge Base 10
- Informatica Documentation 10
- Informatica Product Availability Matrices 11
- Informatica Velocity 11
- Informatica Marketplace 11
- Informatica Global Customer Support 11

Chapter 1: Introduction 12

Provisioning Tool Overview 12

Architecture 13

Prerequisites 13

Key Concepts 14

- Business Entities 14
- Reference Entities 14
- Root Node 14

Logging In to the Provisioning Tool 14

Configure Single Sign-On in MDM Provisioning Tool 15

- Prerequisites 15
- Enabling Single Sign-On 15
- Accessing Single Sign-On from the Login Page 16

Guidelines for Multiple Users to Collaborate 16

Chapter 2: Configuring the Data Director Application 17

Configuring the Data Director Application Overview 17

Creating a Data Director Application 18

Hiding Business Entities from an Application (Optional) 19

Configuring the Search or Query Results Display 20

Configuring Date and Time Formats 20

Chapter 3: Establishing a Root Node 22

Establishing a Root Node Overview 22

- Example of Adding Root Nodes 22
- Prerequisites 23

Root Node Properties 23

Creating a Business Entity Root Node 24

Creating a Reference Entity Root Node 25

Enabling State Management Validation 26

Setting Strategies for Record Deletion and Record Checking. 26

Chapter 4: Defining the Business Entity Model. 28

Defining the Business Entity Model Overview. 28
Example of Configuring the Business Entity Model. 29
Prerequisites. 29
Constraints. 29
Node Types. 30
Creating a Node with a One-to-Many Relationship. 32
Creating a Node with a One-to-One Relationship. 32
Creating a Child Lookup Node with a One-to-One Relationship. 33
Copying a Business Entity. 34

Chapter 5: Configuring Business Entity Properties. 35

Configuring Business Entity Properties Overview. 35
Field Properties. 35
Adding a Field to a Node. 38
Custom Data Types. 39
Predefined Custom Data Types. 39
Adding a Custom Data Type to a Field. 40
Node Labels in Data Director. 40
Configuring Node Labels. 41

Chapter 6: Configuring Reference Entity Properties. 43

Configuring Reference Entity Properties Overview. 43
Lookup Types. 43
Field Properties. 44
Adding a Field to a Node. 45
Adding Field Filters to a Reference Entity. 46
Custom Data Types. 47
Predefined Custom Data Types. 47
Adding a Custom Data Type to a Field. 48
Configuring a Lookup 48
Configuring a Dependent Lookup. 49

Chapter 7: Transforming Business Entities and Views. 51

Transforming Business Entities and Views Overview. 51
Transformation Sources and Targets. 52
Cleanse Transformations. 53
Transformations in Read and Write Processes. 53
Modeling Business Entity Views 55
Creating the Business Entity View Root Node. 55
Adding Fields to the Root and Child Nodes. 56

Adding Child Nodes. . . . 56
Copying a Business Entity View. . . . 56
Configuring Business Entity to View Transformations. . . . 57
Configuring Transformations for One-to-Many Child Nodes. . . . 57
Configuring a Direct Transformation. . . . 57
Configuring a Transformation to Map a View to a Business Entity. . . . 58
Configuring Transformations for Child Nodes. . . . 58
Configuring a Direct Transformation 59
Configuring a Cleanse Transformation. . . . 60
Configuring Business Entity to Business Entity Transformations. . . . 60
Configuring Transformations for Child Nodes. . . . 61
Configuring a Cleanse Transformation. . . . 61
Deleting Transformations. . . . 62

Chapter 8: Configuring Hierarchy and Network Relationships. . . . 63

Configuring Hierarchy and Network Relationships Overview. . . . 63
Hierarchy Relationships. . . . 64
Network Relationships. . . . 65
Prerequisites. . . . 66
Configuring Hierarchy Relationships. . . . 67
Step 1. Create a Hierarchy. . . . 67
Step 2. Configure the Root Node in a Hierarchy. . . . 68
Step 3. Add a Business Entity to a Hierarchy. . . . 68
Step 4. Configure a Hierarchy Relationship. . . . 68
Step 5. Configure the Hierarchy Settings. . . . 69
Step 6. Configure Additional Relationship Attributes (Optional). . . . 70
Configuring Network Relationships. . . . 72
Step 1. Add a Business Entity to the Network. . . . 72
Step 2. Configure a Network Relationship. . . . 72
Step 3. Configure the Network Settings. . . . 73
Step 4. Configure the Relationship Properties (Optional). . . . 74
Configuring Hierarchy Tasks. . . . 77
Step 1. Configuring a Hierarchy Task Type. . . . 77
Step 2. Configuring a Hierarchy Task Template. . . . 78
Step 3. Configuring Hierarchy Workflow Triggers. . . . 79
Configuring Default Hierarchy for the Dashboard. . . . 80
Designing Data Director Interfaces to Manage Relationships. . . . 81

Chapter 9: Creating Match Rule Sets. . . . 82

Creating Match Rule Sets Overview. . . . 82
Configuring the Hub Server Properties. . . . 82
Prerequisites for Creating Match Rule Sets. . . . 83
Creating Match Rule Sets. . . . 84

Step 1. Define a Match Rule Set. . . . 84
Step 2. Import Sample Data Set. . . . 85
Step 3. Train the Match Engine. . . . 86
Step 4. Preview and Save the Match Rule Set. . . . 86

Chapter 10: Configuring Search. . . . 87

Configuring Search Overview. . . . 87
Configure the Elasticsearch Cluster. . . . 88
Configure Elasticsearch Credentials. . . . 89
Create Custom Elasticsearch Index Settings (Optional). . . . 89
Elasticsearch Built-In Tokenizers and Token Filters. . . . 91
Configure the Searchable Fields. . . . 92
Searchable Field Properties. . . . 94
Configure the Search or Query Results Display. . . . 96
Configure the Layout to Display Similar Records (Optional). . . . 97

Chapter 11: Configuring Tasks. . . . 99

Configuring Tasks Overview. . . . 99
Configuring Task Templates. . . . 100
Configuring Workflow Triggers. . . . 101
Step 1. Create a Workflow Trigger. . . . 102
Step 2. Configure Events for a Workflow Trigger. . . . 103
Step 3. Configure Roles for a Workflow Trigger. . . . 104
Configuring Task Types. . . . 104
Step 1. Create a Task Type. . . . 104
Step 2. Configure Roles for a Task Type 105
Step 3. Create Task Action Configurations for a Task Type. . . . 106
Task Administrator Role. . . . 109
Configuring the Task Administrator Role. . . . 110
Descriptive Task Titles. . . . 110
Configuring Descriptive Task Titles. . . . 111
Default Task Configuration. . . . 111
Changing the Approver for Tasks. . . . 113

Chapter 12: Configuring Security and Data Filters for Business Entities. 114

Security and Data Filters for Business Entities Overview. . . . 114
GDPR Example. . . . 115
Entity-level Security. . . . 115
Managing Access to Business Entity Records. . . . 115
Role Privileges for a Business Entity. . . . 117
Role Privileges for a Business Entity View. . . . 118
Field-level Data Filters. . . . 119
Field Filter Rules and User Roles. . . . 119

Adding Field Filters. . . . 120
Role Privileges for Field Filters. . . . 122
Node-level Data Filters. . . . 122
Node Filter Rules. . . . 122
Adding Node Filters. . . . 123
Role Privileges for Nodes Filters. . . . 124
Security and the Data Director User Interface. . . . 125

Chapter 13: Configuring the Content Security Policy. . . . 127

Content Security Policy Overview. . . . 127
Content Security Policy Defaults. . . . 127
Configuring the Content Security Policy. . . . 128

Chapter 14: Integrating Data as a Service 129

Overview. . . . 129
WSDL File for a DaaS Service. . . . 130
Transformations. . . . 130
Transformation of Business Entity to XML. . . . 130
Transformation of XML to Business Entity. . . . 131
SOAP Services. . . . 131
DaaS Providers. . . . 131
DaaS Providers Component. . . . 132
Integrating a DaaS Provider. . . . 132
Uploading a WSDL File. . . . 132
Transforming a Business Entity to XML Format. . . . 133
Transforming XML Data to Business Entity. . . . 135
Registering a SOAP Service. . . . 138
Configuring a DaaS Provider. . . . 139
Creating a DaaS Providers Component. . . . 140
Designing a Record View Layout with DaaS. . . . 141
Deleting Transformations. . . . 141
Deleting WSDL files, SOAP Services, and DaaS Providers. . . . 141

Chapter 15: Configuring External Calls. . . . 142

Overview. . . . 142
Configuring External Calls. . . . 143
Uploading a WSDL File. . . . 143
Registering a SOAP Service. . . . 144
Configuring an External Call. . . . 144

Chapter 16: Designing the Data Director User Interface. . . . 145

Designing the Data Director User Interface Overview. . . . 145
User Interface Design Examples. . . . 146

User Interface Design Prerequisites. . . . 146
User Interface Layouts. . . . 146
Home Page Layouts. . . . 147
Additional Page Layouts. . . . 148
Record View Layouts. . . . 148
Default Record Views. . . . 153
Primary User Interface Components. . . . 154
Task Inbox. . . . 155
Dashboard Reports. . . . 156
Record Details. . . . 157
Matching Records (Advanced). . . . 158
Matching Records. . . . 159
Cross-reference Records. . . . 160
History. . . . 160
Hierarchy Manager. . . . 161
Secondary User Interface Components. . . . 161
Chart. . . . 162
Matched Records. . . . 167
Source Systems. . . . 167
Related Records. . . . 168
Related Tasks. . . . 169
Similar Records. . . . 169
Hierarchy Widget. . . . 172
Custom User Interface Components 172
External Links. . . . 173
How to Design the Data Director User Interface. . . . 175
Creating a Data Director Application. . . . 175
Configuring the Task Inbox Component. . . . 176
Creating the Related Records Component. . . . 179
Creating the Similar Records Component. . . . 180
Creating the External Links Component. . . . 181
Creating the Chart Component. . . . 182
Designing a Home Page Layout. . . . 187
Designing an Additional Page Layout. . . . 189
Designing a Business Entity View Layout. . . . 190
Designing a Matching Records View Layout. . . . 194
Designing a Cross-reference Records View Layout. . . . 196
Designing a History View Layout. . . . 197
Designing a Review Task Layout. . . . 199
Designing a Merge Task Layout. . . . 201
Designing an Unmerge Task Layout. . . . 203
Designing the Quick View Tab for the Hierarchy View. . . . 205

Designing the Quick View Tab for the Network View. 207
Updating Home Page Layouts. 209
Updating Additional Page Layouts. 209
Updating Record View Layouts. 210
Configuring Task Manager. 211
Hiding the Default Record Views. 212

Chapter 17: Localizing Data Director. 214

Localizing Data Director Overview. 214
Localizing Metadata. 214
Localizing Task Actions, Types, and Messages. 215
Localizing Lookup Tables. 216
Localizing Labels and Error Messages. 218

Appendix A: Provisioning Tool Frequently Asked Questions. 219

Provisioning Tool Frequently Asked Questions. 219

Index. 221

Preface

Use the Informatica® *Multidomain MDM Provisioning Tool Guide* to learn how to use the Provisioning tool to configure business entities, business entity views, transformations, Informatica Data Director applications, tasks, relationships, and full-text search. You can create a Data Director application and design pages and views to display master data.

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CHAPTER 1

Introduction

This chapter includes the following topics:

- Provisioning Tool Overview, 12
- Architecture, 13
- Prerequisites, 13
- Key Concepts, 14
- Logging In to the Provisioning Tool, 14
- Configure Single Sign-On in MDM Provisioning Tool, 15
- Guidelines for Multiple Users to Collaborate, 16

Provisioning Tool Overview

You can use the Provisioning tool to define business entity models, tasks, and transformations, and design the user interface for Data Director.

Define

You can create and edit the following definitions:

Definition	Description
Business entities	Define business entity models. Business entities represent entities with significance to an organization. Organizations commonly define business entity types to represent customers, suppliers, employees, products, and accounts.
Business entity views	Define business entity view models. Business entity views represent a condensed version of a business entity.
Reference entities	Define reference entities. Reference entities are business entities that are associated with lookup base objects.
Relationships	Define the relationships between the base objects in the business entities.
Transformations	Define structural transformations and data transformations.

Definition	Description
Task configurations	Define who can accept tasks, define who can launch business processes, and define the default task properties.
Extensions	Define SOAP services and WSDL URLs.

Design

You can create and design the following user interface items:

User Interface Item	Description
Applications	Create an application for use in Data Director. .
Components	Create user interface components to manage business entities.
Layouts	Design user interface layouts.

Architecture

The Provisioning tool reads schema information from an Operational References Store (ORS) in the MDM Hub. You can configure business entities based on the schema information and then publish the configuration. The configuration XML file is saved in the C_REPOS_CO_CS_CONFIG repository table.

As you work on the configuration files, you save your changes to a temporary workspace. The changes are not applied until you publish the changes. If multiple users change the business entity configuration for an ORS simultaneously, the repository is updated with the most recently published configuration.

The Provisioning tool runs on the same application server as the Hub Server.

Prerequisites

To use the Provisioning tool, the Operational Reference Store (ORS) must contain a defined MDM schema. A defined schema contains base objects and constraints based on foreign key relationships between base objects.

Prerequisites when using business entities for web services or when using Data Director

To add nodes to a business entity, you must first define the base objects. Use the MDM Hub Console to define base objects.

To configure the business entity model, you must first define the relationship constraints between the base objects. Use the MDM Hub Console to define foreign key relationship constraints.

Key Concepts

To work effectively in the Provisioning tool, you need a basic understanding of business entities and Multidomain MDM concepts.

Business Entities

Business entities represent entities with significance to an organization. Organizations commonly define business entity types to represent customers, suppliers, employees, products, and accounts. For example, a business entity type might be Person. The customer John Smith is a business entity of type Person.

An organization can also define business entity types for data that is unique to the business. For example, a charity defines donors as a type of business entity. A medical device manufacturer defines unique device identifiers. Many organizations define more than one business entity type. The application developer controls which business entity types exist in a Data Director application.

In the MDM Hub Store, a business entity corresponds to a record in a base object table. The parent record contains identifying information for the business entity. The parent record has a relationship to child records that contain data related to the business entity, such as addresses and telephone numbers.

Reference Entities

Reference entities are business entities that are associated with lookup base objects.

A lookup base object is a table that contains a list of reference data. For example, a lookup table for a phone type field could contain the values Home, Work, and Mobile. The reference entity data can then be used to populate a list of values in Data Director.

Root Node

The root node is the uppermost node in the business entity model and defines the business entity type.

The root node is synonymous with the business entity. For example, a business entity model with a Person root node is considered to be a Person business entity. After you create the root node, you can then define the business entity model. The nodes in the business entity contain information about the attributes of the root node.

Logging In to the Provisioning Tool

To log in, you need the URL for the Provisioning tool and your user credentials. The URL contains the MDM Hub server host name and port number. If you do not have this information, contact your MDM administrator.

1. Open a supported browser.
2. Enter the URL for the Provisioning tool using one of the following formats:

   Secure connections. `https://<MDM Hub Server host name>:<MDM Server port number>/provisioning/`

   Non-secure connections. `http://<MDM Hub Server host name>:<MDM Server port number>/provisioning/`

   The **Log In** page opens.

3. Enter your user name and password.
4. Click **Log In**.
5. Select the database for which you want to configure business entities.

   The Provisioning tool opens and displays the **Home** workspace.

Troubleshooting Tip: If you encounter issues, try the following solutions:

- If the application does not launch, contact your MDM administrator. The administrator might need to restart the application server.
- If you cannot log in to the application and you are already running another MDM application in the browser, open a new browser window to log in to the application. The issue occurs because the application server that runs the MDM Hub uses the same session ID for both applications.

Configure Single Sign-On in MDM Provisioning Tool

You can configure Single Sign-On (SSO) using SAML or OAuth in the MDM Provisioning Tool to enable secure authentication through an external Identity Provider (IdP). Before you begin configuring SSO for the Provisioning Tool, ensure that you have the required information from your IdP/IT team.

Prerequisites

Before you begin Single Sign-On (SSO) configuration for the Provisioning tool, ensure you have the following information:

1. IDP URL for the Provisioning tool.
   Request the IDP URL from your IT team.

   Note: `IDPMetadata.xml` contains the required information.
2. Host Name and Port Number for the Assertion Consumer Service (ACS) URL.
   To access the console, use the following format:

   `http://<host>:<port>/cmx/sso/provisioning/`

Single Sign-On (SSO) operations utilize the ACS URL to post the Security Assertion Markup Language (SAML) response to the Service Provider. After successful authentication, you are redirected to the ACS URL to exchange SAML assertions.

Note: Ensure the name ID in the Provisioning tool matches with the user name of MDM external user.

For details on how to configure SAML SSO using the existing Custom Login Utility, see the Informatica Knowledge Base article Using Custom Log in Utility to Configure SAML SSO.

Enabling Single Sign-On

The following property enables or disables the Single Sign-On (SSO) option on the MDM Provisioning Tool login page:

cmx.prov.enable.mdm.login.with.sso

Enable the property to make the SSO option available on the MDM Provisioning tool login page. Default is false.

Accessing Single Sign-On from the Login Page

Single Sign-On allows users to log in to the MDM Provisioning tool through an Identity Provider using SAML or OAuth-based authentication.

After you enable Single Sign-On in the Provisioning tool and run the application JAR, you can see the SSO option available by default on the IDP login page:



Guidelines for Multiple Users to Collaborate

When multiple users make concurrent changes, plan the tasks for each user to ensure that they don't update the same component. When multiple users update the same component, the tool retains the last published update. To view the changes published by another user, log out and log in to the tool.

When multiple users access the Provisioning tool, consider the following recommendations:

- Ensure that each user has one active session.
- Ensure that multiple users do not work on the same component or fields.
- Ensure that the users save their updates frequently to avoid session timeout error.

CHAPTER 2

Configuring the Data Director Application

This chapter includes the following topics:

- Configuring the Data Director Application Overview, 17
- Creating a Data Director Application, 18
- Hiding Business Entities from an Application (Optional), 19
- Configuring the Search or Query Results Display, 20
- Configuring Date and Time Formats, 20

Configuring the Data Director Application Overview

An application is the main configuration and deployment unit for Data Director. The application is what business users see when they launch and log in to Data Director. Use the Application Editor in the Provisioning tool to add and modify the applications.

You can create an application for each database in the MDM Hub environment. You associate each application with a source system, which the application uses to track data changes. To create a source system, use the Systems and Trust tool in the Hub Console. Ensure that the source system that you use is configured to have the highest level of trust. The trust level guarantees that the changes made by the application users override any other contributing value and ends up in the master record.

You can configure the application to include the search box or query form, or both. You can also configure the business entity view that you want to use to display the results of a search or query.

When you add, modify, or delete an application, to see the changes in Data Director, publish the changes to the MDM Hub.

Note: If you upgraded from a version prior to 10.2, ensure that you create an application in the Provisioning tool with the same name as the existing Data Director application name. Configure all the application properties including the search options that you want to use.

Creating a Data Director Application

The user interface layout that you want to design must be associated with a Data Director application. To create a Data Director application, use the Application Editor.

1. From the **Database** list, select the database to which you want to associate your configurations.
2. Click **Configuration** > **Application Editor**.

   The **Applications** page appears.
3. Click **Create**.
4. In the properties panel, specify the following properties:

Property	Description
Name	Name of the application that appears in the Applications panel.
Display Name	Name of the application that appears in Data Director.
Source System	Source system with which you want to associate the application.
Session Timeout (minutes)	Time in minutes to wait before an idle Data Director session can time out.
Display Default Record Views	Enables the default record views in Data Director. The default record views appear as tabs in opened records.
Enable Draft	Enables the **Drafts** menu option in the Data Director navigation bar. Specifies whether Data Director users can add, edit, and submit business entity records in the draft state.
Display attributes in row order	Indicates whether to display the attributes that you add to a record details component in row order. You can add the attributes in the Record Details Component Designer dialog box. By default, Data Director displays the attributes in column order.
Smart search	Enables the **Search** menu option in the Data Director navigation bar.
Queries	Enables the **Query** menu option in the Data Director navigation bar.
Queries dialog in application	This option appears when you enable both the **Smart search** and the **Queries** options. Enable the **Queries dialog in application** option to allow the use of queries to search for records within a record view, such as **Matching Records** or while using a component, such as **Related Records**. **Note:** Both the **Query** and the **Search** menu options appear in the Data Director navigation bar.

5. Click **Apply**.

   The application that you created appears in the tree view panel and in the **Applications** panel.
6. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.

b. Review the changes or publish without a review.
   - To publish without a review, click **Publish**.
   - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Hiding Business Entities from an Application (Optional)

By default, an application includes all the business entities that are defined for an Operational Reference Store. You can hide specific business entities that you do not want within an application. Users can add records and search for records of the visible business entities.

1. Click **Configuration** > **Application Editor**, and then select the application.
2. In the tree view panel, click the **Business Entities** node.
3. To create a configuration for a business entity within the application, click **Create**.
4. In the properties panel, specify the following properties:

Property	Description
Business Entity	The business entity that you want to configure for the application. From the **Business Entity** list, select the business entity.
Visible	Specifies whether the business entity is visible from the **New** tab and the **Search** box. To hide a business entity in the current application, add the business entity to the application and clear the **Visible** check box.

5. Click **Apply**.

   The configuration is saved to the temporary workspace.
6. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Configuring the Search or Query Results Display

If you use the Elasticsearch engine, for each application, configure the business entity view that you want to use to display the results of a search or query. If you do not configure a business entity view, the search or query results contain the full business entity.

1. Click **Configuration** > **Application Editor**, and then select the application.
2. In the tree view panel, select **Search Configuration**, and click **Create**.
3. In the properties panel, select a business entity and the business entity view that you want to use to display the search or query results.

   If you do not select a business entity view, the search and query results contain all the business entity fields.
4. Optionally, if you configured search, select the filters and configure the display order of the search filters.
   a. Click the **Edit** icon next to **Filter Display Order**.

      The **Edit Filter Display Order** dialog box appears. The dialog box contain filters, which are fields that are configured as filterable in the business entity model.
   b. Drag the filters from the **Available Filters** section to the **Selected Filters** section.
   c. To configure the order, drag and move the filters up or down.
   d. Click **OK**.
5. Click **Apply**.

   The search configuration is saved to the temporary workspace.
6. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Configuring Date and Time Formats

When you create a Data Director application, you can configure custom date and time formats. The default date and time formats are yyyy-MM-dd and yyyy-MM-dd HH:mm:ss.

1. Click **Configuration** > **Application Editor**, and then select the application.
2. In the tree view panel, click the **Default Display Formats** node.
3. In the **Default Display Formats** panel, enter custom formats in the **Date Format** and **Datetime Format** fields.
4. Click **Apply**.

   The date and time formats are saved to the temporary workspace.

5. Publish the changes to the MDM Hub.
    a. Click **Publish**.

       A confirmation dialog box appears that prompts you to publish or review the changes.
    b. Review the changes or publish without a review.
        - To publish without a review, click **Publish**.
        - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

After you configure the date and time formats on the **Applications** page, use the **Modeling** page to set the display format for business entity fields. You set the display format as **Date** or **Datetime** for fields that display a date.

CHAPTER 3

Establishing a Root Node

This chapter includes the following topics:

- Establishing a Root Node Overview, 22
- Root Node Properties, 23
- Creating a Business Entity Root Node, 24
- Creating a Reference Entity Root Node, 25
- Enabling State Management Validation, 26
- Setting Strategies for Record Deletion and Record Checking, 26

Establishing a Root Node Overview

The root node is the uppermost node in the business entity model and defines the business entity type. When you create a business entity model, the first step is to create a root node.

After you establish the root node, you can configure the properties of the root node. After you create the root node, you can define the business entity model according to your business entity model.

You can also configure reference entities. Reference entities are business entities that are associated with lookup base objects. After you establish a lookup root node, you can then configure the properties for the lookup root node.

If you want to use business entities with Data Director, you can generate business entities from subject areas in the IDD Configuration Manager. Business entities created directly in the Provisioning tool do not work in Data Director.

If you do not use Data Director, but use business entity services as web services to directly access business entities, you can use business entities created in the Provisioning tool. You can also edit business entities already defined in your MDM schema.

Example of Adding Root Nodes

You work in Human Resources for a large multinational corporation. You want to add all the employees in your organization. You want to add each employee as a "Person" business entity.

To add the root node for a Person business entity, perform the following steps:

1. Select **Business Entity** > **Modeling**, and then select **Business Entities**.
2. Click **Create**.
3. In the properties panel, select **C_PARTY** in the **Base Object** field.

4. Enter `Person` as the name and label for the root node.
5. Define the model of the business entity in the business entity model pane.
6. Save your changes.

Because employees in your organization live in different countries, you then decide that you want to create the reference entity "LUCountry." This will be associated with the Country lookup base object in the MDM schema.

To add the root node for a LUCountry reference entity, you perform the following steps:

1. Select **Business Entity** > **Modeling**.
2. Click **Create**.
3. In the properties panel, select **C_LU_COUNTRY** in the **Base Object** field.
4. Enter `LUCountry` as the name and label for the root node.
5. Save your changes.

Prerequisites

To add nodes to a business entity, you must first define the base objects. Use the MDM Hub Console to define base objects.

For more information about defining base objects, see the *Multidomain MDM Configuration Guide*.

Root Node Properties

Define the properties of the root node to configure how the business entity appears in Data Director. You can also enable state management for the business entity.

Name

Name of the root node or the business entity model. The name appears only in the Provisioning tool.

Label

Label for the business entity that you want to appear within views in Data Director.

If you do not specify a label or a label format for the business entity, the name appears as the label for the business entity.

If you localize the metadata for Data Director and use the metadata bundle file, `besMetadata.properties`, the labels and label formats in the metadata bundle file take precedence over the labels and label formats configured in the properties panel.

Description

Optional. Description of the root node.

State Management Enabled

Determines if state management is enforced among all nodes in the business entity.

Base Object

The base object in the MDM Hub with which you want to associate the root node.

Label Format

Optional. Label for the business entity that consists of field values instead of hard coded text. The label format that you configure appears in the views in Data Director.

Syntax of the label format is {<field name 1>}, {<field name n>}. For example, use {lastName}, {firstName} to display the values of the last name and the first name as the label.

If you specify a label format and a hard coded label, a label with the specified format appears in all the views in Data Director.

If you do not specify a label or a label format, the business entity name appears as a label for the business entity in the views in Data Director.

If you localize the metadata for Data Director and use the metadata bundle file, `besMetadata.properties`, the labels and label formats in the metadata bundle file take precedence over the labels and label formats configured in the properties panel.

Color

The color that represents the business entity in Data Director. For example, the color appears as the background color for business entity icons in the **New** dialog box and record views, such as the business entity view, **Hierarchy** view, and **Network** view.

Business Entity Icon

Optional. The icon that represents the business entity in Data Director. For example, the icon appears in the **New** dialog box and record views, such as the business entity view, **Hierarchy** view, and **Network** view.

Business Entity Record Icon

Optional. A field that Data Director users can specify an image URL or attach an image to represent the record. The icon appears in record views, such as the business entity view, **Hierarchy** view, and **Network** view. For more information the image URL or file attachment data types, see "Predefined Custom Data Types" on page 39.

If you do not specify a field for the business entity record icon, then Data Director uses the business entity icon.

Creating a Business Entity Root Node

To create a business entity model, the first step is to create the root node for the business entity. For example, you can create a **Person** business entity associated with the base object C_PARTY in the MDM schema.

1. Click **Business Entity** > **Modeling**, and then select **Business Entities**.
2. Click **Create**.

   A new root node appears among the business entities.
3. In the **Name** field, enter a name for the business entity.
   For example, enter `Person`.
4. In the **Label** field, enter a display name for the business entity.
   For example, enter `Person`.
5. Optionally, enter a description of the business entity.
6. Optionally, enable the **State Management Enabled** check box.

When this check box is enabled, the Repository Manager validation checks that all base objects in the business entity are enabled for state management, except referenced lookups.

7. In the properties panel, in the **Base Object** field, select a base object from the MDM schema to associate with the business entity.
   For example, select the base object C_PARTY.

   Note: The base objects are created in the MDM Hub before you add root nodes to the business entity.

8. Optionally, configure the colors and icons for the business entity and individual records.

   a. To configure a color to represent the business entity, in the **Color** field, click the color swatch icon, and then select a color.

   b. To configure an icon to represent the business entity, in the **Business Entity Icon** field, select an icon source, and then configure the icon.

   c. To specify a field where users can specify an icon to represent the record, in the **Business Entity Record Icon** field, select the field.

      You must have a field in the business entity that uses a data type that supports images. For more information, see "Predefined Custom Data Types" on page 39.

9. Click **Apply**.

   The changes are saved but are not published to the MDM Hub.

Related Topics:

- "Defining the Business Entity Model Overview" on page 28

Creating a Reference Entity Root Node

Reference entities are business entities that are associated with lookup base objects. As with a business entity, the first step is to create the root node for the reference entity. For example, you can create a reference entity associated with the lookup base object LUCountry in the MDM schema.

Before you begin, create the lookup base object in the MDM Hub Console. To avoid potential case sensitivity issues, ensure that all the codes and values in the lookup base object follow a consistent capitalization strategy. For example, all values start with an uppercase letter or all codes are in uppercase.

1. Click **Business Entity** > **Modeling**, and then select **Reference Entities**.
2. Click **Create**.

   A new root node appears among the reference entities.

3. In the **Name** field, enter a name for the reference entity.
   For example, enter `LUCountry`.
4. In the **Label** field, enter a display name for the reference entity.
   For example, enter `LUCountry`.
5. Optionally, enter a description of the reference entity.
6. Optionally, enable the **State Management Enabled** check box.

   When this check box is enabled, the Repository Manager validation checks that all base objects in the reference entity are enabled for state management, except referenced lookups.

7. In the **Base Object** field and select a lookup base object from the MDM schema to associate with the reference entity.
   For example, select the base object C_LU_COUNTRY.

8. Click **Apply**.

   The changes are saved but are not published to the MDM Hub.

Related Topics:

- "Defining the Business Entity Model Overview" on page 28

Enabling State Management Validation

To enable the Repository Manager to check that state management is enabled consistently for all base objects in a business entity, set the attribute `stateManagementEnabled` to `true` in the root element. The value of `stateManagementEnabled` is `false` by default.

You can enable state management through the **State Management Enabled** check box.

When `stateManagementEnabled` is `true` the Repository Manager validation checks that all base objects in the business entity are enabled for state management, except referenced lookups.

When `stateManagementEnabled` is `false` the Repository Manager does not verify that the base objects in the business entity are all state management enabled. The business entity can contain base objects that are state management enabled and base objects that are not state management enabled.

The value of `stateManagementEnabled` affects the Repository Manager validation check. The value of `stateManagementEnabled` does not affect run-time operations. When you change `stateManagementEnabled` from `true` to `false`, state management is not disabled for base objects.

Setting Strategies for Record Deletion and Record Checking

By default, the MDM Hub always identifies existing records by the ROWID_OBJECT value and by the primary key and source system. However, you can set additional strategies to determine how the MDM Hub deletes and identifies child base object records.

Delete Strategy

Determines the behavior when you delete a child record.

When you enable the **Delete Strategy** check box, select one of the following values:

Values	Description
DISASSOCIATE	Records remain active but the foreign key is set to null. Default is DISASSOCIATE.
SOFT_DELETE	Soft-deletes the record. The MDM Hub sets the HUB_STATE_IND to -1.
DELETE_PARENT	Deletes the record from the relationship table. The child record is not deleted.

Identifying Strategy

Determines how the MDM Hub checks for an existing child record. When you add a child record to a business entity, the MDM Hub determines if the record is new or already exists.

When you enable the **Identifying strategy** check box, configure the following options:

Option	Description
One Row	Determines the behavior when a record exists. Select one of the following values: - Accept. The MDM Hub does not create a child record. Default is Accept. - Create. The MDM Hub creates a child record - Error. The MDM Hub does not create a child record. An error occurs.
No Row	Determines the behavior when a record does not exist. Select one of the following values: - Create. The MDM Hub creates a child record - Error. The MDM Hub does not create a child record. An error occurs.
Unique Field	Field by which the MDM Hub identifies whether a child record is unique.
Label Format	Specifies the format of labels shown in the user interface. A label can be configured to include a subset of fields from the business entity record. For example, the Person: {lastname}, {firstname} will be shown as follows in the user interface: Person: John, Smith.

CHAPTER 4

Defining the Business Entity Model

This chapter includes the following topics:

- Defining the Business Entity Model Overview, 28
- Prerequisites, 29
- Constraints, 29
- Node Types, 30
- Creating a Node with a One-to-Many Relationship, 32
- Creating a Node with a One-to-One Relationship, 32
- Creating a Child Lookup Node with a One-to-One Relationship, 33
- Copying a Business Entity, 34

Defining the Business Entity Model Overview

The business entity model is a tree model of nodes. Each node corresponds to a base object table in the MDM Hub. Each field corresponds to a column of a base object table in the MDM Hub.

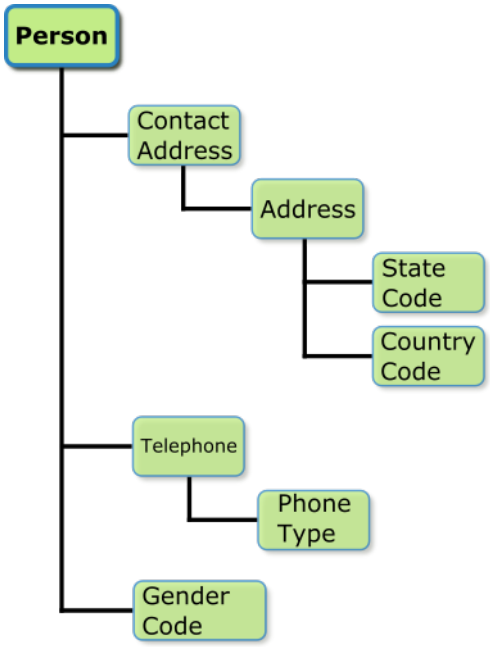
The business entity model is based upon a root node. The root node is synonymous with the business entity. For example, a business entity model with a Person root node is considered to be a Person business entity.

After establishing the root node, use the tree panel to create child nodes that have a one-to-one or one-to-many relationship with their parent nodes.

Example of Configuring the Business Entity Model

You are configuring the business entity model for the Person business entity. The Person business entity contains data for the employees in your company. Under the Person root node, you want to add nodes for the employee details such as gender, address, and phone.

The following image shows the model of the business entity example:



Prerequisites

To configure the business entity model, you must first define the relationship constraints between the base objects. Use the MDM Hub Console to define foreign key relationship constraints.

To add nodes to a business entity, you must first define the base objects. Use the MDM Hub Console to define base objects.

For more information about defining foreign key relationships and defining base objects, see the *Multidomain MDM Configuration Guide*.

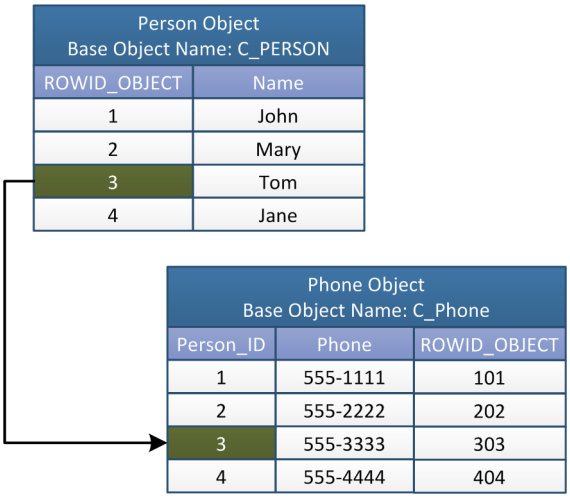
Constraints

The MDM Hub uses foreign key constraints to maintain relationships between child objects and parent objects. When you create a node in the business entity model, you must specify the MDM Hub constraint that defines the relationship between the child node and the parent node.

A foreign key is a field in a relational table that matches the primary key column of another table. In a foreign key relationship, the child object has a foreign key column and the parent object has a primary key column.

The MDM Hub uses the keys to associate a parent record with a child record. The foreign key column of the child record contains the value of the associate parent primary key.

The following image shows a Phone child object linked to a Person parent object by a foreign key relationship:



Person Object
Base Object Name: C_PERSON

ROWID_OBJECT	Name
1	John
2	Mary
3	Tom
4	Jane

Phone Object
Base Object Name: C_Phone

Person_ID	Phone	ROWID_OBJECT
1	555-1111	101
2	555-2222	202
3	555-3333	303
4	555-4444	404

The Person object has a primary key column called ROWID_OBJECT and the Phone object has a foreign key column called Person_ID. The primary key value of 3 links the record for Tom to the record for the phone number 555-3333.

The constraint attribute requires the following syntax:

```
<child base object name>(<foreign key column name>).<parent base object name>(<zprimary key column name>)
```

The following code shows the constraint attribute for the example:

```
constraint="C_PHONE(PERSON_ID).C_PERSON(ROWID_OBJECT)"
```

Node Types

You build the business entity model by adding nodes of different types. Each node type defines the nature of the relationship between the parent node and child node.

The business entity model can contain the following node types:

Business entity name

The name you give the business entity establishes the root node of the business entity.

one

Specifies a one-to-one relationship between a parent node and a child node.

For example, a one-to-one relationship between a Contact Address node and an Address node means that each contact address can only have one address associated with it. A person can have a home contact address and a work contact address, but only one address can be associated with the home contact address and only one address can associated with the work contact address.

many

Specifies a one-to-many relationship between a parent node and a child node.

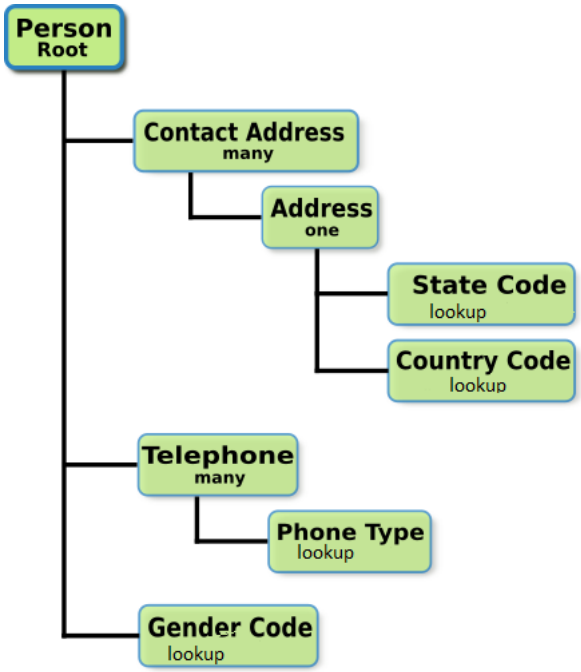
For example, a one-to-many relationship between a Person node and a Telephone node means that a person record can have many telephone number records associated with it.

lookup (referenceOne)

Specifies a one-to-one relationship between a parent node and a child reference entity node.

For example, a referenceOne relationship between a Person node and a Gender node means that a person record can be associated with only one gender value. The gender values reside in a lookup table.

The following image shows the node types for each node in the business entity example:



You can also configure the lookup business entity field to point to another business entity, instead of a reference entity. For example, the Person business entity can refer to the Automobile or Organization business entity.

The Data Director user interface depends on the object that you select. The following criteria affect the Data Director user interface:

- If the selected object is a reference entity, the business entity data view in edit mode renders this lookup field as a drop-down field.
- If the selected object is a business entity, the business entity view shows a link to the referenced business entity.

Both fields appear as text fields in any read-only view. For example, they appear as text fields in the History, Match and Merge, and Cross-reference Records views.

Creating a Node with a One-to-Many Relationship

Create a node with a one-to-many relationship when the parent node can contain multiple child nodes. You create a telephone child node with a one-to-many relationship to a person node because one person can have more than one telephone number.

1. Click **Business Entity** > **Modeling**, and then select **Business Entities**.
2. Select the business entity.
3. In the tree view panel, select **One to Many** under the parent node, and then click **Create**.
4. In the properties panel, select the base object that contains the child node data.

   For example, select the base object C_PARTY_PHONE.

   Note: The base objects are created in the MDM Hub before you configure the business entity model.
5. Select the constraint reference that defines the MDM Hub relationship between the base object of the parent node and the base object of the child node. The constraints are created in the MDM Hub before you configure the business entity model.

   For example, select the constraint C_PARTY_PHONE(PARTY_ID).C_PARTY(ROWID_OBJECT).

   Note: The Data Director user interface depends on the object that you select in the Provisioning tool. If the selected object is a reference entity, the business entity data view in edit mode renders this lookup field as a drop-down field. If the selected object is a business entity, the business entity view shows a link to the referenced business entity.
6. Enter a node name and a node label name. Optionally, enter a node description.

   For example, enter `Telephone` for the node name and node display name.
7. In the **Lookup Display Field** list, select a field name to display for the lookup table in Data Director.

   For example, select the lookup table display name **genderDisp.**

   Note: The lookup field and the lookup display field name are created in the Provisioning tool.
8. Click **Apply**.

   The changes are saved but are not published to the MDM Hub.

Creating a Node with a One-to-One Relationship

Create a child lookup node with a one-to-one relationship when the parent node contains one child lookup node. You create an address child node with a one-to-one relationship to a contact address node because one person can only have one address that is a home contact address.

1. Click **Business Entity** > **Modeling**, and then select **Business Entities**.
2. Select the business entity.
3. In the tree view panel, select **One to One**, and then click **Create**.
4. In the properties panel, select the base object that contains the child node data.

   For example, select the base object C_ADDRESS.

   Note: The base objects are created in the MDM Hub before you configure the business entity model.
5. Select the constraint reference that defines the MDM Hub relationship between the base object of the parent node and the base object of the child node.

For example, select the constraint C_RL_PARTY_ADDRESS(ADDRESS_ID).C_ADDRESS(ROWID_OBJECT).

Note: The constraints are created in the MDM Hub before you configure the business entity model.

6. Enter a node name and a node display name. Optionally, enter a node description.

   For example, enter `Address` for the node name and node label name.

7. Click **Apply**.

   The changes are saved but are not published to the MDM Hub.

Creating a Child Lookup Node with a One-to-One Relationship

Create a child lookup node with a one-to-one relationship when the parent node contains one child lookup node. You create a child reference node for a gender lookup with a one-to-one relationship to a person.

Before you begin, in the MDM Hub, ensure that you configured the lookup base object and the relationship between the base object of the parent node and the lookup base object.

1. Click **Business Entity** > **Modeling**, and then select **Business Entities**.
2. Select the business entity.
3. In the tree view panel, select **Lookup Fields**, and then click **Create**.
4. In the properties panel, enter a node name and a label.

   For example, enter `Gender` for the name and label.

5. From the **Object** list, select the base object that contains the lookup child node data.

   For example, select the lookup base object `LUGender`.

6. From the **Constraint** list, select the relationship between the base object of the parent node and the base object of the lookup node.

   For example, select the constraint `C_PARTY(GENDER_CD).C_LU_GENDER(GENDER_CODE).`

7. Optionally, configure default values for user roles.

   Note: The picklist in Data Director defaults to this value for the specified user roles.

   a. Beside the **Default Value** heading, click the **Add** icon.

   b. Enter a default value that matches a value in the selected lookup base object. When entering dates, use the GMT format, such as `2018-06-06T09:34:25.801-04:00`.

      Note: The Provisioning tool does not validate the values. You can test the default values from Data Director.

   c. From the **User Role** list, select user roles and then click the **Add** icon. To specify all user roles, select the asterisk (*).

      The selected user roles appear. To remove a user role, click the **Delete** icon beside the role.

   d. Optionally, to use this default value with the user roles that have no other default values configured for this field, select **Apply this value to the user roles that have no default value assigned**.

   e. To configure more default values, repeat these substeps.

8. Click **Apply**.

   The changes are saved but are not published to the MDM Hub.

Copying a Business Entity

After you define a business entity, you can copy the business entity and use it as a template to quickly create a similar business entity.

1. Click **Business Entity** > **Modeling**, and then select **Business Entities**.
2. Select the business entity and click **Copy**.

   The **Copy Business Entity** dialog box appears.
3. Enter a unique name.
4. Select the nodes to copy and click **Save & Publish**.

   A copy of the business entity is created and the changes are published to the MDM Hub.
5. To change the business entity, in the tree view panel, edit the fields and child nodes as required.

CHAPTER 5

Configuring Business Entity Properties

This chapter includes the following topics:

- Configuring Business Entity Properties Overview, 35
- Field Properties, 35
- Custom Data Types, 39
- Node Labels in Data Director, 40

Configuring Business Entity Properties Overview

Configure the business entity properties in the right-hand panel of the Provisioning tool.

In the business entity properties panel, you can perform the following tasks:

- Add filters to a field.
- Configure node labels.
- Associate a base object with a node. You configure the root node properties when you create a root node. For more information, see the chapter Establishing a Root Node.
- Associate a constraint with a child node. You configure the child node properties when you define the business entity model. For more information, see Chapter 4, “Defining the Business Entity Model” on page 28.
- Enable fields to be searchable and configure search properties. For more information about configuring searchable fields and configuring search properties, see Chapter 10, “Configuring Search” on page 87
- Assign default values to fields and associate the default values with user roles.

Field Properties

You can configure the field properties to associate a field with a base object column, specify the data type, mark the field as read-only, mark the field as required, and make the field searchable.

You can configure the following field properties:

Name

Name of the field that you want to appear in the tree view panel.

Label

Label for the field that you want to appear within views in Data Director.

Read Only

Defines whether the field is editable. If this option is clear, end users can edit the field in a record view if they have create or update privileges for the base object field in the MDM Hub.

If the user has create or update privileges for the base object column in the MDM Hub, the field is editable in Data Director.

If the user does not have create or update for the base object column in the MDM Hub, an error occurs when the user tries to save changes in Data Director.

For information about configuring resource privileges, see the *Multidomain MDM Configuration Guide*.

Required

Defines if users must enter data in the field when a business entity is created or updated.

URI

Defines the namespace where data types are declared. The data type that you want to configure depends on the URI that you select.

To configure basic data types, such as string, integer, and boolean, select `commonj.sdo`. To configure Informatica data types, such as image URL, hyperlink, and file attachment, select `urn:co-types.informatica.mdm`.

Type

Data type of the field. The available data types for configuration depends on the URI that you select.

You can configure the following data types:

Boolean, Date, Decimal, File attachment, Hyperlink, Image URL, Integer, String, Textarea, and Rich text editor.

By default, the data type of a business entity field is as close as possible to the data type of the base object column that the field is associated with. For example, a string column in a base object might contain information for an image, so you configure the image URL data type for the business entity field. If you want a field to display web URL, file URI, FTP link, and email address as hyperlinks, configure the hyperlink data type for the field.

Note: You cannot open file URIs with Google Chrome and Mozilla Firefox because of security reasons.

Display Format

The display format for the date fields. Before you specify the display format for the date fields, specify the URI and set the Type property to **Date**.

You can select the **Date** or **Datetime** format for business entity fields that display a date. The display format that you configure is applied to the date field in all the Data Director views.

If a display format is not set for a date field, the date format of the field is the same as the format of the associated base object column.

Filter

Defines a static filter to regulate the data that users can or cannot enter in a field.

When you enable the Filter property, specify the following fields:

- **Operator**. Controls whether the specified values are allowed or not allowed in the field.

Option	Description
In	Displays the values to the user. If you add a list of comma-separated values as allowed values, Data Director users can choose a value from a lookup list. **Note:** If you also configure a field filter on the same field, the displayed values reflect the intersection of the static filter and the field filter. For more information about field filters, see "Field Filter Rules and User Roles" on page 119.
Not In	Does not display the values to the user.

- **Value**. Add values that match the data type of the field.

Requirement	Description
Add multiple values	Comma-separated list of field values. Based on the operator you select, Data Director determines whether to display these values. For example, `Person,SecurePerson,Employee.`
Allow null value	To allow NULL data values for fields within a particular business entity, enter empty double quotes ("") as a value. For example, Person, "". For example, you can configure the firstName field of the Person business entity to match NULL values. When a Data Director user performs a search, the search results include the records where the First Name field is empty.
Allow date and time values	Specifies whether records with the field that matches the date and time appear in Data Director search results. Add the date and time values in the following format: `yyyy-MM-ddTHH:mm:ssTZD.` For example, `2021-06-18T00:00:00Z`

Note: The Provisioning tool does not validate the values. If you add a value of the wrong data type, the value does not appear in Data Director.

Searchable

A search request searches only the fields that you configure as searchable fields. The scope of a search request increases with the increased number of searchable fields. Multiple searchable fields might affect the performance of the search requests, so configure only the significant fields as searchable fields. For example, configure fields that contain full names, organization names, or email addresses as searchable fields rather than fields that contain country codes, gender codes, or address types.

When you enable the Searchable property, select one or more of the following additional properties for search:

- Search Analyzer
- Suggester
- Sortable
- Filterable
- Facet Range
- Facet

- Displayable

Optionally, specify the language for the field values. The default language is English. For information about the search properties, see the chapter about search in the *Multidomain MDM Configuration Guide*.

Column

The base object column associated with the field. You associate a field with a base object column when you add a field to the node.

Related Topics:

- "Custom Data Types" on page 39

Adding a Field to a Node

After you create a node in the business entity model, you can add fields to the node. The fields correspond to columns in the base object that is associated with the node. For example, you want to add the City field to the Address node. The Address node is associated with the C_ADDRESS base object, which has a CITY_NAME column.

1. Click **Business Entity** > **Modeling**, and then select **Business Entities**.
2. Select the business entity.
3. In the tree view panel, select **Fields**, and then click **Create**.
4. In the properties panel, select the base object column to associate with the field from the **Column** list.

   For example, select `City_Name`.

   Note: You created the base object columns in the MDM Hub.
5. Enter a field name and a field label name.

   For example, enter `City` for the field name and field label name.
6. Optionally, configure the other field properties. For more information, see "Field Properties" on page 35.
7. Optionally, configure default values for user roles.

   Note: If you also configured a filter with allowed values, the picklist in Data Director defaults to this value for the specified user roles.

   a. Beside the **Default Value** heading, click the **Add** icon.

   b. Enter a default value that matches the data type of the field. If you configured filters with allowed values, the default value must match one of the allowed values. When entering dates, use the GMT format, such as `2018-06-06T09:34:25.801-04:00`.

      Note: The Provisioning tool does not validate the values. You can test the default values from Data Director.

   c. From the **User Role** list, select user roles.

      To select a user role, select the role and then click the **Add** icon.

      To select multiple user roles, hold the CTRL key, select the roles, and then click the **Add** icon.

      To specify all user roles, select the asterisk (*) and then click the **Add** icon.

      To remove a user role, click the **Delete** icon beside the role.

   d. Optionally, to use this default value with the user roles that have no other default values configured for this field, select **Apply this value to the user roles that have no default value assigned**.

   e. To configure more default values, repeat these substeps.

8. Optionally, configure field filters by user role. See "Adding Field Filters" on page 120.
9. Click **Apply**.

   The changes are saved but are not published to the MDM Hub.

Custom Data Types

By default, the data type for a field is the same as the data type of the base object column that the field is associated with. You can define custom data types for fields. For example, a string column in a base object might contain information for an image so you configure a custom Image data type for the fields.

You define custom data types in a namespace external to the Provisioning tool. When you associate a custom data type with a field, you must enter the location of the namespace where your custom data types are defined.

Predefined Custom Data Types

The MDM Hub includes some predefined custom data types. You can find custom data types from the `co-types.xsd` schema in the `urn:co-types.informatica.mdm` namespace.

You can use the following predefined custom data types:

Image URL

Stores the URLs to access images.

Hyperlink

Converts text into hyperlinks.

The following types of text in entity fields are converted into hyperlinks:

- Web URL
- File URI
- FTP link
- Email address

File Attachment

Stores the files that you attach to the records. The field that you associate with the File Attachment data type shows a browse button to select the document.

You can update the following properties related to the File Attachment data type in the `cmxserver.properties` file:

- `cmx.file.max_file_size_mb`. Maximum size of the file that you can attach in megabytes. Default is 20.
- `cmx.file.allowed_file_extensions`. Lists the extensions of the files that you can store. By default, you can store the .pdf and .jpg files.

For more information about the `cmx.file.max_file_size_mb` and `cmx.file.allowed_file_extensions` properties, see the *Multidomain MDM Configuration Guide*.

Adding a Custom Data Type to a Field

You can associate custom data types with business entity fields. When you associate a custom data type with a field, you enter the location of the resource where your custom data types are defined. For example, you can associate the File Attachment, ImageURL, or Hyperlink data type with a field.

1. Click **Business Entity** > **Modeling**, and then select **Business Entities**.
2. Select the business entity.
3. In the tree view panel, select **Fields**, and then click **Create**.

   The **New Field** form appears in the properties panel.
4. In the **New Field** form, enter a field name and a field label name.

   For example, enter `Picture` for an image, or `Document` for an attachment as the field name and field label name.
5. Select the URI where your custom data types are defined.
6. Select a custom data type.

   Note: Before you add a data type to a field, create the custom data types in the namespace.
7. To associate the field with a base object column, select a base object column.

   Note: Before you add a field to a node, an MDM administrator must create the base object column in the MDM Hub.
8. Optionally, select the **Read Only**, **Required**, **Searchable**, and **Filter** check boxes based on your requirement.

   If you select **Searchable**, additional search properties appear. For information about the search properties, see Chapter 10, “Configuring Search” on page 87.
9. Click **Apply**.

   The changes are saved but are not published to the MDM Hub.

Node Labels in Data Director

You can configure the node labels to define the business entity titles in Data Director.

By default, the node name is the value for the node labels. If you do not configure the node labels, the node name appears as the node label.

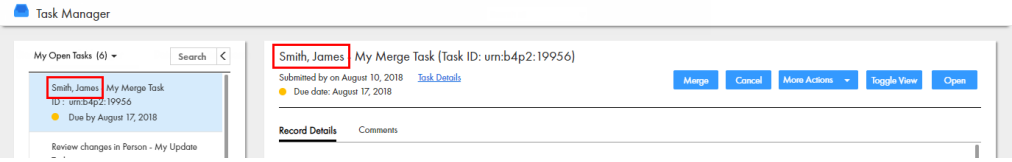
You can hard code text for the labels or you can display field values. To display field values, enter the field name in parenthesis. For example, use `{lastName},{firstName}` to display the values of the last name and first name in the title.

You can configure the following node labels:

Task format

The task title.

The following image shows the Task Manager with a red square around the portion of the task title that you can configure:

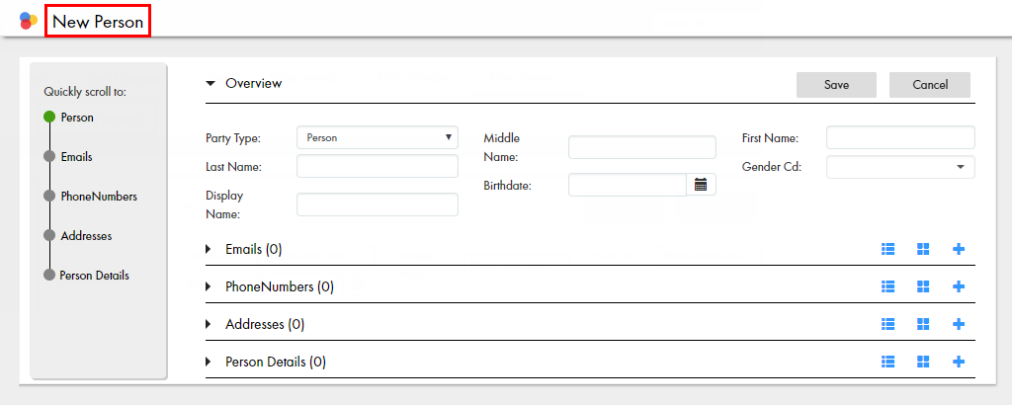


The task format value for this example is `{lastName},{firstName}`.

No fields format

The title of the page that appears when you create a new entity in Data Director.

The following image shows the entity creation page with a red square around the title:



The new format value for this example is `New Person`.

Exists format

The title to display if the field you configured for a title does not contain a value.

Configuring Node Labels

To configure the labels for a node, configure the label formats in the properties panel.

1. Click **Business Entity** > **Modeling**, and then select **Business Entities**.
2. Select the business entity.
3. In the properties panel, select the **Label Format** check box.
4. Enter the node labels.
   a. In the **Task Format** field, enter the format of the task title.

      For example, to display the last name followed by the first name in the task title, enter `{lastName}, {firstName}`. The values in these fields are displayed in the task title separated by a comma.

   b. In the **Exists Format** field, enter the format of the title to display if the field you configured for a title does not contain a value.

      For example, enter `Person`.

c. In the **No Fields Format** field, enter the title that appears when you create a new business entity record.

   For example, enter `New Person`.

5. Click **Apply**.

   The changes are saved but are not published to the MDM Hub.

CHAPTER 6

Configuring Reference Entity Properties

This chapter includes the following topics:

- Configuring Reference Entity Properties Overview, 43
- Lookup Types, 43
- Field Properties, 44
- Custom Data Types, 47
- Configuring a Lookup , 48

Configuring Reference Entity Properties Overview

Configure the reference entity properties in the right-hand panel of the Provisioning tool.

For many users, reference entities are synonymous with lookups.

In the reference entities properties panel, you can perform the following tasks:

- Associate a lookup base object with a node. You configure the root node properties when you create a root node. For more information, see Chapter 3, “Establishing a Root Node” on page 22.
- Enable fields to be searchable and configure search properties. For more information about configuring searchable fields and configuring search properties, see Chapter 10, “Configuring Search” on page 87.

Lookup Types

A lookup is a predefined list of values that are available for selection in a user interface. For business entity models, a regular lookup is linked to a lookup base object that contains a list of reference data. For example, a gender lookup is associated with the C_LU_GENDER base object.

Use dependent lookups when a dependency exists between two lookup tables, such as between a type table and a subtype table. The list of values that display in the subtype table depends on the selection in the type table. Another example of a dependent lookup is when a user populates a list of values for state or province based on the selected country.

Field Properties

You can configure the field properties to associate a field with a lookup base object column, specify the data type, mark the field as read-only, mark the field as required, and make the field searchable.

You can configure the following field properties:

Name

The field name in the reference entity configuration file.

Label

The field name as it appears in the Provisioning tool tree view panel.

Read Only

Defines if the field is editable. If this option is clear, end users can edit the field in a record view if they have create or update privileges for the base object field in the MDM Hub.

If the user has create or update privileges for the base object column in the MDM Hub, the field is editable in Data Director.

If the user does not have create or update privileges for the base object column in the MDM Hub, an error occurs when the user tries to save changes in Data Director.

For information about configuring resource privileges, see the *Multidomain MDM Configuration Guide*.

Required

Defines if users must enter data in the field when a business entity is created or updated.

URI

Defines the namespace where data types are declared. The data type that you want to configure depends on the URI that you select.

To configure basic data types, such as string, integer, and boolean, select `commonj.sdo`. To configure Informatica data types, such as image URL, hyperlink, file attachment, rich text editor, and text area, select `urn:co-types.informatica.mdm`.

Type

The data type of the field. The data type that you can configure depends on the URI that you select. You can configure the following data types: string, integer, decimal, date, boolean, image URL, hyperlink, file attachment, rich text editor, and text area.

By default, the data type of a business entity field is as close as possible to the data type of the lookup base object column that the field is associated with. For example, a string column in a base object might contain information for an image, so you configure the image URL data type for the business entity field. If you want a field to display web URL, file URI, FTP link, and email address as hyperlinks, configure the hyperlink data type for the field.

Note: You cannot open file URIs with Google Chrome and Mozilla Firefox because of security reasons.

Display Format

The display format for the date fields. Before you specify the display format for the date fields, specify the URI and set the Type property to **Date**.

You can select the **Date** or **Datetime** format for reference entity fields that display a date. The display format that you configure is applied to the date field in all the Data Director views.

If a display format is not set for a date field, the date format of the field is the same as the format of the associated base object column.

Filter

Defines a static filter to regulate the data that users can or cannot select in a field.

When you enable the Filter property, specify the following fields:

- **Operator**. Controls whether the specified values are allowed or not allowed in the field.

Option	Description
In	Display the values to the user. If you add a list of comma-separated values as allowed values, Data Director users can choose a value from a picklist.
Not In	Do not display the values to the user.

- **Value**. Add one or more values that exist in the selected lookup base object. To add multiple values, enter a comma-separated list of values. To allow users to leave the field blank, add empty quotes ("") as a value.

  Note: The Provisioning tool does not validate the values. If you add a value that is not in the lookup base object, the value does not appear in Data Director.

Searchable

A search request searches only the fields that you configure as searchable fields. The scope of a search request increases with the increased number of searchable fields. Multiple searchable fields might affect the performance of the search requests, so configure only the significant fields as searchable fields. For example, configure fields that contain full names, organization names, or email addresses as searchable fields rather than fields that contain country codes, gender codes, or address types.

When you enable the Searchable property, select one or more of the following additional properties for search:

- Search Analyzer
- Suggester
- Sortable
- Filterable
- Facet
- Displayable

Optionally, specify the language for the field values. The default language is English. For information about the search properties, see the chapter about search in the *Multidomain MDM Configuration Guide*.

Column

The lookup base object column associated with the field. You associate a field with a base object column when you add a field to the node.

Adding a Field to a Node

After you create a node in the reference entity structure, you can add fields to the node. The fields correspond to columns in the lookup base object that is associated with the node. For example, you want to add the Gender Code field to the LU Gender node. The LU Gender node is associated with the C_LU_GENDER lookup base object, which has a Gender_Code column.

1. Click **Business Entity** > **Modeling**, and then select **Reference Entities**.
2. Select the reference entity.

3. In the tree view panel, select **Fields**, and then click **Create**.
4. In the properties panel, select the base object column to associate with the field from the **Column** list.

   For example, select `Gender_Code`.

   Note: You created the base object columns in the MDM Hub.
5. Enter a field name and a field label name.

   For example, enter `Gender Code`.
6. Optionally, configure the other field properties. For more information, see "Field Properties" on page 44.
7. Optionally, configure field filters by user role. For more information, see "Adding Field Filters to a Reference Entity" on page 46.
8. Click **Apply**.

   The changes are saved but are not published to the MDM Hub.

Adding Field Filters to a Reference Entity

To control which values appear in a lookup list by user role, create a field filter on the reference entity that contains the list. Field filters are case insensitive. Field filters affect lookups, but not dependent lookups.

Note: If you implement field filters on a reference entity, and then in the business entity you add a field filter on a field that uses that reference entity, the field filter on the business entity takes precedence.

1. Click **Business Entity** > **Modeling**, and then select **Reference Entities**.
2. Select the reference entity.
3. In the tree view panel, select **Field Filters**, and then click **Create**.

   A **[New fieldFilter]** option appears under **Field Filters**, and a form opens in the properties panel.
4. In the properties panel, enter a name for the filter.
5. Select a field.
   a. Beside the **Field** field, click the **Browse** button.

      A dialog box displays a list of fields.
   b. Select the field, and then click **Select**.

      The selected field appears in the **Field** field.
6. To define the filter in terms of who *cannot* see values, create deny rules.
   a. Beside the **Deny** heading, click the **Add** icon.
   b. In the **Value** field, enter a value that is in the lookup list.

      Note: When you enter dates, use the date format that your database environment requires.
   c. From the **User Role** list, select the user roles, and then click the **Add** icon.
   d. To make this rule a default for any unassigned user roles in the **Deny** section, select **Apply this rule to the user roles that are not specified in any Deny rule**.
   e. Repeat to configure more filter rules.
   f. Add user roles to the **Remaining Values** rule.
7. To define the filter in terms of who *can* see values, create allow rules.
   a. Beside the **Allow** heading, click the **Add** icon.
   b. In the **Value** field, enter a value that matches the data type of the field.

Note: When entering dates, use the date format required by your database environment.

c. From the **User Role** list, select the user roles, and then click the **Add** icon.

d. To make this rule a default for any unassigned user roles in the **Allow** section, click **Apply this rule to the user roles that are not specified in any Allow rule**.

e. Repeat to configure more filter rules.

f. Add user roles to the **Remaining Values** rule.

8. Click **Apply**.

   The changes are saved but are not published to the MDM Hub.

Custom Data Types

By default, the data type for a field is the same as the data type of the base object column that the field is associated with. You can define custom data types for fields. For example, a string column in a base object might contain information for an image so you configure a custom Image data type for the fields.

You define custom data types in a namespace external to the Provisioning tool. When you associate a custom data type with a field, you must enter the location of the namespace where your custom data types are defined.

Predefined Custom Data Types

The MDM Hub includes some predefined custom data types. You can find custom data types from the `co-types.xsd` schema in the `urn:co-types.informatica.mdm` namespace.

You can use the following predefined custom data types:

Image URL

Stores the URLs to access images.

Hyperlink

Converts text into hyperlinks.

The following types of text in entity fields are converted into hyperlinks:

- Web URL
- File URI
- FTP link
- Email address

File Attachment

Stores the files that you attach to the records. The field that you associate with the File Attachment data type shows a browse button to select the document.

You can update the following properties related to the File Attachment data type in the `cmxserver.properties` file:

- `cmx.file.max_file_size_mb`. Maximum size of the file that you can attach in megabytes. Default is 20.
- `cmx.file.allowed_file_extensions`. Lists the extensions of the files that you can store. By default, you can store the .pdf and .jpg files.

For more information about the `cmx.file.max_file_size_mb` and `cmx.file.allowed_file_extensions` properties, see the *Multidomain MDM Configuration Guide*.

Adding a Custom Data Type to a Field

You can associate custom data types with reference entity fields. When you associate a custom data type with a field, you enter the location of the resource where your custom data types are defined. For example, you can attach files, include hyperlinks, and provide image URLs.

1. Click **Business Entity** > **Modeling**, and then select **Reference Entities**.
2. Select the reference entity.
3. In the tree view panel, select **Fields**, and then click **Create**.

   The **New Field** form appears in the properties panel.
4. In the **New Field** form, enter a field name and a field label name.

   For example, enter `Picture` for an image, `Email` for a hyperlink, or `Document` for an attachment as the field name and field label name.
5. Select the URI, `urn:co-types.informatica.mdm`, where your custom data types are defined.
6. Select a custom data type.

   Note: Before you add a data type to a field, create the custom data types in the namespace.
7. To associate the field with a base object column, select a base object column.

   Note: Before you add a field to a node, an MDM administrator must create the base object column in the MDM Hub.
8. Optionally, select the **Read Only**, **Required**, **Searchable**, and **Filter** check boxes based on your requirement.

   If you select **Searchable**, additional search properties appear. For information about the search properties, see Chapter 10, “Configuring Search” on page 87.
9. Click **Apply**.

   The changes are saved but are not published to the MDM Hub.

Configuring a Lookup

To configure a lookup, create a new reference entity in the tree view panel and then associate the root node with a lookup base object. For example, you can create a gender lookup reference entity related to the C_LU_GENDER lookup base object.

1. Click **Business Entity** > **Modeling**, and then select **Reference Entities**.
2. Select the reference entity.
3. In the tree view panel, select **Fields**, and then click **Create**.
4. From the New Field form in the properties panel, select the lookup base object column to associate with the field from the **Column** list.

   For example, select `Gender_Code` from the **Column** list.

   Note: The lookup base object column is created in the MDM Hub before you add a field to a node.
5. Enter a field name and a field display name.

For example, enter `Gender Code` for the field name and field label name.

6. Optionally, configure the Read Only, Required, URI, Type, Display Format, Filter, and Searchable properties for the field you created.

   If you enable the Searchable property, additional search properties appear. For information about the search properties, see the *Multidomain MDM Configuration Guide*.

7. Click **Apply**.

   The field you created appears in the tree view panel.

8. Create another field. In this example, select `Gender_Disp` from the **Column** list.
9. Enter a field name and a field display name. In this example, enter `Gender Disp` for the field name and field label name.
10. Click **Apply**.

    The changes are saved but are not published to the MDM Hub.

Configuring a Dependent Lookup

To configure a dependent lookup, create a new reference entity in the tree view panel and then associate the root node with a lookup base object. To create a one-to-many relationship within the lookup, in the tree view panel, select **One to Many** under the parent node and then click **Create**. For example, you can create a country lookup reference entity related to the C_LU_COUNTRY lookup base object.

1. Click **Business Entity** > **Modeling**, and then select **Reference Entities**.
2. Select the reference entity.
3. In the tree view panel, select **Fields**, and then click **Create**.
4. From the New Field form in the properties panel, select the base object column to associate with the field from the **Column** list.

   For example, select `Country_Code` from the **Column** list.

   Note: The base object column is created in the MDM Hub before you add a field to a node.

5. Enter a field name and a field display name.

   For example, enter `Country Code` for the field name and field display name.

6. Optionally, configure the Read Only, Required, URI, Type, Display Format, Filter, and Searchable properties for the field you created.

   If you enable the Searchable property, additional search properties appear. For information about the search properties, see the *Multidomain MDM Configuration Guide*.

7. Click **Apply**.

   The field you created appears in the tree view panel.

8. Create another field. In this example, select `Country_Name_Disp` from the **Column** list.
9. Enter a field name and a field display name. In this example, enter `Country Name Disp` for the field name and field label name.
10. Select **One to Many** under the parent node, and then click **Create**.
11. In the properties panel, select the base object that contains the lookup child node data.
    For example, select the lookup base object C_LU_State.
12. Select the constraint reference that defines the MDM Hub relationship between the base object of the parent node and the base object of the child node.
    For example, select the constraint C_LU_STATE(COUNTRY_CODE).C_LU_COUNTRY(COUNTRY_CODE).

Note: The constraints are created in the MDM Hub before you configure the business entity structure.

13. Optionally, enter a node name, a node label name, and a node description.
    For example, enter LU State for the node name and node label name.
14. Click **Apply**.

    The changes are saved but are not published to the MDM Hub.

CHAPTER 7

Transforming Business Entities and Views

This chapter includes the following topics:

- Transforming Business Entities and Views Overview, 51
- Transformation Sources and Targets, 52
- Cleanse Transformations, 53
- Transformations in Read and Write Processes, 53
- Modeling Business Entity Views , 55
- Copying a Business Entity View, 56
- Configuring Business Entity to View Transformations, 57
- Configuring a Transformation to Map a View to a Business Entity, 58
- Configuring Business Entity to Business Entity Transformations, 60
- Deleting Transformations, 62

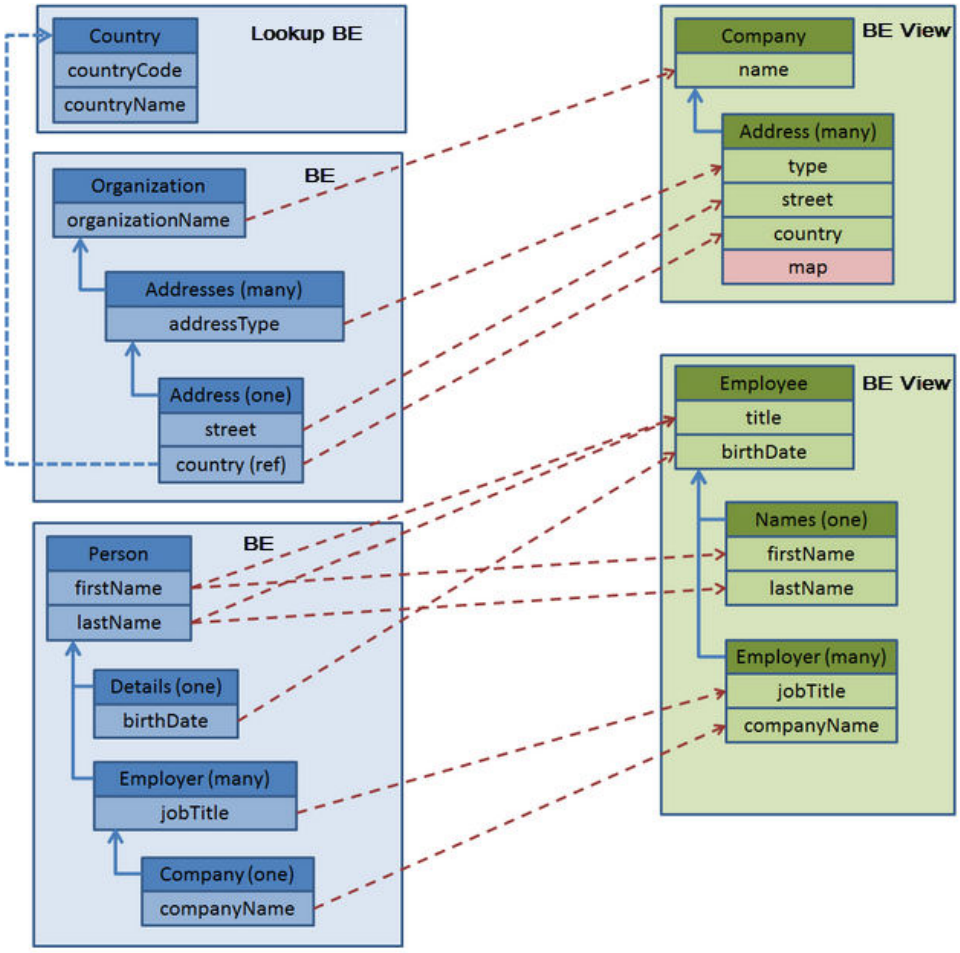
Transforming Business Entities and Views Overview

A business entity represents a comprehensive, universal definition of an entity for your organization. However, users and departments in your organization might only require a condensed version of the business entity. You can transform a business entity into a condensed version called a business entity view.

If Data Director or a web application reads from or writes to a business entity view, the read or write service performs the transformation. If you use business entity views in Data Director or a web application, you must configure business entity to view transformations for read operations, and view to business entity transformations for write events.

You can configure cleanse transformations to validate and cleanse the data during a transformation. You can also configure transformations to leverage DaaS to enrich the data in a business entity.

The following image shows two business entities transformed into two business entity views that have a simpler structure:



To configure a business entity to business entity view, first create the business entity view, and then map the transformations. To create a business entity to a view, model the structure of the view. After you model the view, map the transformations from the business entity to the business entity view.

Transformation Sources and Targets

You can configure the transformations for the following source-to-target combinations:

Business Entity to View

For configuring a condensed version of a business entity as a business entity view to create an object that is easier to consume. For example, you can configure a view that only includes the fields required by a business process, user, or department. The business entity to view transformation is required when you search for and access a business entity.

View to Business Entity

For configuring business entity view to business entity transformation to make data updates to a view. The data is stored in the business entity, so the view to business entity transformation propagates any

data changes made to the view. The view to business entity transformation is required when you create a business entity.

Business Entity to Business Entity

For configuring validation and data cleansing for a business entity. For a business entity to business entity transformation, the cleanse transformations on the fields in the Fields branch are applied during the write process.

Note: The cleanse transformations do not apply to the fields in the One to Many branch or the Lookup Fields branch of the business entity.

Business Entity to XML

For DaaS configuration.

XML to Business Entity

For DaaS configuration.

Cleanse Transformations

You can configure a cleanse-type transformation to use a cleanse function to cleanse, standardize, and validate data.

The following examples show how you can use cleanse transformations:

- Concatenate a given name and a surname name
- Standardize all instances of Doctor to Dr.
- Validate a postal address

For more information about cleanse functions, see the *Multidomain MDM Configuration Guide*.

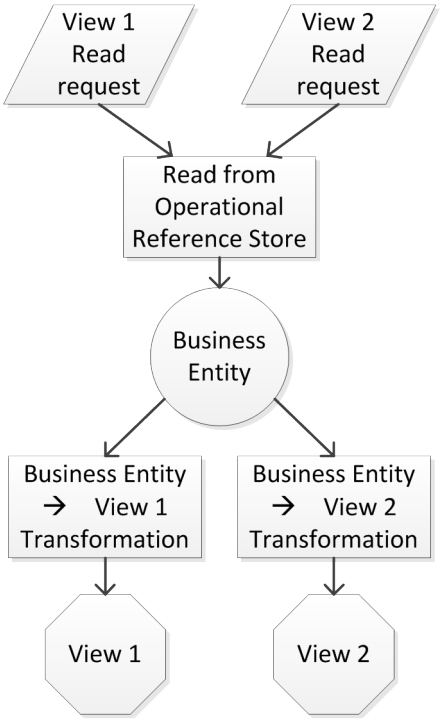
Transformations in Read and Write Processes

The business entity to view and business entity to business entity transformations participate in the business entity read process. View to business entity and business entity to business entity transformations participate in write processes.

The following read process occurs when you open a business entity view:

1. A business entity service request reads data with the database.
2. The business entity transforms to a business entity view.

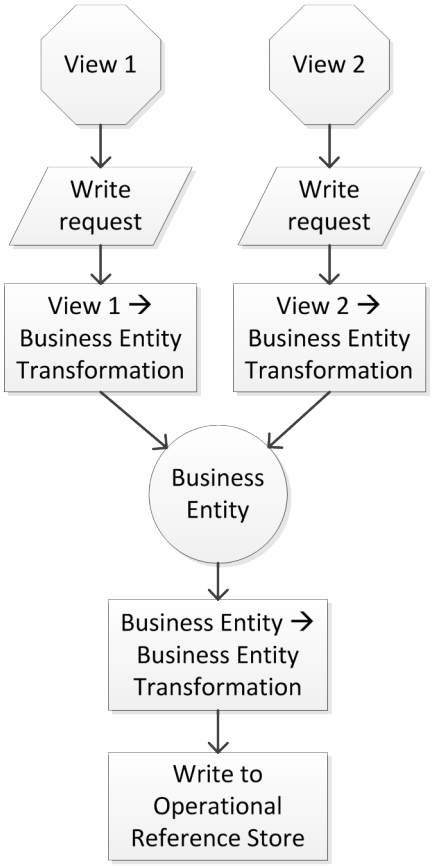
The following image shows the transformations in the business entity read process:



The following process occurs when data changes during a write process:

1. A view transforms into a business entity.
2. The business entity validates and cleanses the data to business entity transformation.
3. The process writes the data to the database, which is the Operational Reference Store.

The following image shows the transformations in the business entity write process:



Modeling Business Entity Views

Before you configure transformations, you must create and configure business entity views. You can define the structure to model a business entity view.

To configure a business entity view, perform the following steps:

1. Create the business entity view root note.
2. Add fields to the root and child nodes.
3. Add child nodes.

Related Topics:

- "Role Privileges for a Business Entity View" on page 118

Creating the Business Entity View Root Node

When you create a business entity view, you first establish a root node.

1. Click **Business Entity** > **Modeling**, and then select **Business Entity Views**.
2. Click **Create**.

A root node for the business entity view appears.

3. In the properties panel, enter a view name and label, and then select a business entity to associate with the view.

   The business entity you select is the entity that you will transform into the view.

4. Click **Apply**.

Adding Fields to the Root and Child Nodes

After you establish a root node, add the root node fields from the business entity that you want to transform into the view.

1. Add fields to the root node.
   a. In the tree view panel, select **Fields** under the root node, and then click **Create**.
   b. In the properties panel, enter the field properties.
   c. Click **Apply**.
2. Add fields to the child nodes.
   a. In the tree view panel, select **Fields** under the child node, and then click **Create**.
   b. In the properties panel, enter the field properties.
   c. Click **Apply**.
3. In the properties panel, enter a field name.

   All other fields and selections are optional. For more information, see "Field Properties" on page 35.

4. Click **Apply**.

   The child node is created.

Adding Child Nodes

You can add a child node that has a one-to-one or one-to-many relationship with the parent.

1. Select the type of relationship the child node has with the parent node.
   - To add a child with a one-to-one relationship, expand **One to One**, and then click **Create**.
   - To add a child with a one-to-many relationship, expand **One to Many**, and then click **Create**.
2. Enter a name, and optionally a label for the child node.
3. Click **Apply**.

   The child node is created.

Copying a Business Entity View

After you define a business entity view, you can copy the business entity view and use it as a template to quickly create a similar business entity view.

1. Click **Business Entity** > **Modeling**, and then select **Business Entity Views**.
2. Select the business entity view and click **Copy**.

   The **Copy Business Entity View** dialog box appears.

3. Enter a unique name.
4. Select the nodes to copy and click **Save & Publish**.

   A copy of the business entity view is created and the changes are published to the MDM Hub.
5. To change the business entity view, in the tree view panel, edit the fields and child nodes as required.

Configuring Business Entity to View Transformations

Configure transformations to map business entity fields to business entity view fields.

To map the transformations, perform the following steps:

- Create the transformation groups to contain the one-to-many child transformations.
- Add direct transformations to map the business entity fields to the business entity view fields.

Configuring Transformations for One-to-Many Child Nodes

To configure transformations for child nodes, add the node to a group, and then configure the transformations.

1. Click **Business Entity** > **Transformations**.
2. Create the business entity to view transformation.
   a. From the **Transformations** list, select **Business Entity to View.**
   b. Click **Create**.
   c. In the properties panel, enter the transformation properties.
   d. Click **Apply**.
3. In the tree view panel, select **Groups**, and then click **Create**.
4. In the properties panel, specify the following properties:

Property	Description
Name	Unique name for the node.
Source	Specifies the child node of the business entity.
Target	Specifies the child node of the view.

5. Repeat step 1 to step 4 for each child node.
6. Click **Apply**.

   You can now configure the transformations for the child nodes.

Configuring a Direct Transformation

You can directly map data from a business entity field to a view field without changing the data.

1. Click **Business Entity** > **Transformations**, and then select **Business Entity to View**.
2. Select a business entity to view transformation.

3. In the tree view panel, select **Transformations**, and then click **Create**.
4. In the properties panel, specify the following properties:

Property	Description
Name	Unique name for the transformation.
Status Output	Not applicable for a direct transformation. Output column in the cleanse function that indicates whether the cleanse function validated the data from the business entity. Use to validate the output of the cleanse function.
Status Success	Not applicable for a direct transformation. Specifies the value that the cleanse function returns when the data in the status output column is successfully validated. If the status output and status success values are provided, any value in the status output column other than the status success value is displayed to the end user as an error message. Use this property to validate the output of the cleanse function.
Type	The type of transformation. Select **direct**.
Input Fields	Specifies the business entity fields to map to the view.
Output Fields	Specifies the view fields to which the business entity fields map to.
Value	Optional. Used to enter an input constant when the input field or output field are not defined.

5. Click **Apply**.

Configuring a Transformation to Map a View to a Business Entity

Configure transformations to map a business entity view to a business entity.

You can configure a transformation so the data maps to the view unchanged. You can also transform the data with a cleanse function so that the data that was entered into a business entity view can be cleansed and validated before moving to the business entity.

To map the transformations, perform the following steps:

- Create the transformation groups to contain the one-to-many child transformations.
- Add transformations to map the view fields to the business entity fields. You can configure the following types of transformations:
  - Direct transformations to move the data unchanged.
  - Cleanse transformations to transform the data before it populates the business entity.

Configuring Transformations for Child Nodes

To configure transformations for child nodes, add the node to a group, and then configure the transformations.

1. Click **Business Entity** > **Transformations**, and then select **View to Business Entity**.

2. Select the view to business entity transformation.
3. In the tree view panel, select **Groups**, and then click **Create**.
4. In the properties panel, specify the following properties:

Property	Description
Name	Unique name for the node.
Source	Specifies the child node of the view.
Target	Specifies the child node of the business entity.

5. Repeat step 3 to step 4 for each child node.
6. Click **Apply**.

   You can now configure the transformations for the child node.

Configuring a Direct Transformation

You can directly map data from a view field to a business entity field without transforming the data.

1. Click **Business Entity** > **Transformations**, and then select **View to Business Entity**.
2. Select a view to business entity transformation.
3. In the tree view panel, select **Transformations**, and then click **Create**.
4. In the properties panel, specify the following properties:

Property	Description
Name	Unique name for the transformation.
Status Output	Not applicable for a direct transformation. Output column in the cleanse function that indicates whether the cleanse function validated the data from the business entity. Use to validate the output of the cleanse function.
Status Success	Not applicable for a direct transformation. Specifies the value that the cleanse function returns when the data in the status output column is successfully validated. If the status output and status success values are provided, any value in the status output column other than the status success value is displayed to the end user as an error message. Use this property to validate the output of the cleanse function.
Type	The type of transformation. Select **direct**.
Input Fields	Specifies the business entity fields to map to the view.
Output Fields	Specifies the view fields to which the business entity fields map to.
Value	Optional. Used to enter an input constant when the input field or output field are not defined.

5. Click **Apply**.

Configuring a Cleanse Transformation

When you map a business entity field to a view field, you can choose to cleanse the data with a cleanse transformation. The cleanse transformation cleanses data for write operations on that particular business entity view.

1. Click **Business Entity** > **Transformations**, and then select **View to Business Entity**.
2. Select the view to business entity transformation.
3. In the tree view panel, select **Transformations**, and then click **Create**.
4. In the properties panel, specify the following properties:

Property	Description
Name	Unique name for the transformation.
Status Output	Output column in the cleanse function that indicates whether the cleanse function validated the data from the business entity. Use to validate the output of the cleanse function.
Status Success	Specifies the value that the cleanse function returns when the data in the status output column is successfully validated. If the status output and status success values are provided, any value in the status output column other than the status success value is displayed to the end user as an error message. Use this property to validate the output of the cleanse function.
Type	The type of transformation. Select **cleanse**.
Cleanse Library	Specifies the cleanse library that contains the cleanse function that you want to use to transform the data.
Function	Specifies the cleanse function that you want to use to transform the data.
Input Parameters	Specifies the view field to use as an input to a business entity field. To specify the view field, click the **Browse** button next to the **input** field.
Output Parameters	Specifies the business entity field to receive the transformed data. To specify the business entity field, click the **Browse** button next to the **output** field.
Value	Optional. Can be used to enter an input constant to use as a value separator when you use functions such as **Concatenate**.

5. Click **Apply**.

Configuring Business Entity to Business Entity Transformations

Configure business entity to business entity transformations to cleanse the entity data.

To configure business entity to business entity transformations, perform the following steps:

- Create the transformation groups to contain the one-to-many child transformations.
- Add cleanse transformations to map the business entity fields to the business entity fields.

Configuring Transformations for Child Nodes

To configure transformations for child nodes, add the node to a group, and then configure the transformations.

1. Click **Business Entity** > **Transformations**, and then select **Business Entity to Business Entity**.
2. Select the business entity to business entity transformation.
3. In the tree view panel, select **Groups**, and then click **Create**.
4. In the properties panel, specify the following properties:

Property	Description
Name	Unique name for the node.
Source	Specifies the child node of the business entity.
Target	Specifies the same child node of the business entity selected in the **Source** field.

5. Click **Apply**.

   You can now configure the transformations for the child nodes.

Configuring a Cleanse Transformation

You can apply a cleanse transformation to a business entity field that you map to itself. The cleanse transformation cleanses data for write operations to all business entity views.

1. Click **Business Entity** > **Transformations**, and then select **Business Entity to Business Entity**.
2. Select the business entity to business entity transformation.
3. In the tree view panel, select **Transformations**, and then click **Create**.
4. In the properties panel, specify the following properties:

Property	Description
Name	Unique name for the transformation.
Status Output	Output column in the cleanse function that indicates whether the cleanse function validated the data from the business entity. Use to validate the output of the cleanse function.
Status Success	Specifies the value that the cleanse function returns when the data in the status output column is successfully validated. If the status output and status success values are provided, any value in the status output column other than the status success value is displayed to the end user as an error message. Use this property to validate the output of the cleanse function.
Type	The type of transformation. Select **cleanse**.
Cleanse Library	Specifies the cleanse library that contains the cleanse function that you want to use to transform the data.
Function	Specifies the cleanse function that you want to use to transform the data.

Property	Description
Input Parameters	Specifies the view field to use as an input to a business entity field. To specify the view field, click the **Browse** button next to the **input** field.
Output Parameters	Specifies the business entity field to receive the transformed data. To specify the business entity field, click the **Browse** button next to the **output** field.
Value	Optional. Can be used to enter an input constant to use as a value separator when you use functions such as **Concatenate**.

5. Click **Apply**.

Deleting Transformations

You can delete transformations. If you delete a transformation, the components or extension that depend on the transformation are also deleted.

CHAPTER 8

Configuring Hierarchy and Network Relationships

This chapter includes the following topics:

- Configuring Hierarchy and Network Relationships Overview, 63
- Hierarchy Relationships, 64
- Network Relationships, 65
- Prerequisites, 66
- Configuring Hierarchy Relationships, 67
- Configuring Network Relationships, 72
- Configuring Hierarchy Tasks, 77
- Configuring Default Hierarchy for the Dashboard, 80
- Designing Data Director Interfaces to Manage Relationships, 81

Configuring Hierarchy and Network Relationships Overview

A relationship is a connection between two business entities. Use the Provisioning tool to configure hierarchy relationships and network relationships between business entities. Then Data Director users can use the relationships between business entities to create relationships between records.

You can create the following relationships:

hierarchy relationship

A connection between two business entities that is hierarchical. For example, the Organization business entity the parent of the Person business entity.

network relationship

A connection between two related business entities. For example, the Person business entity is related to the Credit Card business entity.

Important: If you have relationships in Hierarchy Manager that you want to continue to use, you must upgrade your relationships and create them as hierarchy relationships or network relationships. For instructions about upgrading hierarchies, see the *Multidomain MDM Upgrade Guide*.

Hierarchy Relationships

A hierarchy consists of a root business entity node and hierarchy relationships that define parent-child relationships between business entities. When you create a hierarchy, you must specify a business entity as the root node. Then you add a business entity to the hierarchy and configure a hierarchy relationship from the root business entity to the business entity.

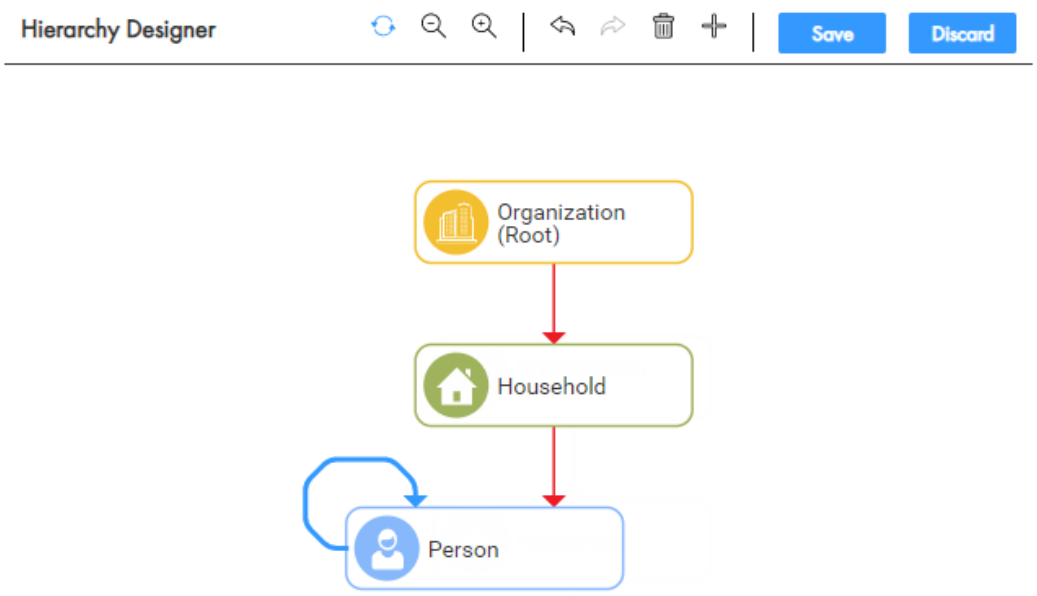
You can add additional business entities and configure hierarchy relationships between the business entities. Each hierarchy relationship in a hierarchy must have a direct or indirect relationship to the root business entity. You can create as many hierarchy relationships as you need to represent the hierarchy model.

For example, your organization needs to track customers by households. You might create the following hierarchy and hierarchy relationships:

1. Create an Organization hierarchy.
2. Add the Organization business entity as the root business entity.
3. Create a relationship from the Organization business entity to the Household business entity.
4. Create a relationship from the Household business entity to the Person business entity.

Then in Data Director, users can create a hierarchy relationship from the Informatica record to the Smith record. Create a relationship from the Smith record to the John Smith record, and then create a relationship from the Smith record to the Jane Smith record.

The following image shows a sample hierarchy in the Hierarchy Designer:



Network Relationships

The network is a collection of network relationships between business entities. A network relationship is a connection between related business entities.

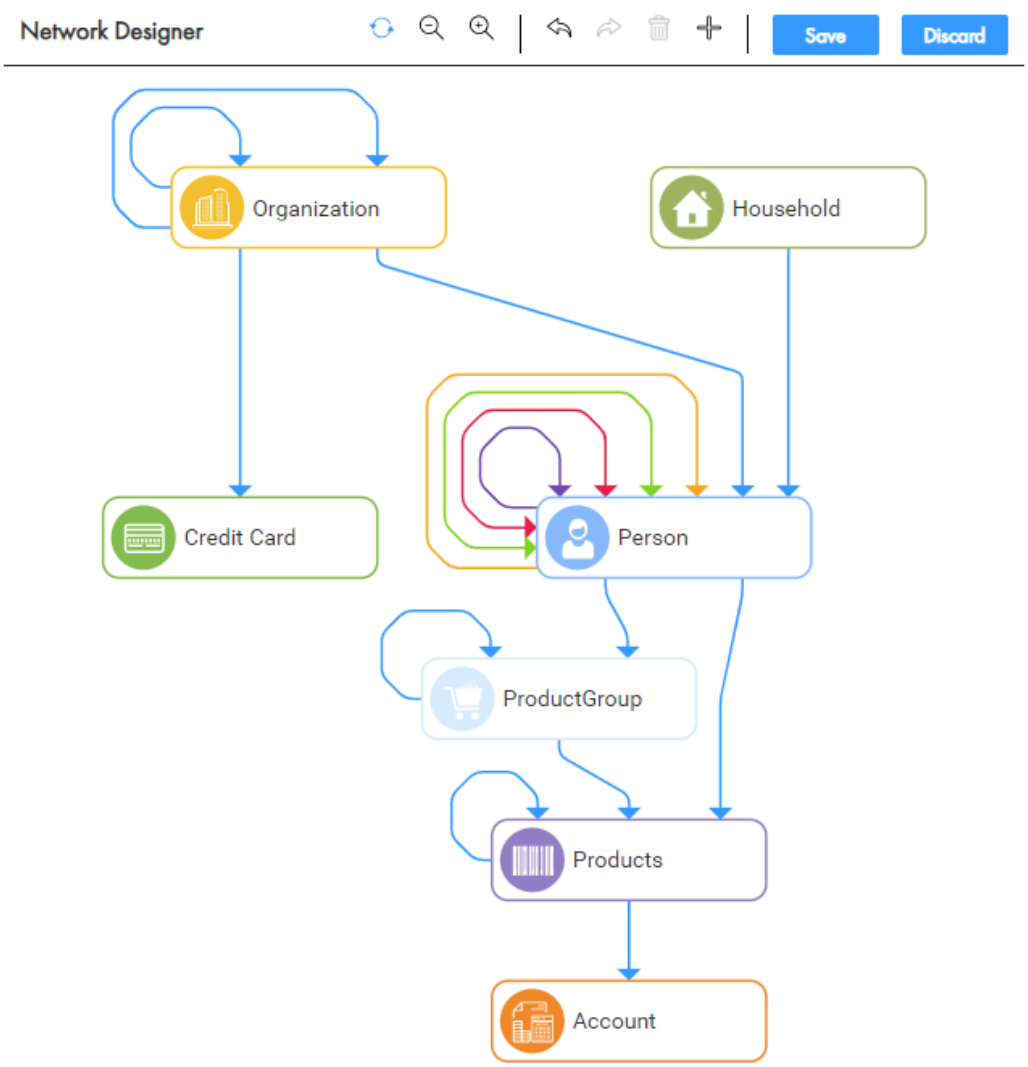
When you edit the network, you add business entities and configure network relationships between the business entities. Each network relationship becomes part of the network of relationships. The network does not contain a root node.

For example, your organization needs to track related business entities. You might create the following network relationships:

1. Create a relationship from the Person business entity to the Products business entity.
2. Create a relationship from the Organization business entity to the Person business entity.
3. Create a relationship from the Organization business entity to the Credit Card business entity.
4. Create a relationship from the Household business entity to the Person business entity.

Then in Data Director, users can create a network relationship between the John Smith record and the X100 Keyboard record. Users can also create a network relationship from the Smiths record to the John Smith record.

The following image shows a sample network in the Network Designer:



Prerequisites

Before you configure the relationships, perform the following prerequisite tasks:

1. Create base objects in the MDM Hub.
2. Define the business entity structure.
3. If you want to add additional attributes for a relationship, ensure that the required columns are present in the base object that maintains the relationship.

Configuring Hierarchy Relationships

Configure a hierarchy by adding a root business entity node, adding business entities, and then configuring hierarchy relationships between business entities.

To configure hierarchy relationships, perform the following steps:

1. Create a hierarchy.
2. Configure a root node in a hierarchy.
3. Add a business entity to a hierarchy.
4. Configure a hierarchy relationship.
5. Configure the hierarchy settings.
6. Optionally, configure additional relationship attributes.

Tip: You can configure icons that represent business entities and business entity records in Data Director. When Data Director users view business entities in the **Hierarchy** view, the icon for each record appears. For more information configuring icons, see "Creating a Business Entity Root Node" on page 24.

Step 1. Create a Hierarchy

Create a hierarchy to contain a collection of hierarchy relationships. For example, you want to configure the organization hierarchy for your business.

1. Click **Business Entity** > **Modeling**, and then select **Hierarchies**.
2. Click **Create** > **New Hierarchy**.
3. In the properties panel, specify the following hierarchy properties:

Property	Description
Name	Unique name for the hierarchy.
Label	Label for the hierarchy.
Description	A meaningful description for the hierarchy.
Search type	Algorithm to search for records in the hierarchy. Select one of the following search algorithms for the hierarchy: - BFS. An algorithm that starts the search from the root node and moves from left to right along the hierarchy tree. The algorithm explores all nodes on one child level before proceeding to the next level. BFS uses the queue data structure to find the shortest path. Use this algorithm in a narrower hierarchy tree with many child levels. - DFS. An algorithm that starts the search from the root node and moves from top to bottom of the hierarchy tree. The algorithm explores each branch of the tree until it reaches the bottom node before proceeding onto the next branch. DFS uses the stack data structure to find the shortest path. Use this algorithm in a wider hierarchy tree with fewer child levels.

4. Click **Apply**.

The hierarchy name appears in the modeling panel and the hierarchy structure appears in the tree view panel.

Step 2. Configure the Root Node in a Hierarchy

Configure the root node in the hierarchy. For example, you have an Organization Structure hierarchy. You might define the Organization business entity as the root node.

1. Click **Edit Hierarchy**.

   The Hierarchy Designer opens with an undefined root node in the canvas.
2. Select the **Root Business Entity (Undefined)** node.
3. In the properties panel, select the business entity that you want to specify as the root node of the hierarchy.
   For example, select Organization.
4. Click **Apply**.

Step 3. Add a Business Entity to a Hierarchy

Add a business entity to a hierarchy. For example, you have an Organization Structure hierarchy. You configured the Organization business entity as the root node. You want to add the Person business entity so that you can create a relationship from the Organization business entity to the Person business entity.

1. Click the add icon.

   An undefined node appears on the canvas.
2. In the properties panel, select a business entity.
   For example, select **Person**.
3. Click **Apply**.

Step 4. Configure a Hierarchy Relationship

Configure the relationship between two business entities in a hierarchy. For example, you have an Organization Structure hierarchy with the Organization business entity as the root node and the Person business entity as a node. You want to configure the relationship between the Organization business entity and the Person business entity. The relationship might be that Organization employs Person.

1. Hover over the business entity from which you want to create a relationship.

   The add icon and arrow icon appear.
2. Click the arrow icon, and then release the arrow over the node to which you want to create a relationship.

   For example, on the Organization business entity, click the arrow icon, and then release the arrow over the Person business entity.
3. To define an relationship between nodes, perform the following actions:
   a. Select the red arrow between two business entity nodes.

      The relationship properties appear in the properties panel.

b. In the properties panel, specify the following relationship properties:

Property	Description
Name	Unique name for the relationship.
Description	A meaningful description for the relationship.
Data-driven Relationship based on	Lookup field that dynamically defines the relationship between records. A data-driven relationship uses data in lookup tables to define relationships between records in Data Director. For example, you create a relationship from the Organization business entity to the Person business entity. The relationship is data-driven based on the relName field. Then in Data Director, when users create a relationship from the Informatica record to the John Smith record, the relationship is defined as Employee. Or, when users create a relationship from the Informatica record to the Jane Smith record, the relationship is defined as CEO.
Label	Label for the relationship that appears in the Data Director user interface.
Label Format	Indicates whether you want to show labels on the user interface.
Exists Format	Title to display when you view the relationship.
No Fields Format	Title to display when all the fields in the Exists Format have NULL values.
Parent Constraint	The column that the foreign key points to.
Child Constraint	The column that the foreign key points to.

c. Click **Apply**.

4. To copy relationship properties from system-suggested similar relationship, perform the following actions:
   a. Select the red arrow between two business entity nodes.

      The relationship properties appear in the properties panel.
   b. In the properties panel, hover over the bulb icon and select the **Similar Relationships Identified** link.

      The Copy Properties window appears.
   c. In the **Similar Relationships** panel, select a relationship with properties that you want to copy.
   d. In the **Select Properties to Copy from** panel, select the relationship properties that you want to copy.
   e. Click **OK**.
   f. Click **Apply**.

Step 5. Configure the Hierarchy Settings

Configure the display and export settings for a hierarchy. For example, when users view a hierarchy in Data Director, you might want 10 child relationships to appear by default. Or, when users export a hierarchy, you might want to include the same fields in the business entity views.

1. Click **Business Entity** > **Modeling**, and then select **Hierarchies**.
2. Select the hierarchy you want to configure.

3. In the tree view panel, click the settings icon.

   The **Hierarchy Settings** dialog box appears.
4. Optionally, on the **Display** tab, enter the following values:
   - Number of expanded child relationships in a hierarchy that you want to appear by default.
   - Number of returned paths in the hierarchy search. The number of returned paths represents the connected paths between the root node and the leaf node including all intermediate nodes. For example, if you specify three as the number of returned paths, the search returns three connected paths from the root node consisting of two child nodes and the leaf node.
5. Click **Export**.

   The **Export** tab opens.
6. Optionally, select the business entity views that contain the fields that you want to include in exports.
7. Click **OK**.
8. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Step 6. Configure Additional Relationship Attributes (Optional)

You can configure additional relationship attributes for a relationship. For example, a relationship between an organization and person can have an additional attributes, such as salary and designation of employee. A relationship between a person and automobile can have additional attributes, such as purchase date, mileage, and price. The fields appear in the Data Director user interface when creating or viewing the relationship.

Before you begin, you must configure hierarchy relationships. Also, the base object that maintains the relationship must have the columns that correspond to the attributes you want to add.

1. Click **Business Entity** > **Modeling**, and then select **Hierarchies**.
2. Select the hierarchy you want to configure.
3. In the tree view panel, expand the relationship for which you want to configure relationship attributes.
4. In the tree view panel, select **Fields**, and then click **Create**.
5. Specify the following properties:

Property	Description
Name	Name of the field.
Label	Display name of the field in the user interface.
Read Only	Indicates whether the field is read only.
Required	Indicates whether you must specify the field when you create a relationship.

<table>
<tr><th>Property</th><th>Description</th></tr>
<tr><td>URI</td><td>URI for where your custom data types are defined, if you want to associate a custom data type with the business entity field,</td></tr>
<tr><td>Type</td><td>Name of the custom data type that you define in the URI. Used with the URI property.</td></tr>
<tr><td>Display Format</td><td>The display format for the date fields.</td></tr>
<tr><td>Filter</td><td>Regulates the data that users can enter in the field. If you enable the Filter property, the following properties appear:
- Operator. Controls whether the specified values are allowed or not allowed in the field.
<table><tr><th>Option</th><th>Description</th></tr><tr><td>In</td><td>Displays the values to the user. If you add a list of comma-separated values as allowed values, Data Director users can choose a value from a lookup list.
Note: If you also configure a field filter on the same field, the displayed values reflect the intersection of the static filter and the field filter. For more information about field filters, see "Field Filter Rules and User Roles" on page 119.</td></tr><tr><td>Not In</td><td>Does not display the values to the user.</td></tr></table>- Value. Add values that match the data type of the field.
<table><tr><th>Requirement</th><th>Description</th></tr><tr><td>Add multiple values</td><td>Comma-separated list of field values. Based on the operator you select, Data Director determines whether to display these values. For example, <code>Person,SecurePerson,Employee.</code></td></tr><tr><td>Allow null value</td><td>To allow NULL data values for fields within a particular business entity, enter empty double quotes ("") as a value. For example, Person, "".
For example, you can configure the firstName field of the Person business entity to match NULL values. When a Data Director user performs a search, the search results include the records where the First Name field is empty.</td></tr><tr><td>Allow date and time values</td><td>Specifies whether records with the field that matches the date and time appear in Data Director search results. Add the date and time values in the following format: <code>yyyy-MM-ddTHH:mm:ssTZD.</code>
For example, <code>2021-06-18T00:00:00Z</code></td></tr></table>Note: The Provisioning tool does not validate the values. If you add a value of the wrong data type, the value does not appear in Data Director.</td></tr>
<tr><td>Column</td><td>Business entity field.</td></tr>
</table>

6. Click **Apply**.
7. Repeat step 4 to 6 for each additional relationship attribute that you want to configure.

Configuring Network Relationships

Configure the network by adding business entities to the network and then configuring network relationships between business entities.

To configure the network, perform the following steps:

1. Add a business entity to the network.
2. Configure a network relationship.
3. Configure the network settings.
4. Optionally, configure additional relationship properties.

Tip: You can configure icons that represent business entities and business entity records in Data Director. When Data Director users view business entities in the **Network** view, the icon for each record appears. For more information configuring icons, see "Creating a Business Entity Root Node" on page 24.

Step 1. Add a Business Entity to the Network

Add a business entity to the network. For example, you want to define a network relationship between the Person business entity and the Product business entity.

1. Click **Business Entity** > **Modeling**, and then select **Network**.
2. Click **Edit Network**.

   The Network Designer opens.
3. Click the add icon.

   An undefined node appears on the canvas.
4. In the properties panel, select a business entity.
   For example, select **Person**.
5. Click **Apply**.
6. Repeat step 3 to 5 to add additional business entities.

Step 2. Configure a Network Relationship

Configure the relationship between two business entities in a network. For example, you want to define the relationship between the Person business entity and the Product business entity. The relationship might be that Person owns Product.

1. Hover over the business entity from which you want to create a relationship.

   The add icon and arrow icon appear.
2. Click the arrow icon, and then release the arrow over the node to which you want to create a relationship.

   For example, on the Person business entity, click the arrow icon, and then release the arrow over the Product business entity.
3. To define an relationship between nodes, perform the following actions:
   a. Select the red arrow between two business entity nodes.

      The relationship properties appear in the properties panel.

b. In the properties panel, specify the following relationship properties:

Property	Description
Name	Unique name for the relationship.
Description	A meaningful description for the relationship.
Data-driven Relationship based on	Lookup field that dynamically defines the relationship between records. A data-driven relationship uses data in lookup tables to define relationships between records in Data Director. For example, you create a relationship from the Organization business entity to the Person business entity. The relationship is data-driven based on the relName field. Then in Data Director, when users create a relationship from the Informatica record to the John Smith record, the relationship is defined as Employee. Or, when users create a relationship from the Informatica record to the Jane Smith record, the relationship is defined as CEO.
Label	Label for the relationship that appears in the Data Director user interface.
Label Format	Indicates whether you want to show labels on the user interface.
Exists Format	Title to display when you view the relationship.
No Fields Format	Title to display when all the fields in the Exists Format have NULL values.
Parent Constraint	The column that the foreign key points to.
Child Constraint	The column that the foreign key points to.

c. Click **Apply**.

4. To copy relationship properties from system-suggested similar relationship, perform the following actions:
   a. Select the red arrow between two business entity nodes.

      The relationship properties appear in the properties panel.
   b. In the properties panel, hover over the bulb icon and select the **Similar Relationships Identified** link.

      The Copy Properties window appears.
   c. In the **Similar Relationships** panel, select a relationship with properties that you want to copy.
   d. In the **Select Properties to Copy from** panel, select the relationship properties that you want to copy.
   e. Click **OK**.
   f. Click **Apply**.

You configured a network relationship between two business entities. You can add additional business entities and configure network relationships between the business entities.

Step 3. Configure the Network Settings

Configure the display and export settings for the network. For example, when users view a network in Data Director, you might want 10 relationships to initially appear and three relationships to appear when expanding

a node. Or, when users export a network, you might want to include the same fields in the business entity views.

1. Click **Business Entity** > **Modeling**, and then select **Network**.
2. Select the network relationship you want to configure.
3. In the tree view panel, click the settings icon.

   The **Network Settings** dialog box appears.
4. In the **Display** tab, specify the following display settings:

Setting	Description
Number of Initial Relationships	The initial number of relationships that appear in a network.
Number of Expanded Relationships	The number of relationships that appear when expanding a node in a network.

5. Click **Export**.

   The **Export** tab opens.
6. Optionally, select the business entity views that contain the fields that you want to include in exports.
7. Click **OK**.
8. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Step 4. Configure the Relationship Properties (Optional)

You can configure the relationship properties, such as the relationship line color and additional relationship attributes for a relationship. The relationship line color represents the relationship in a network. The relationship attributes are fields that appear in the Data Director user interface when creating or viewing the relationship.

Before you begin, you must configure a network with network relationships between business entities.

1. Click **Business Entity** > **Modeling**, and then select **Network**.
2. Select the network relationship you want to configure.
3. In the properties panel, from the **Line Color** field, select the icon, and then select a color.

   A preview of the color appears in the icon.
4. In the tree view panel, select **Fields**, and then click **Create**.

5. Specify the following properties:

Property	Description
Name	Name of the field.
Label	Display name of the field in the user interface.
Read Only	Indicates whether the field is read only.
Required	Indicates whether you must specify the field when you create a relationship.
URI	URI for where your custom data types are defined, if you want to associate a custom data type with the business entity field,
Type	Name of the custom data type that you define in the URI. Used with the URI property.
Display Format	The display format for the date fields.

<table>
<tr><th>Property</th><th>Description</th></tr>
<tr><td>Filter</td><td>Regulates the data that users can enter in the field. If you enable the Filter property, the following properties appear:
- Operator. Controls whether the specified values are allowed or not allowed in the field.
<table><tr><th>Option</th><th>Description</th></tr><tr><td>In</td><td>Displays the values to the user. If you add a list of comma-separated values as allowed values, Data Director users can choose a value from a lookup list.
Note: If you also configure a field filter on the same field, the displayed values reflect the intersection of the static filter and the field filter. For more information about field filters, see “Field Filter Rules and User Roles” on page 119.</td></tr><tr><td>Not In</td><td>Does not display the values to the user.</td></tr></table>- Value. Add values that match the data type of the field.
<table><tr><th>Requirement</th><th>Description</th></tr><tr><td>Add multiple values</td><td>Comma-separated list of field values. Based on the operator you select, Data Director determines whether to display these values. For example, <code>Person,SecurePerson,Employee.</code></td></tr><tr><td>Allow null value</td><td>To allow NULL data values for fields within a particular business entity, enter empty double quotes (“”) as a value. For example, Person, “”.
For example, you can configure the firstName field of the Person business entity to match NULL values. When a Data Director user performs a search, the search results include the records where the First Name field is empty.</td></tr><tr><td>Allow date and time values</td><td>Specifies whether records with the field that matches the date and time appear in Data Director search results. Add the date and time values in the following format: <code>yyyy-MM-ddTHH:mm:ssTZD.</code>
For example, <code>2021-06-18T00:00:00Z</code></td></tr></table>Note: The Provisioning tool does not validate the values. If you add a value of the wrong data type, the value does not appear in Data Director.</td></tr>
<tr><td>Column</td><td>Business entity field.</td></tr>
</table>

6. Click **Apply**.
7. Repeat step 3 to 6 for each additional relationship attribute that you want to configure.

Configuring Hierarchy Tasks

Hierarchy tasks ensures changes to hierarchies move through an approval workflow. To use hierarchy tasks, you configure the task type, workflow triggers, and a task template. Then when Data Director users edit or create hierarchies, ActiveVOS creates hierarchy tasks with specific properties.

Hierarchy tasks are a type of task. For more information about general tasks, see "Configuring Tasks Overview" on page 99.

To configure hierarchy tasks, perform the following steps:

1. Configure a hierarchy task type.
2. Configure a hierarchy task template.
3. Configure hierarchy workflow triggers.

Step 1. Configuring a Hierarchy Task Type

Create the hierarchy task type and configure which users can claim or be assigned tasks. For example, you might want Senior Managers to review hierarchy tasks.

1. Click **Business Entity** > **Tasks**, and then select **Task Types**.
2. Click **Create**.
3. In the properties panel, in the **Name** field, enter **HierarchyFinalReview**.
4. Optionally, in the **Label** field, enter a label for the task type.
5. To support attachments when users edit the task details, select the **Enable File Attachments in Task Details** check box.
6. Click **Apply**.

   The **Roles** and **Task Action Configuration** folders appear in the tree view panel.
7. Select roles that can claim or be assigned the task.
   a. In the tree view panel, select **Roles**, and then click **Create**.
   b. In the properties panel, from the **Role** list, select an MDM Hub user role.
   c. Click **Apply**.

      The role appears in the tree view panel.
   d. Optionally, create additional roles.
8. Optionally, configure the hierarchies that the role can access.

   Tip: If you do not configure hierarchies, users assigned the selected roles can manage tasks for all hierarchies.
   a. In the tree view panel, select a role.
   b. Select **Hierarchy**, and then click **Create**.
   c. In the properties panel, from the **Hierarchy** list, select a hierarchy.
   d. Click **Apply**.
   e. Optionally, configure additional hierarchies.
9. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.

b. Review the changes or publish without a review.
   - To publish without a review, click **Publish**.
   - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Step 2. Configuring a Hierarchy Task Template

To configure a task template for hierarchy tasks, set the default task properties, such as the task title, due date, and priority. When ActiveVOS creates hierarchy tasks, the tasks are created with the specified properties.

1. Click **Business Entity** > **Tasks**, and then select **Templates**.
2. Click **Create**.
3. In the properties panel, enter the template properties.
   a. In the **Name** field, enter **HierarchyApprovalTemplate**.
   b. In the **Title** field, enter the title format for task title.

      Title Example

      You want the task title to be `Review hierarchy changes in <record label>`.

      Enter the following value for title: `Review hierarchy changes in {taskRecord[0].label}`.

      Where

      - `taskRecord[0]` is the target record.
      - `label` is the value of the task format in the business entity label format configuration.

   c. Select a task priority.

      The following table describes the possible task priorities:

Priority	Description
HIGH	The task has a high priority.
NORMAL	The task has a medium priority. Default is NORMAL.
LOW	The task has a low priority.

   d. In the **Due Date** field, enter the amount of time from the day that the task is created that the task is due.

The following table describes the due date syntax:

Type of Parameter	Description	Values
sign	Determines if the due date is set to a date after the task is created or before the task is created.	If `+`, the due date is determined by adding time to the date the task is triggered. If `-`, the due date is determined by subtracting time from the date the task is triggered.
number	The amount of units to add or subtract from the current date when establishing the due date.	Any integer value.
unit	The number of days, weeks, or months that the due date is from the date the task is triggered.	If `d`, the specified number of days determines the due date. If `w`, the specified number of weeks determines the due date. If `m`, the specified number of months determines the due date.

e. Optionally, in the **Comment** field, enter the text that populates the task comment field.

f. Optional, in the **Status** field, select the status of the task.

The following table describes the possible task statuses:

Status	Description
OPEN	The task status is Open. Default is OPEN.

4. Click **Apply**.

Step 3. Configuring Hierarchy Workflow Triggers

Create a trigger that launches the hierarchy workflow and creates a hierarchy task. For example, you might want changes that Data Stewards make to a hierarchy to trigger a hierarchy task.

1. Configure a workflow trigger.
   a. Click **Business Entity** > **Tasks**, and then select **Triggers**.
   b. Click **Create**.
   c. In the properties panel, enter **HierarchyOneStepApproval**.

   The Repository Manager refers to the name when reporting trigger validation errors.

   d. Select **Start Workflow** so that the event trigger launches a workflow.

e. Configure the following parameters:

Parameter	Input
Template	HierarchyApprovalTemplate
Task Type	HierarchyFinalReview
Task Kind	Hierarchies
Process	HierarchyOneStepApprovalTask

f. Click **Apply**.

2. Configure events for a workflow trigger.
   a. In the tree view panel, select **Events**, and then click **Create**.
   b. In the properties panel, from the **Event** list, select **ModifyHierarchy**.
   c. Click **Apply**.
3. Configure roles for a workflow trigger.
   a. In the tree view panel, select **Roles**, and then click **Create**.
   b. In the properties panel, from the **Role** list, select the MDM Hub role.
   For example, select Data Steward if you want changes that Data Stewards make to a hierarchy to trigger a hierarchy task.
   c. Click **Apply**.
   d. Optionally, configure additional roles.
4. Optionally, configure the hierarchies that trigger hierarchy tasks.

   Tip: If you do not configure hierarchies, hierarchy tasks are triggered for all hierarchies.
   a. In the tree view panel, select a role.
   b. Select **Hierarchy**, and then click **Create**.
   c. In the properties panel, from the **Hierarchy** list, select a hierarchy.
   d. Click **Apply**.
   e. Optionally, configure additional hierarchies.

Configuring Default Hierarchy for the Dashboard

In the customer dashboard, the Hierarchies panel displays how a record is related to other records in a hierarchical format. A record can belong to multiple hierarchies. If you define a default hierarchy for the hierarchy component, the Hierarchies panel loads the default hierarchy for the record. Otherwise, the Hierarchies panel loads the list of hierarchies to which the record belongs.

1. In the Provisioning tool, click **Configuration** > **Component Editor**.

   The **Component Editor** appears.
2. In the **Component Editor** list, select **Hierarchy Widget**.

   The **Properties** panel appears.

3. Specify the following hierarchy properties:

Property	Description
Name	Unique name for the relationship.
Label	Label for the relationship that appears in the Data Director user interface.
Default Hierarchy	List of available hierarchies. Set one of the hierarchies as the default hierarchy.
Widget Height in px	Display height of the widget in pixels.

4. Click **Apply**.

   A confirmation dialog box appears confirming that the changes were applied successfully.

5. Click **Publish**

   A confirmation dialog box appears that prompts you to confirm or review the changes.

6. Review the changes or publish without a review.
   - To publish without a review, click **Publish**.
   - to publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Designing Data Director Interfaces to Manage Relationships

By default, the **Hierarchy** view and **Network** view are available to all Data Director users. Data Director users can use the **Hierarchy** view to view, create, and manage hierarchy relationships. Data Director users can use the **Network** view to view, create, and manage network relationship and to view hierarchy relationships.

You can design a **Quick View** tab for the **Hierarchy** view to provide a summary of record details for the selected record in the hierarchy. You can also design a **Quick View** tab for the **Network** view to provide a summary of record details for the selected record in the network.

For more information about design the Data Director, see "Designing the Quick View Tab for the Hierarchy View" on page 205 and "Designing the Quick View Tab for the Network View" on page 207.

CHAPTER 9

Creating Match Rule Sets

This chapter includes the following topics:

- Creating Match Rule Sets Overview, 82
- Configuring the Hub Server Properties, 82
- Prerequisites for Creating Match Rule Sets, 83
- Creating Match Rule Sets, 84

Creating Match Rule Sets Overview

A match rule set is a logical collection of match rules. Match rules determine whether a record matches with another record.

You can use the Provisioning tool to create match rule sets based on a sample data and the match training. The match engine matches the records in the sample data set and identifies 10 pairs of records for you to review. Based on your review, the match engine creates the match rules and provides a confidence score for the match rules. If the confidence score is low, retrain the match engine until you get a high match confidence score. Use the match rules to run the match process.

The quality of the match rules depends on the data set that you use. Ensure that the data set contains a combination of matching records, partially matching records, and unique records to create accurate match rules.

Note: Informatica recommends that you create match rule sets in non-production environments only.

Configuring the Hub Server Properties

Before you create match rule sets, ensure that you configure the Hub Server properties related to match training in the `cmxserver.properties` file.

Configure the following properties for match training:

cmx.match.training.confidence.threshold

Optional. Minimum match confidence score required to create a match rule set in the Provisioning tool. Default is 85.

cmx.match.training.data.encoding

Optional. Configures encoding for match training in the Provisioning tool. Set to 1 to enable encoding for match training. Default is 0.

Ensure that you use the same value that the `cmx.server.match.server_encoding` property uses.

For more information about the Hub Server properties, see the *Multidomain MDM Configuration Guide*.

Prerequisites for Creating Match Rule Sets

You can create a match rule set for a business entity based on a sample data set.

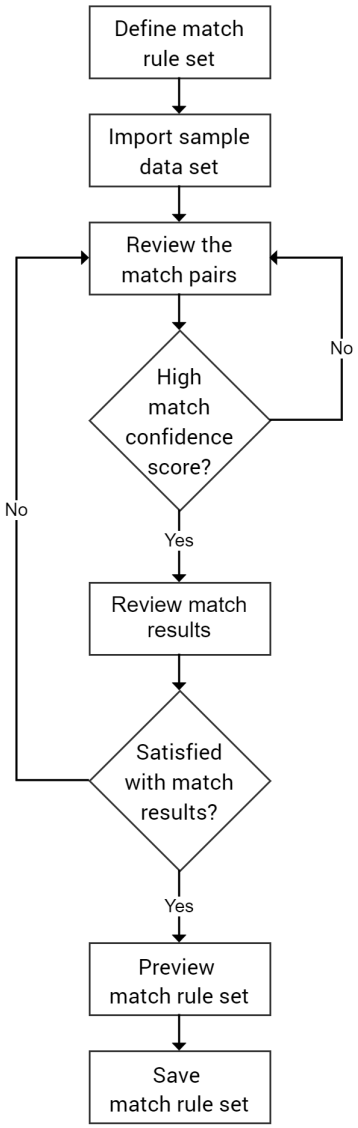
Before you create a match rule set for a business entity, verify that your environment meets the following prerequisites:

- No match rule sets exist for the business entity.
- No match rule sets exist for the root base object of the business entity.
- No records exist for the root base object of the business entity.
- The data set contains a minimum of 1000 records and a maximum of 10,000 records.
- Your user role has permission to import bulk data. For more information about assigning permissions, see the *Multidomain MDM Security Guide*.

Creating Match Rule Sets

Creating match rule sets using the Provisioning tool is easier and faster. Before you create a rule set, ensure that you have created an optimal data set.

The following flow chart shows the process of creating a match rule set:



Step 1. Define a Match Rule Set

When you define a data set, you must specify the data population. A data population contains intelligence about name, address, and other identification information that is typical for a given population. Use an appropriate data population to improve the match accuracy.

1. In the Provisioning tool, click **Configuration** > **Match Rule Sets**.
2. Click **New Match Rule Set**.

   The **New Match Rule Set** page appears.

3. Enter the name of the match rule set.
4. Optionally, enter a description for the match rule set.
5. Select a business entity for which no match rule set exists.
6. Select a data population.

Step 2. Import Sample Data Set

Upload a data set that contains a minimum of 1000 records and a maximum of 10,000 records. The match engine automatically maps the fields in the sample data with the existing business entity fields.

1. Under **Match Training Data Set**, click **Upload**.

   The **Match Training Data Set Import** dialog box appears.
2. To upload a file, drag a CSV or Excel file to the **Upload File** box, or click **Browse** and select a file.

   The match training data set preview and the file properties appear.

   Note: Click **Change File** if you want to change the sample data set.
3. In the **File Properties** panel, specify the following file properties:

Property	Description
Delimiter	A character that represents the break between data values in the import file.
Text Qualifier	Symbol used in the file to enclose a string.
Code Page	Character encoding used in the file. Character encoding is the mapping of the characters from a language or group of languages to hexadecimal code.
Import Data From Line	Line from where you want to import data.
Has column headers	Indicates whether the uploaded file includes column headers.
Column Headers in Row	Number of the row that serves as the column header in the file.
Regional settings	Select the check box to update the following settings: - Date Pattern: Date format for the date fields in the import file. - Decimal Separator: Symbol used to separate decimal part of a number. - Thousand Separator: Symbol used to separate groups of thousands.

4. Click **Next**.

   The match engine maps the column headers in the uploaded file with the business entity fields.
5. To update any of the auto-generated field mapping, drag the source column headers to the relevant target business entity fields.
6. Click **Save**.

   A confirmation message to delete any existing records in the root base object that the selected business entity uses appears.
7. To continue, click **Yes**.

   A message to confirm the delete operation appears.

8. Click **Yes** to continue with match training.

   The **Match Training** page appears.

Step 3. Train the Match Engine

Review the data pairs. After you review all the data pairs, the match engine displays the match results based on the match rule set that identified the match pairs. The match engine requires at least eight confirmed review inputs to display the match results.

1. On the **Match Training** page, select a key type to use for key generation and click **Continue**.

   The first data pair appears for review.
2. Review the data pair, and click one of the following options:
   - **Match**. If the records are same.
   - **Not Sure**. If you are not sure whether the records are same.
   - **Not a Match**. If the records are not same.
3. After you complete reviewing 10 data pairs, click **Show Match Results**.

   The match confidence meter and the match results appear.

Step 4. Preview and Save the Match Rule Set

Review the match results and match confidence score and decide whether to continue training, or save the match rule set.

A low match confidence score indicates that the prediction of the match engine did not match with your inputs for the match pairs. You must continue the match training until you get a high confidence score.

A high match confidence score indicates that the prediction of the match engine almost matched with your inputs for the match pairs. To be sure, check the match results. If you are satisfied with the match results, proceed to preview and save the match rule set.

1. If the match confidence score is low, click **Continue Training**.
2. If the match confidence score is high, do the following tasks:
   a. To review the match results, select a training session from the list of match trainings.
   b. To view the details of a match pair, click the match score.
   c. Select the best training session, and click **Next**.
   d. Review the match rules, and click **Save**.

      The match rule set appears in the **Match Rule Sets** page.

CHAPTER 10

Configuring Search

This chapter includes the following topics:

- Configuring Search Overview, 87
- Configure the Elasticsearch Cluster, 88
- Configure Elasticsearch Credentials, 89
- Create Custom Elasticsearch Index Settings (Optional), 89
- Configure the Searchable Fields, 92
- Configure the Search or Query Results Display, 96
- Configure the Layout to Display Similar Records (Optional), 97

Configuring Search Overview

You can use the Data Director application or a custom application to search for data within a specific business entity. You must configure search with Elasticsearch.

Elasticsearch is an open-source, full-text search engine. You can set up Elasticsearch as a single node cluster or as a multi-node cluster to provide distributed indexing and search. Before you configure search in the Provisioning tool, install and set up Elasticsearch and configure the MDM Hub properties for search.

To configure search, perform the following tasks:

1. Configure the Elasticsearch cluster.
2. Optionally, create custom Elasticsearch index settings.
3. Configure the searchable fields.
4. Configure the search and query results display.
5. Optionally, configure the layout to display similar records.

After you configure search, you must validate the Operational Reference Store (ORS) and index the search data.

For information about tasks that you must perform before and after you configure search in the Provisioning tool, see the *Multidomain MDM Configuration Guide*.

Configure the Elasticsearch Cluster

Use the Provisioning tool to configure the Elasticsearch cluster for the MDM applications. The search APIs use the configuration. The Data Director application and any custom applications use the search APIs.

Note: When you configure the Elasticsearch cluster, only master nodes of the cluster must be specified.

1. Open a supported browser, and enter the following URL:

   `https://<MDM Hub Server host name>:<MDM Hub Server port number>/provisioning/`

   The **Login** page appears.
2. Enter the user name and password, and click **Log In**.
3. From the **Database** list, select the database for which you want to configure the Elasticsearch cluster.
4. Click **Configuration** > **Infrastructure Settings**.

   The **Infrastructure Settings** page appears.
5. Select **Elasticsearch Cluster** from the list, and then click **ESCluster**.

   The **ESCluster** appears in the tree view panel.
6. To configure an Elasticsearch cluster node, select **esNode** in the tree view panel, and then click **Create**.
7. Specify the following properties of the configured Elasticsearch cluster:

Property	Description
Name	Name of the master node in the Elasticsearch cluster.
URL	URL of the master node in the Elasticsearch cluster. The URL format is `https://<host name>:<port>`.
Update Elasticsearch Node	Indicates whether to update the node with the index details when the existing node continues to manage search requests. After you manually run the Initially Index Smart Search Data batch jobs so that the indexes are added to the new node, you can remove the existing node. You can then clear the check box so that the new node can manage search requests. **Note:** Applies when you upgrade from Elasticsearch Version 6 to Version 7. Elasticsearch authentication support is valid only for Version 7.x. For more information about installing and configuring Elasticsearch, see the *Multidomain MDM Configuration Guide*.

8. Click **Apply**.
9. If you want to create additional master nodes, repeat steps 6 through 8.
10. Publish the changes to the MDM Hub.
    a. Click **Publish**.

       A confirmation dialog box appears that prompts you to publish or review the changes.
    b. Review the changes or publish without a review.
       - To publish without a review, click **Publish**.

- To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

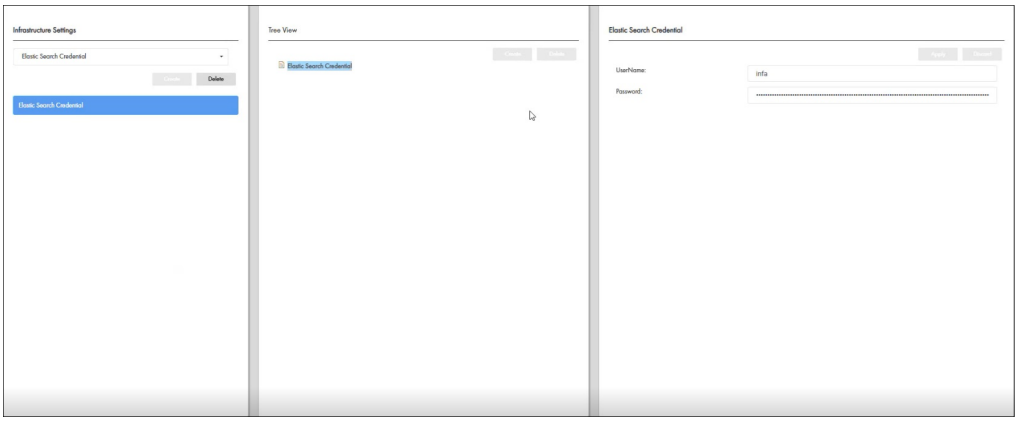
Configure Elasticsearch Credentials

Provide user credentials to securely connect and interact with the Elasticsearch cluster.

1. Log in to the Provisioning tool.
2. From the **Database** list, select the database for to configure the Elasticsearch credentials.
3. Go to**Configuration** > **Infrastructure Settings**.

   The **Infrastructure Settings** page appears.
4. From the infrastructure settings list, go to **Elastic Search Credential**, and select **Elastic Search Credential**.

   The **Elastic Search Credential** appears in the tree view panel.



5. In the **UserName** field, enter the username configured for Elastic Search.
6. In the **Password** field, enter the password for the Elastic Search user account.

   When you type the password, a series a dots appears. Multidomain MDM application captures the input accurately and encrypts the password characters to hide from view.

Create Custom Elasticsearch Index Settings (Optional)

If the Elasticsearch index settings that Informatica provides do not meet your requirements, you can create custom index settings. The custom index settings must include analyzers, which convert text into tokens or terms that are added to the inverted index for searching.

An analyzer must have only one tokenizer, and can have zero or more character filters and token filters. Tokenizers receive a stream of characters that are converted into tokens. Token filters receive a stream of tokens generated by a tokenizer and might add, remove, or change tokens. Character filters receive a stream of characters and can add, remove, or change characters in the stream.

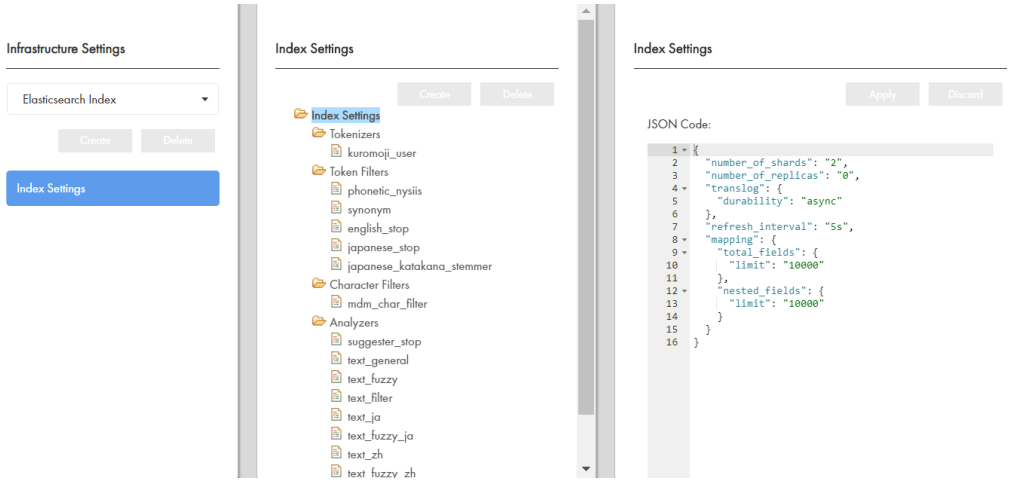
The tokenizer, token filters, and character filters that you use in a custom analyzer can be the Informatica default, custom, or Elasticsearch built-in components. You cannot edit the default settings. When you configure an analyzer, the Elasticsearch built-in tokenizers and token filters are available for selection.

For more information about Elasticsearch index settings, see the Elasticsearch documentation.

1. Log in to the Provisioning tool.
2. From the **Database** list, select the database for which you want to configure the Elasticsearch index settings.
3. Click **Configuration** > **Infrastructure Settings**.

   The **Infrastructure Settings** page appears.
4. From the infrastructure settings list, select **Elasticsearch Index**, and click **Index Settings**.

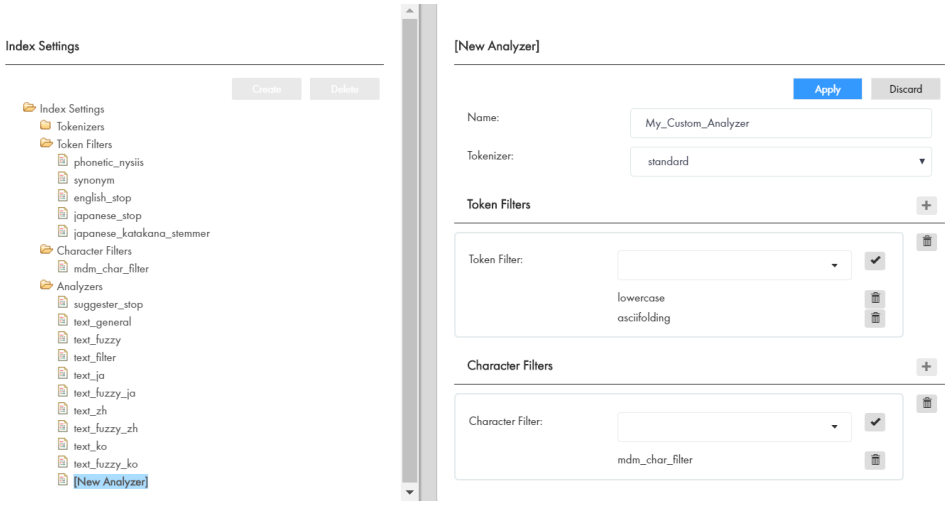
   The **Index Settings** appear in the tree view panel, and the **JSON Code** box for the index settings appears in the properties panel. If the index settings were not changed, the page displays the default settings.



5. In the **JSON Code** box, enter index settings for modules other than the `Analysis` module. Also, enter the index settings that are not associated with any specific index module, such as the number of shards, number of replicas, and refresh interval.
6. Configure the analyzer components, such as a tokenizer, token filters, and character filters.
   a. In the tree view panel, select the component that you want to configure, and click **Create**.
   b. In the properties panel, enter the name and JSON code for the component.
   c. Click **Apply**.
7. Configure an analyzer.
   a. In the tree view panel, select **Analyzers**, and then click **Create**.
   b. In the properties panel, specify the name, tokenizer, token filters, and character filters for the analyzer.

      Ensure that you specify the token filters in the order in which you want the analyzer to use them.

The following image shows an example of a custom analyzer configuration:



c. Click **Apply**.

8. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.

   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.
9. Review the application server logs for any validation errors related to the index settings and make changes.

Elasticsearch Built-In Tokenizers and Token Filters

You can select the available Elasticsearch built-in tokenizers and token filters for your custom analyzers.

The following Elasticsearch built-in tokenizers are available for the custom analyzers:

- standard
- letter
- lowercase
- whitespace
- uax_url_email
- classic
- thai
- keyword

The following Elasticsearch built-in token filters are available for the custom analyzers:

- asciifolding
- standard
- lowercase

- uppercase
- porter_stem
- trim
- cjk_width
- cjk_bigram
- classic
- apostrophe
- kuromoji_baseform

For more information about the custom and built-in Elasticsearch analyzer components, see the Elasticsearch documentation.

Configure the Searchable Fields

You can use the Provisioning tool to configure a field as a searchable field and set the field properties. A search request searches only the fields that you configure as searchable fields.

Multiple searchable fields might affect the performance of the search requests, so configure only the significant fields as searchable fields. For example, configure fields that contain full names, organization names, or email addresses as searchable fields rather than fields that contain country codes, gender codes, or address types.

1. Log in to the Provisioning tool.
2. From the **Database** list, select the database for which you want to configure the fields.
3. Click **Business Entity** > **Modeling**.

   The **Modeling** page appears.
4. Select **Business Entities** from the list, and select the business entity for which you want to configure the searchable fields.
5. In the tree view panel, select **Fields**, and click **Create**.
6. Configure the following properties based on your requirement:

   Name

   Name of the field that that you want to appear in the tree view panel.

   Label

   Label for the field that you want to appear within views in Data Director.

   Read Only

   Optional. Indicates whether you can edit the field in the Entity View. To configure the field as a non-editable field, select the property.

   Required

   Optional. Indicates whether the field is a required field. To configure the field as a required field, select **Required**. By default, the field is not a required field.

URI

Defines the namespace where data types are declared. The data type that you want to configure depends on the URI that you select.

To configure basic data types, such as string, integer, and boolean, select `commonj.sdo`. To configure Informatica data types, such as image URL, hyperlink, and file attachment, select `urn:co-types.informatica.mdm`.

Type

The data type of the field. The data type that you can configure depends on the URI that you select. You can configure the following data types: string, integer, decimal, date, boolean, image URL, hyperlink, and file attachment.

By default, the data type of a business entity field is as close as possible to the data type of the base object column that the field is associated with. For example, a string column in a base object might contain information for an image, so you configure the image URL data type for the business entity field. If you want a field to display web URL, file URI, FTP link, and email address as hyperlinks, configure the hyperlink data type for the field.

Note: You cannot open file URIs with Google Chrome and Mozilla Firefox because of security reasons.

Display Format

The display format for the date fields. Before you specify the display format for the date fields, specify the URI and set the Type property to **Date**.

You can select the **Date** or **Datetime** format for business entity fields that display a date. The display format that you configure is applied to the date field in all the Data Director views.

If a display format is not set for a date field, the date format of the field is the same as the format of the associated base object column.

Filter

Defines a static filter to regulate the data that users can or cannot enter in a field.
When you enable the Filter property, specify the following fields:

- **Operator**. Controls whether the specified values are allowed or not allowed in the field.

Option	Description
In	Displays the values to the user. If you add a list of comma-separated values as allowed values, Data Director users can choose a value from a lookup list. If you also configure a field filter on the same field, the displayed values reflect the intersection of the static filter and the field filter.
Not In	Does not display the values to the user.

- **Value**. Add values that match the data type of the field. To add multiple values, enter a comma-separated list of values. To allow users to leave the field blank, add empty quotes ("") as a value.

  Note: The Provisioning tool does not validate the values. If you add a value of the wrong data type, the value does not appear in Data Director.

Column

Name of the base object column that you want to associate with the field.

7. Select **Searchable**.

Additional field properties appear.

8. Based on your requirement, select one or more of the following properties:
   - Search Analyzer
   - Suggester
   - Sortable
   - Filterable
   - Facet Range
   - Facet
   - Displayable
9. Optionally, if you select **Facet**, in the **Facet Range** field, specify the range for the numeric or date fields that you configure as facets in the following format:

   ```
   <Start Value>,<End Value>,<Frequency>
   ```

   For example, `1000,2000,50`

   Note: The facet ranges are not visible in the Data Director application. When you use the REST web service to perform the search, the response can return the facet ranges.
10. Click **Apply**.
11. Publish the changes to the MDM Hub.
    a. Click **Publish**.

       A confirmation dialog box appears that prompts you to publish or review the changes.

    b. Review the changes or publish without a review.
       - To publish without a review, click **Publish**.
       - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Searchable Field Properties

To configure the searchable field properties, you can use the Provisioning tool or apply a change list to the repository.

When you enable a searchable reference entity field as a filterable facet, the Data Director record view displays the filter field label in the following format:

<Business entity lookup field label> - <Reference entity lookup field label>

If you configure the Filterable property for a reference entity, for filtering to work, ensure that you configure the Filterable property for all the dependent reference entities.

The following table describes the searchable field properties:

Property	Description
Searchable	Indicates whether a search request can search the field for a search string. To include the field in search requests, enable the property. If you do not want to include the field in search requests, disable the property. When you enable the Searchable property, you can configure additional properties for search. **Note:** System columns cannot be configured as searchable except for the consolidationInd and hubStateInd columns. The following additional properties are available for configuration: - Search Analyzer - Suggester - Sortable - Filterable - Facet Range - Facet - Displayable
Search Analyzer	Specifies the custom search analyzer that you want to use for the field. Based on the type of data that the field will contain, decide on a suitable search analyzer.
Suggester	Indicates whether you want to suggest the values of the field as the search string in the Data Director application. To suggest the values of the field as the search string, enable the property. If you do not want to suggest the values of the field as the search string, disable the property. **Important:** To ensure data security, do not enable the Suggester property for fields that contain sensitive data.
Sortable	Do not use this property.
Filterable	Indicates whether you want to enable filtering on a field. The Data Director application displays the filterable fields as filters in the Search workspace. To configure the field as a filter, enable the property. If you do not want to configure the field as a filter, disable the property.

Facet Range	Indicates the range for the numeric or date fields that you configure as facets. Use the following format to specify the range:
	`<Start Value>,<End Value>,<Frequency>`
	The start value is inclusive, and the end value is exclusive in the range. For example, if you set the facet range to `1000,2000,500` for an integer field, a search request returns the following ranges:
	`[1000 to 1500]`
	`[1500 to 2000]`
	The range `1000 to 1500` includes the values from 1000 to 1499, and the range `1500 to 2000` includes the values from 1500 to 1999.
	Ensure that you set a valid minimum and maximum value for the range and an offset that limits the number of ranges to 10.
	Facets cannot be configured for negative numbers, but a search request still displays negative values.
	For a date field, suffix `Y|M|D` to the frequency, where `Y` indicates year, `M` indicates month, and `D` indicates day. For example, `2M` indicates 2 months.

Property	Description
Facet	Indicates whether you want to set the field as a facet. A facet field groups the search result values and shows the count of each group. The Data Director application displays the facet fields, the field values grouped based on the search results, and the count of each group in the Search workspace. When a child record field is set as a facet field, the Data Director application displays a tooltip for the facet field. The format of the tooltip text is <Child record name>/<Child record field name>. The Facet property functions in conjunction with the Filterable property, so enable the Filterable property if you want to configure the field as a facet. If you do not want to configure the field as a facet, disable the Facet property. **Note:** When a data security filter is configured for a facet field, faceting is disabled for the field to protect sensitive data. Enable the Facet property, if you want to configure the **Facet Label Format** menu options, used for the facet values on the search page. The following options are available: - Lowercase - Displays the facet values in lowercase. For example, john autumn. - Uppercase - Displays the facet values in uppercase. For example, JOHN AUTUMN. - Title Case - Displays the facet values title case. For example, John Autumn.
Displayable	Do not use this property.

Configure the Search or Query Results Display

You can use the Provisioning tool to configure the business entity views that you want to use for search. A search result includes only the fields that are part of the business entity view that you configure for search results. You can also configure the order in which the search filters appear.

Before you configure the searchable views, create the business entity views that you want to use for the search results.

Note: To display child record fields of a business entity in the search results, use a business entity view that is transformed from a business entity. Ensure that the view includes child record fields at the root record level.

1. Log in to the Provisioning tool.
2. From the **Database** list, select the database with which your application is associated.
3. Click **Configuration** > **Application Editor**.

   The **Applications** page appears.
4. From the **Applications** list, select the application for which you want to configure search.

   If you do not have an application, create one before you can configure search.
5. In the tree view panel, select **Search Configuration**, and click **Create**.
6. In the properties panel, select a business entity and the business entity view that you want to use to display the search or query results.

   If you do not select a business entity view, the search and query results contain all the business entity fields.

7. Optionally, if you configured search, select the filters and configure the display order of the search filters.
    a. Click the **Edit** icon next to **Filter Display Order**.

        The **Edit Filter Display Order** dialog box appears. The dialog box contain filters, which are fields that are configured as filterable in the business entity model.
    b. Drag the filters from the **Available Filters** section to the **Selected Filters** section.
    c. To configure the order, drag and move the filters up or down.
    d. Click **OK**.
8. Click **Apply**.

    The search configuration is saved to the temporary workspace.
9. Publish the changes to the MDM Hub.
    a. Click **Publish**.

        A confirmation dialog box appears that prompts you to publish or review the changes.
    b. Review the changes or publish without a review.
        - To publish without a review, click **Publish**.
        - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Configure the Layout to Display Similar Records (Optional)

When you enter data in the Data Director application to create a record, you can view similar records that are retrieved based on the data that you enter. To view the similar records, you must configure the layout to define the fields based on which you want to search for the similar records.

1. Log in to the Provisioning tool.
2. From the **Database** list, select the database for which you want to configure the application.
3. Click **Configuration** > **Component Editor**.

    The **Component Editor** appears.
4. From the Component type list, select **Similar Records**, and click **Create**.
5. In the properties panel, enter a name for the Similar Records component.
6. In the **XML** field, enter an XML configuration that includes a list of fields to search for similar records:

The following table describes the XML elements that you can use to configure the Similar Records component:

Element	Description
searchableFields	Specifies one or more fields on which you want to base the search. The `searchableFields` element is the parent of the `field name` element.
field name	Specifies the name of a field on which you want to base the search for similar records. The `field name` element is a child of the `searchableFields` element. You can configure multiple `field name` elements.
searchType	Specifies the type of search to perform. The `searchType` element can contain the following child elements: - `smartSearch` - `searchMatch`
smartSearch	Specifies that you want to use search to find similar records.
searchMatch	Specifies that you want to use queries to find similar records. The `searchMatch` element can contain the following child elements: - `fuzzy` - `matchRuleSet`
fuzzy	Specifies whether you want to perform a fuzzy search. To perform a fuzzy search, set to `true`. The `fuzzy` element is a child of the `searchMatch` element. If you do not add this element, exact search is performed.
matchRuleSet	Specifies the name of the match rule set to use for finding similar records. The `matchRuleSet` element is a child of the `searchMatch` element.
label	Specifies the label format for the search field values. To configure the label format, use the `existsFormat` attribute.
column	Specifies a single column to use in the label format. To configure the column for the label, use the `columnUid` attribute, which is the unique identifier of the column. The `column` element is a child of the label element. You can specify more than one column for the label.

For sample configurations, see the *Multidomain MDM Provisioning Tool Guide*.

7. Click **Apply**.

   The Similar Records component that you created appears in the **Component Editor** panel and in the tree view panel.

8. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.

   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

CHAPTER 11

Configuring Tasks

This chapter includes the following topics:

- Configuring Tasks Overview, 99
- Configuring Task Templates, 100
- Configuring Workflow Triggers, 101
- Configuring Task Types, 104
- Task Administrator Role, 109
- Descriptive Task Titles, 110
- Default Task Configuration, 111
- Changing the Approver for Tasks, 113

Configuring Tasks Overview

You can configure tasks to ensure changes to records and hierarchies move through an approval workflow. You can configure the task template, workflow triggers, task types to meet your business requirements. For example, you might want changes by Data Stewards to trigger a workflow, or you might want to create tasks to review new records and edited records. Also, you might want to specify the default title or priority for tasks.

You can perform the following task configurations:

Task template

You can configure task templates so that a task is created with specific properties. For example, you can specify that when a trigger launches a workflow, the tasks have a particular title, priority, due date, and task status.

Workflow triggers

You configure workflow triggers so the appropriate ActiveVOS® workflow is launched after certain events in Data Director. You can configure whether users must add comments or attachments when they trigger workflows.

Task types

You can configure task types so that specific user roles can claim or be assigned tasks. You can create task action configurations for each task type. You can configure whether users must add a comment, attach a file, or reassign tasks when they perform task actions.

Task Administrator role

You can configure the Task Administrator role and map the MDM Hub role with privileges to manage all tasks to the Task Administrator role. For example, if you use the default ActiveVOS workflows, map the abAdmin MDM Hub role to the Task Administrator role.

Configuring Task Templates

To configure a task template, set the default task properties, such as the task title, due date, and priority. When ActiveVOS creates tasks, the tasks are created with the specified properties.

1. Click **Business Entity** > **Tasks**, and then select **Templates**.
2. Click **Create**.
3. In the properties panel, enter the template properties.
   a. In the **Name** field, enter the template name.
   b. In the **Title** field, enter the title format for task title.

      Title Example 1

      You want the task title to be `<business entity name>: <source record display name><source record row ID> merges to <target record display name><target record row ID>`.

      Enter the following value for title: `Merge {taskRecord[1].label} into {taskRecord[0].label}`.

      Where

      - `taskRecord[N]` is the (N+1)th business entity node in the TaskData.getTaskRecords() list
      - `taskRecord[1]` is the source record
      - `taskRecord[0]` is the target record
      - `label` is the value of the task format in the business entity label format configuration.

      Title Example 2

      You want the task title to be `Review changes in <record label>`.

      Enter the following value for title: `Review changes in {taskRecord[0].label}`.

      Where

      - `taskRecord[0]` is the target record.
      - `label` is the value of the task format in the business entity label format configuration.

   c. Select a task priority.

The following table describes the possible task priorities:

Priority	Description
HIGH	The task has a high priority.
NORMAL	The task has a medium priority. Default is NORMAL.
LOW	The task has a low priority.

d. In the **Due Date** field, enter the amount of time from the day that the task is created that the task is due.

The following table describes the due date syntax:

Type of Parameter	Description	Values
sign	Determines if the due date is set to a date after the task is created or before the task is created.	If `+`, the due date is determined by adding time to the date the task is triggered. If `-`, the due date is determined by subtracting time from the date the task is triggered.
number	The amount of units to add or subtract from the current date when establishing the due date.	Any integer value.
unit	The number of days, weeks, or months that the due date is from the date the task is triggered.	If `d`, the specified number of days determines the due date. If `w`, the specified number of weeks determines the due date. If `m`, the specified number of months determines the due date.

e. Optionally, in the **Comment** field, enter the text that populates the task comment field.

f. Optional, in the **Status** field, select the status of the task.

The following table describes the possible task statuses:

Status	Description
OPEN	The task status is Open. Default is OPEN.

4. Click **Apply**.

Configuring Workflow Triggers

Workflow triggers determine which task workflows are launched after create, update, or merge events.

Note: If you configure a role to trigger an unmerge task, do not assign the same user role as reviewer.

To configure workflow triggers, perform the following steps:

1. Create a workflow trigger, and configure whether users must add a comment or attachment to the triggered task.
2. Configure events for the workflow trigger.
3. Configure which roles and business entities trigger workflows.

Step 1. Create a Workflow Trigger

To create a trigger to launch a workflow, specify a name for the trigger, the workflow to start, and the trigger properties. If you use Data Director with business entities, you can also specify whether users are prompted for comments or attachments.

1. Click **Business Entity** > **Tasks**, and then select **Triggers**.
2. Click **Create**.
3. In the properties panel, enter the trigger name.

   The Repository Manager refers to the name when reporting trigger validation errors.
4. Select **Start Workflow** so that the event trigger launches a workflow. If you do not select **Start Workflow**, changes are made directly to the data without first going through a review process.
5. Configure the following parameters:

<table>
<tr><th>Parameter</th><th>Description</th></tr>
<tr><td>Template</td><td>The name of the task template to use when the task is created.</td></tr>
<tr><td>Task Type</td><td>The first People Activity in the workflow.</td></tr>
<tr><td>Task Kind</td><td>Can be Merge, Unmerge, Hierarchies, or Review.</td></tr>
<tr><td>Process</td><td>The name of the ActiveVOS service. Corresponds to the Partner Links Service property in the ActiveVOS PDD file.
The following tables shows the service names for the default business entity workflows:
<table><tr><th>Business Entity Workflow</th><th>Service</th></tr><tr><td>BeMergeWorkflow</td><td>BeMergeTask</td></tr><tr><td>BeUnmerge</td><td>BeUnmergeTask</td></tr><tr><td>BeOneStepApproval</td><td>BeOneStepApprovalTask</td></tr><tr><td>BeTwoStepApproval</td><td>BeTwoStepApprovalTask</td></tr><tr><td>BeUpdateWithApproval</td><td>BeUpdateWithApprovalTask</td></tr><tr><td>HierarchyOneStepApproval</td><td>HierarchyOneStepApprovalTask</td></tr></table></td></tr>
</table>

6. To control whether users must attach a file and whether users are prompted for attachments, select one of the following options from the **Attachment** field:

Option	Description
Disabled - Do not prompt users (Default)	Users are not prompted for attachments. Attachments are disabled.
Optional - Prompt users	Users are prompted for attachments, but they can choose whether to attach a file.
Required - Prompt users	Users are prompted for attachments and must attach a file.

Note: When using Data Director with subject areas, users are never prompted for attachments.

7. To control whether users must add a comment and whether users are prompted for comments, select one of the following options from the **Comment** field:

Option	Description
Optional - Do not prompt users (Default)	Users are not prompted for comments. To add a comment, users can open the **Task Details** dialog box.
Optional - Prompt users	Users are prompted for comments, but they can choose whether to add a comment.
Required - Prompt users	Users are prompted for comments and must add a comment.

Note: When using Data Director with subject areas, users are never prompted for comments.

8. Click **Apply**.

Step 2. Configure Events for a Workflow Trigger

Associate the event in Data Director with the workflow trigger. You can configure more than one event for a trigger.

1. Click **Business Entity** > **Tasks**, and then select **Triggers**.
2. Select the trigger.
3. In the tree view panel, select **Events**, and then click **Create**.
4. In the properties panel, from the **Event** list, select one of the following events:

Event name	Description
CreateBE	Occurs when a user creates a record.
UpdateBE	Occurs when a user updates a record.
MatchedBE	Occurs when the MDM Hub identifies two records as a match.
MergeBE	Occurs when a user merges records.

Event name	Description
UnMergeBE	Occurs when a user unmerges records.
ModifyHierarchy	Occurs when a user modifies a hierarchy.

5. Click **Apply**.
6. Repeat step 3 to 5 for additional events.

Step 3. Configure Roles for a Workflow Trigger

The trigger that applies to events depends on the role of the user who performs the triggering action.

1. Click **Business Entity** > **Tasks**, and then select **Triggers**.
2. Select the trigger.
3. In the tree view panel, select **Roles**, and then click **Create**.
4. In the properties panel, from the **Role** list, select the MDM Hub role.
5. Click **Apply**.
6. Optionally, configure additional roles.

Configuring Task Types

You can create task types and configure which user roles can claim or be assigned tasks. For each task type, you can create task action configurations. Task action configurations control whether users must add a comment, attach a file, or reassign the task when performing a task action. Also, you can configure custom messages for task actions that appear when task actions are triggered.

To configure task types, perform the following actions:

1. Create a task type, and configure whether users can attach files when they edit the task details.
2. Configure the user roles that can claim or be assigned tasks.
3. Create task action configurations, and configure the comment, attachment, reassignment settings, and custom message for the task action.

Step 1. Create a Task Type

Create a task type and configure whether users can add an attachment when they edit the task details.

1. Click **Business Entity** > **Tasks**, and then select **Task Types**.
2. Click **Create**.
3. In the properties panel, enter the name of an ActiveVOS task in the **Name** field.

   The name must match the `mdmavxsd:name` value defined in the `mdmavxsd:taskType` expression of the setup script that is before a People Activity in the workflow `.bpel` file.

The following table describes the task names for the default business entity workflows:

Task Name (mdmavxsd:name)	People Activity Description	Associated Workflows
AVOSBeMerge	Data steward reviews the candidates and either merges the records or cancels the action.	Merge
AVOSBeUnMerge	Data steward reviews the candidate and either unmerges the record or cancels the action.	Unmerge
AVOSBeNotification	If approved, the record is flagged for promotion and the data steward is notified.	Update with approval One step approval Two step approval
AVOSBeReviewNoApprove	Manager reviews an update and either rejects it or sends it on for Final Review.	Update with approval Two step approval
AVOSBeFinalReview	Senior manager reviews the update and either rejects it or approves it.	Update with approval One step approval Two step approval
AVOSBeUpdate	Data steward reviews the update and either cancels the update or sends it through a two-step approval process.	Update with approval One step approval Two step approval
HierarchyFinalReview	Senior manager reviews the update and either rejects it or approves it.	Hierarchy one step approval

4. Optionally, in the **Label** field, enter a label for the task type.
5. To support attachments when users edit the task details, select the **Enable File Attachments in Task Details** check box.
6. Click **Apply**.

   The **Roles** and **Task Action Configuration** folders appear in the tree view panel.
7. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Step 2. Configure Roles for a Task Type

You can configure the user roles that are allowed to review tasks and the type of business entity records that a role can review.

User roles must have the proper MDM Hub privileges to allow users to process tasks. For more information about granting privileges to roles, see the *Multidomain MDM Security Guide*.

Note: If you configure a role to trigger an unmerge task, do not assign the same user role as reviewer.

1. Click **Business Entity** > **Tasks**, and then select **Task Types**.
2. Select the task type.
3. Select roles that can claim or be assigned the task.
   a. In the tree view panel, select **Roles**, and then click **Create**.
   b. In the properties panel, from the **Role** list, select an MDM Hub user role.
   c. Click **Apply**.
      The role appears in the tree view panel.
   d. Optionally, create additional roles.
4. Configure the type of business entity records that a role can access.
   a. In the tree view panel, select a role.
   b. Select **Business Entities**, and then click **Create**.
   c. In the properties panel, from the **Business Entity** list, select a business entity.
   d. Click **Apply**.
      The business entity appears in the Roles folder.
   e. Optionally, configure additional business entities for the role.
5. Publish the changes to the MDM Hub.
   a. Click **Publish**.
      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Step 3. Create Task Action Configurations for a Task Type

You can configure whether users must add a comment, attach a file, or reassign the task when they perform a task action. You can create multiple task action configurations with different comment, attachment, and reassignment settings for each configuration.

1. Click **Business Entity** > **Tasks**, and then select **Task Types**.
2. Select the task type.
3. In the tree view panel, select **Task Action Configuration**, and then click **Create**.
4. Specify a name for the task action configuration and configure whether users are prompted for comments or attachments.
   a. In the properties panel, enter a name for the task action configuration in the **Name** field.
   For example, to design a task action configuration for the Disclaim action, enter `Disclaim`.

b. To control whether users must attach a file and whether users are prompted for attachments, select one of the following options from the **Attachment** field:

Option	Description
Disabled - Do not prompt users (Default)	Users are not prompted for attachments. Attachments are disabled.
Optional - Prompt users	Users are prompted for attachments, but they can choose whether to attach a file.
Required - Prompt users	Users are prompted for attachments and must attach a file.

c. To control whether users must add a comment and whether users are prompted for comments, select one of the following options from the **Comment** field:

Option	Description
Optional - Do not prompt users (Default)	Users are not prompted for comments. To add a comment, users can open the **Task Details** dialog box.
Optional - Prompt users	Users are prompted for comments, but they can choose whether to add a comment.
Required - Prompt users	Users are prompted for comments and must add a comment.

d. Click **Apply**.

The subfolder for the task action configuration appears in the tree view panel.

5. Optionally, configure task actions.

Note: If you do not configure task actions, the task action configuration applies to all task actions.

a. In the tree view panel, select the subfolder for the task action configuration.

The **Business Entities**, **Roles**, and **Task Actions** nodes appear.

b. Select **Task Actions**, and then click **Create**.

c. In the properties panel, enter the name of a task action in the **Task Action** field. The name must match the task action names defined in the ActiveVOS workflow.
For example, to associate the Disclaim action with the task action configuration, enter `Disclaim`.

The following table describes task actions for the default task types:

Task Actions	Description	Task Type
Approve	User agrees with the proposed changes.	AVOSBeFinalReview
Cancel	User disagrees with the proposed changes.	AVOSBeMerge AVOSBeUpdate AVOSBeUnMerge

Task Actions	Description	Task Type
Disclaim	User can no longer work on the task.	AVOSBeFinalReview AVOSBeMerge AVOSBeReviewNoApprove AVOSBeUpdate AVOSBeUnMerge
Escalate	User agrees with the proposed changes and wants to send them for approval.	AVOSBeReviewNoApprove
Merge	User agrees with the merge preview.	AVOSBeMerge
Reject	User disagrees with the proposed changes.	AVOSBeFinalReview AVOSBeReviewNoApprove
Submit for Approval	User agrees with the proposed changes and submits the proposed changes for approval.	AVOSBeUpdate
Unmerge	User agrees with the unmerge preview.	AVOSBeUnMerge

d. Optionally, in the Action Message field, enter a message that appears when the task action is triggered.

   If you do not enter a message, the default message associated with the task action appears when the task action is triggered.

   The default messages have the following format:

   You are about to <task action name> this task.

   Example: You are about to cancel this task.

e. Click **Apply**.

   The task action node appears.

f. Optionally, configure additional task actions.

6. Optionally, select business entities to associate with the task action.

   Note: If you do not select business entities, the task action configuration applies to all types of business entity records.

   a. In the tree view panel, select **Business Entities**, and then click **Create**.

   b. In the properties panel, from the **Business Entity** list, select a type of business entity.

   c. Optionally, select additional types of business entities.

   d. Click **Apply**.

      The business entity node appears.

7. Optionally, select user roles to associate with the task action.

   Note: If you do not select roles, the task action configuration applies to all roles.

   a. In the tree view panel, select **Roles**, and then click **Create**.

   b. In the properties panel, in the **Roles** list, select a role.

   c. Click **Apply**.

The role node appears.

d. Optionally, create additional roles.

8. Publish the changes to the MDM Hub.

a. Click **Publish**.

A confirmation dialog box appears that prompts you to publish or review the changes.

b. Review the changes or publish without a review.

- To publish without a review, click **Publish**.
- To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Task Administrator Role

Data Director users can manage multiple tasks by performing task administration actions, such as assigning, releasing, or editing tasks. All users can manage tasks available to their user role, while users assigned the Task Administrator role can manage all tasks. For example, users can release their assigned tasks, while task administrators can release all assigned tasks.

Task administrators cannot work on tasks or view task details. If you assign users multiple roles, they might be able to manage tasks, work on tasks, and view task details. For example, you can assign an user the following MDM Hub roles: Data Steward, abAdmin, and abadmin. The user can work on and manage tasks available to Data Stewards and manage all tasks as a task administrator. For more information about the MDM Hub roles, see the *Multidomain MDM Configuration Guide*.

To use the Task Administrator role, you enable the role and specify the MDM Hub role that you want to map to the Task Administrator role.

The following table lists the task administration actions available to user roles:

User Role	Task Administration Actions
Task Administrator	Users can perform the following task administration actions on all tasks: - Assign - Release - Edit
Other user roles	Users can perform the following task administration actions on tasks available to their user role: - Claim Users can perform the following task administration actions on tasks based on whether they are assigned the tasks: - Release - Edit - Task actions, such as Accept, Reject, or Disclaim **Note:** The available task actions depend on the selected tasks and your ActiveVOS workflows.

Configuring the Task Administrator Role

You can use the Provisioning tool to configure the Task Administrator role and specify the MDM Hub role that you want to map to the Task Administrator role.

Before you begin, if you are using the default ActiveVOS workflows, you must create the abAdmin and abadmin MDM Hub roles in the MDM Hub.

Important: To ensure users can perform task administration actions on all tasks for all workflows, you must assign both the abAdmin and abadmin MDM Hub roles to users.

1. Log in to the Provisioning tool.
2. From the **Database** list, select the database to which you want to associate your configurations.
3. Click **Business Entity** > **Tasks**.
4. In the Tasks panel, select **Task Administrator Role**, and then click **Create**.
5. In the properties panel, enter `TaskAdministrator` in the **Name** field.
6. Select the **Enable Task Administrator role** check box.
7. From the **MDM Hub Role** list, select the MDM Hub role that you want to map to the Task Administrator role.

   If you are using the default ActiveVOS workflows, select **abAdmin**. If you are using custom ActiveVOS workflows, select the MDM Hub role that you use for business administration in ActiveVOS.
8. Click **Apply**.

   The changes are saved but are not published to the MDM Hub.
9. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

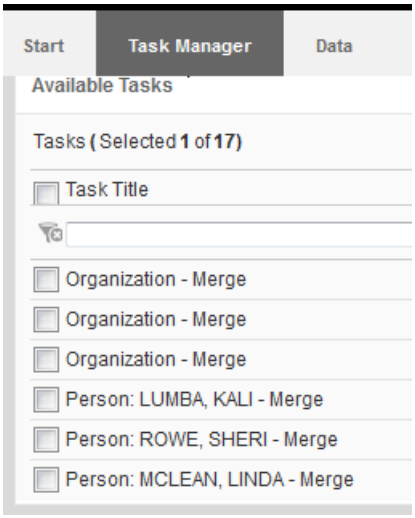
You configured the Task Administrator role. Users assigned the mapped MDM Hub role can perform task administration actions on all tasks in Data Director.

Descriptive Task Titles

You can configure descriptive task titles for each business entity on the Modeling page.

For example, you can configure a task title that displays the business entity type, the display name of the business entity record, and the task type. Data stewards and business managers can more easily differentiate between tasks when the tasks have a descriptive task title.

The following image shows generic task titles for the Organization business entity and descriptive task titles for the Person business entity:



Configuring Descriptive Task Titles

To configure descriptive task titles, edit the `Task Format` business entity attribute.

1. Click **Business Entity** > **Modeling**, and then select **Business Entities**.
2. Select the business entity.

   For example, click **Person**.
3. In the tree view panel, expand **labelFormat** and then click **LabelFormatType**.
4. In the properties panel, enter the descriptive task title in the **Task Format** field.

   For example, enter `{label}: {lastName}, {firstName}.`
5. Click **Apply**.

   The changes are saved but are not published to the MDM Hub.

Default Task Configuration

A default task configuration generates if you do not create a task configuration in the Provisioning tool. You can modify the default task configuration to meet your business needs.

The following task configuration generates by default:

Task type configuration

The following table describes the default comment and attachment settings and roles assigned to process each task type:

Task Type	Role	Comment	Attachment
AVOSBeUnMerge	DataSteward	Optional - Do not prompt users	Disabled - Do not prompt users
AVOSBeMerge	DataSteward	Optional - Do not prompt users	Disabled - Do not prompt users
AVOSBeFinalReview	SrManager	Optional - Do not prompt users	Disabled - Do not prompt users
AVOSBeNotification	DataSteward	Optional - Do not prompt users	Disabled - Do not prompt users
AVOSBeUpdate	DataSteward	Optional - Do not prompt users	Disabled - Do not prompt users
AVOSBeReviewNoApprove	Manager	Optional - Do not prompt users	Disabled - Do not prompt users

Trigger configuration

The following table describes the default triggers:

Trigger	Event	Role	Other Information
DefaultApproval	CreateBE and UpdateBE	All roles	-
Default Merge	MergeBE	All roles	-
Default UnMerge	UnMergeBE	All roles	-
Matched	UnMergeBE	System	start workflow process: BeMergeTask task kind: MERGE task template: MergeTaskGenerator first task type: AVOSBeMerge

Task template configuration

The following table describes the configuration for the default task templates:

Task Type	Title	Priority	Due Date	Comment	Status
DefaultApproval	Review changes in {taskRecord[0].label}	Normal	+7d	-	Open
MergeTaskGenerator	{taskRecord[0].label} - Merge	Normal	+7d	Merge task autogenerated by Hub.	Open

Changing the Approver for Tasks

If you change the role of the approver for tasks, the change applies to new tasks only. To process existing tasks, configure a default trigger for the previous approver.

1. Click **Business Entity** > **Tasks**, and then select **Triggers**.
2. Click **Create**.
3. In the properties panel, enter `DefaultApproval` in the **Name** field, and then click **Apply**.
4. Create an event.
   a. In the tree view panel, select **Events**, and then click **Create**.
   b. In the properties panel, select an event from the **Event** list, and then click **Apply**.
5. Associate the previous role with the DefaultApproval trigger.
   a. In the tree view panel, select **Roles**, and then click **Create**.
   b. In the properties panel, select the role from the **Role** list, and then click **Apply**.
6. Optionally, specify the business entities to which the trigger applies.
   a. In the tree view panel, expand the **Role** node.
   b. Expand the node for the previous role.
   c. Select **Business Entities**, and then click **Create**.
   d. In the properties panel, select the business entity to which the trigger applies from the **Business Entity** field.
   e. Repeat steps 4 to 5 for any other business entities to which the trigger applies.
7. Click **Apply**.

   The changes are saved but are not published to the MDM Hub.
8. Click **Publish**.

CHAPTER 12

Configuring Security and Data Filters for Business Entities

This chapter includes the following topics:

- Security and Data Filters for Business Entities Overview, 114
- Entity-level Security, 115
- Field-level Data Filters, 119
- Node-level Data Filters, 122
- Security and the Data Director User Interface, 125

Security and Data Filters for Business Entities Overview

You can control access to business entity records by implementing the MDM Hub user roles.

You can control access to records at the following levels:

- Entity-level security. Controls whether users have access to all or none of the records for a business entity.
- Field-level data filters. Controls whether users have access to a record based on the value that appears in a field in the record.
- Node-level data filters. Controls whether users have access to a parent record when a node has no child records.

Tip: Start by granting role privileges at the entity level, and then you can use field filters and node filters to refine that access.

A business entity view contains a subset of nodes and fields from a business entity. The view inherits the data filters from the business entity. Consider implementing business entity views to hide some fields from users. For more information, see the "Transforming Business Entities and Views" chapter.

The Data Director user interface adapts to the user role privileges. For example, if a user role cannot create a record based on the Person business entity, the user cannot select the business entity type when creating a record.

GDPR Example

Your organization has customers in the European Union and other countries. You must implement the General Data Protection Regulation (GDPR) to ensure that only EU data stewards have access to the records of European Union citizens. You have other data stewards who have access to the records of non-European Union citizens. In this example, the customer data includes a GDPR field.

The following table summarizes the requirements for role access based on the value of the GDPR field:

GDPR Value in Data	DataSteward-EU Role	DataSteward-NonEU Role
GDPR="true"	Allow access	Deny access
GDPR="false"	Deny access	Allow access

The following process outlines one way to implement the GDPR example:

1. In the Hub Console, create the following user roles: DataSteward-EU and DataSteward-NonEU.
2. Grant all permissions to each role for the Customer base object tables, child tables, lookup tables, and cleanse functions.
3. In the Provisioning tool, open the Customer business entity.
4. Create a field filter for the GDPR field.
5. Create a deny rule, and set the value to true and the role to DataSteward-NonEU.
   This rule translates to "When GDPR is selected, don't allow the non-EU data stewards to access the records."
6. In the Remaining Values rule, set the role to DataSteward-EU.
   This rule translate to "When GDPR is not selected, don't allow the EU data steward to access the records."
7. Publish the changes.

Entity-level Security

You can allow or deny user access to all the base object tables and other resources on which a business entity is based. For example, you might assign all permissions to a data steward user role, but assign only read and update permissions to a manager user role.

To assign privileges to user roles, use the Security Access Manager in the Hub Console.

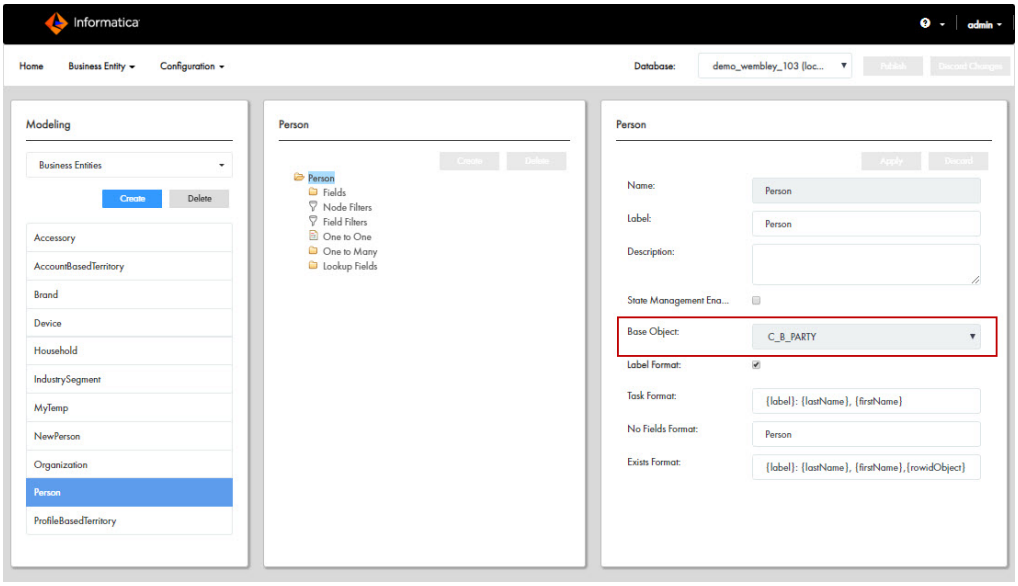
Managing Access to Business Entity Records

Ensure that the user role has permission to the root base object for the business entity and any related resources. In the Provisioning tool, find the root base object and other resources, such as lookup tables. In the Hub Console, verify the user role permissions for the root base object and resources.

1. Click **Business Entity** > **Modeling**, and then select **Business Entities**.
2. Select the business entity.

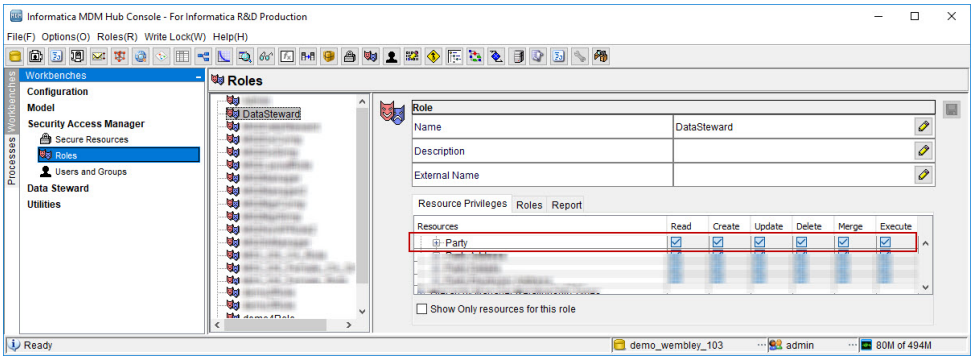
   The business entity opens with the root of the tree selected in the tree view panel.
3. Make a note of the name of the table that appears in the **Base Object** field of the properties panel.

The following image shows a Person business entity that is based on the C_B_PARTY table:



4. Log in to the Hub Console.
5. Acquire a write lock.
6. From the **Security Access Manager** workbench, click **Roles**.
7. For each user role that requires access to the business entity, ensure that the role has appropriate permissions to base objects and other resources. For a summary of permissions, see "Role Privileges for a Business Entity" on page 117.
   a. Select a role that requires access to the business entity.
      For example, select Data Steward.
   b. In the **Resource Privileges** tab, find the base object with the name that you noted in the Provisioning tool.
      For example, look for Party.
   c. Set the permissions for the business entity by selecting the corresponding permissions for the base object.
      For example, if you want the Data Steward role to have full access, select all the permissions for the Party base object.

      The following image shows the Hub Console with the Data Steward role selected and the Party base object highlighted:



d. Set the permissions for any related resources, such as child records, lookup tables, and cleanse functions.

8. Click **Save**.

Role Privileges for a Business Entity

To gain access to a business entity, your user role must have the correct resource privileges on the base objects and other resources.

For example, to be able to create a record for a Person business entity, a user role requires the create and read privileges on the root base object, the read privilege on the lookup base objects, and the execute privilege on the cleanse functions.

The following table lists the minimum privileges that are required for the ORS resources to create a record:

ORS Resource	Resource Privilege
Root base object	Create and Read
Lookup base object	Read
Cleanse functions used in transformations	Execute

The following table lists the minimum privileges that are required for the ORS resources to read a record:

ORS Resource	Resource Privilege
Root base object	Read
Lookup base objects	Read

The following table lists the minimum privileges that are required for the ORS resources to update a record:

ORS Resource	Resource Privilege
Root base object	Read and Update
Lookup base objects	Read
Child: many	Create, Update, or Delete
Child: one	Update
Cleanse functions used in transformations	Execute

The following table lists the minimum privileges that are required for the ORS resources to delete a record:

ORS Resource	Resource Privilege
Root base object	Delete
XREF object	Delete

The following table lists the minimum privileges that are required for the ORS to search for records:

ORS Resource	Resource Privilege
One or more searchable fields	Read

The following table lists the merge and unmerge actions that you can perform for a business entity and the minimum privileges that are required for the ORS resources:

Business Entity Actions	ORS Resource	Resource Privilege
Merge records	One or more nodes	Update and Merge
Start a merge task	One or more nodes	Read and Update
Unmerge a record	One or more nodes	Merge
Start an unmerge task	Root base object	Read
Start an unmerge task	One or more nodes	Read

Note: If a role is allowed to perform an action without going through a workflow, remember to set a matching workflow trigger for that role.

Role Privileges for a Business Entity View

A business entity view is a subset of nodes in a business entity. To create a business entity view, you create a transformation from the business entity to the business entity view. The business entity view inherits the role privileges from the business entity.

The following table lists the actions that you can perform for a business entity view and the minimum privileges that are required on the nodes in the business entity:

Actions in the Business Entity View	Privileges on the Nodes in the Business Entity
Create a record	All nodes that appear in the view must have the create privilege.
Read a record	When at least one node that appears in the view has the read privilege, then the entire view can be read.
Update a record	When at least one node that appears in the view has the update privilege, then the entire view can be updated.
Delete a record	All nodes that appear in the view must have the delete privilege.
Merge records	All nodes that appear in the view must have the merge privilege.
Unmerge a record	All nodes that appear in the view must have the unmerge privilege.

Related Topics:

- "Modeling Business Entity Views " on page 55

Field-level Data Filters

You can control access to records based on a value in a field. For example, for compliance with the General Data Protection Regulation (GDPR), you want only the EU data stewards to access records that have an address within the European Union.

You can create field filters on business entities. In a field filter, you define access permissions in terms of deny rules, allow rules, or a mix of both types of rules. A rule is made up of a value for the field and the applicable user roles. For example, you might create a field filter on a GDPR field. Create an allow rule on the field with the value of true and select the DataSteward-EU role. Use the Remaining Values rule to assign false to all other user roles.

Field filters in the Provisioning Tool restrict task visibility to business entity records that the role can access. The restriction applies to tasks, such as Merge, Unmerge, Match, Escalate, Cancel, and Notification.

Note: When you update field filters, existing tasks remain unchanged. Updated filters apply to new tasks.

The field filters on a business entity override the field filters on a reference entity. A user role might be allowed to see a subset of lookup values based on the field filter in the reference entity, but if a field filter on the business entity denies access to some of those values, the user role cannot see the records with those values.

Field Filter Rules and User Roles

When you create a field filter for a business entity, the records that a user can see are constrained by the rules in the field filter and the assigned user role.

Before you create field filters, decide how you want to manage rules. You can define rules in terms of who is denied access, who is allowed access, or both. Familiarize yourself with the behaviors of each approach and create an overall plan. For maintenance purposes, it can be more straightforward to use only one type of rule.

Important: Avoid creating an excessive number of field filters. The more filters, the longer it takes to process API requests. If you have performance issues after adding field filters, consider reducing the number of field filters.

Deny rules only

Deny rules are joined by a logical OR operator. If you create at least one deny rule, you must also fill in the Remaining Values rule in the Deny section. If the Remaining Values rule is empty, and records contain a value that is not covered by the other rules, users can see the records with the other values.

If you define only deny rules for a field filter, the following behaviors apply:

- When a user role is not assigned to any deny rules, the user can see any record.
- When a user role is assigned to all the deny rules, the user cannot see any record.
- When a user role is assigned to one deny rule, and a record satisfies the rule, the user cannot see the record.
- When a user role is assigned to multiple deny rules, and a record satisfies **any** of the rules, the user cannot see the record.

- When a user role is not assigned to any deny rules, but one or more deny rules have the **Apply this rule** option selected, and a record satisfies any of these rules, the user cannot see the record.

Allow rules only

Allow rules are joined by a logical AND operator. If you create at least one allow rule, you must also fill in the Remaining Values rule in the Allow section. If the Remaining Values rule is empty, and records contain a value that is not covered by the other rules, users cannot see the records with the other values.

If you define only allow rules for a field filter, the following behaviors apply:

- When a user role is not assigned to any allow rules, the user cannot see any record.
- When a user role is assigned to all the allow rules, the user can see any record.
- When a user role is assigned to one allow rule, and a record satisfies the rule, the user can see the record.
- When a user role is assigned to multiple allow rules, and a record satisfies **all** the rules, the user can see the record.
- When a user role is assigned to multiple allow rules, and a record does not satisfy all the rules, the user cannot see the record.
- When a user role is not assigned to any allow rules but one or more allow rules have the **Apply this rule** option selected, and a record satisfies all of these rules, the user can see the record.

Both deny and allow rules

If you define both deny rules and allow rules for a field filter, the following behaviors apply:

- If a user role is not assigned to any deny rules or allow rules, the user cannot see any records.
- If a user role is assigned to one or more deny and allow rules, and a record satisfies any of the deny rules, the allow rules are ignored and the user cannot see the record.
- If a record does not satisfy a deny rule, the allow rules are processed as described in the **Allow rules only** section.

Multiple field filters exist for the same business entity

When a user role is assigned to more than one field filter, the following behaviors apply:

- If a record satisfies a deny rule in any of the field filters, the user cannot see the record. If an allow rule is assigned to the same user role, the allow rule is ignored.
- If a record does not satisfy any deny rules, the allow rules are processed as described in the **Allow rules only** section.

Adding Field Filters

Before you begin, create the business entities. To allow or deny access to records based on a value in a field, create field filters.

Note: Field filters are case insensitive.

1. Click **Business Entity** > **Modeling**, and then select **Business Entities**.
2. Select the business entity.
3. In the tree view panel, select **Field Filters**, and then click **Create**.

   A **[New fieldFilter]** option appears under **Field Filters**, and a form opens in the properties panel.
4. In the properties panel, enter a name for the filter.

5. Select a field.
   a. Beside the **Field** field, click the **Browse** button.

      A list of fields opens in a dialog box.
   b. Select the field, and then click **Select**.

      The selected field appears in the **Field** field.
6. To define the filter in terms of who *cannot* access the business entity when the data matches the value, create deny rules.
   a. Beside the **Deny** heading, click the **Add** icon.
   b. In the **Value** field, enter a value that matches the data type of the field.

      Note: When you enter dates, use the date format that your database environment requires.
   c. From the **User Role** list, select the user roles, and then click the **Add** icon.
   d. To make this rule a default for any unassigned user roles in the **Deny** section, select **Apply this rule to the user roles that are not specified in any Deny rule**.
   e. Repeat to configure more filter rules by clicking the **Add** icon beside the **Deny** heading.
   f. Add user roles to the **Remaining Values** rule.

      Caution: If the Remaining Values rule is empty, and a record contains a value that is not covered by the other rules, users can see the records.
7. To define the filter in terms of who *can* access the business entity when the data matches the value, create allow rules.
   a. Beside the **Allow** heading, click the **Add** icon.
   b. In the **Value** field, enter a value that matches the data type of the field.

      Note: When you enter dates, use the date format that your database environment requires.
   c. From the **User Role** list, select the user roles, and then click the **Add** icon.
   d. To make this rule a default for any unassigned user roles in the **Allow** section, select **Apply this rule to the user roles that are not specified in any Allow rule**.
   e. Repeat to configure more filter rules by clicking the **Add** icon beside the **Allow** heading.
   f. Add user roles to the **Remaining Values** rule.

      Caution: If the Remaining Values rule is empty, and a record contains a value that is not covered by the other rules, users cannot see the records.
8. Click **Apply**.

   The field filter is saved but not published to the MDM Hub.

Role Privileges for Field Filters

For each user role that you assigned in a field filter, ensure that the user role has the required privilege on the base object column.

The following table lists the actions that you can perform for a business entity field and the minimum privileges that are required on the ORS resource:

Business Entity Field Action	ORS Resource	Resource Privilege
Create a value	Base object column	Create
Read a value	Base object column	Read
Update a value	Base object column	Update

Node-level Data Filters

If you added field filters to fields in child nodes, you can add node filters. The node filters define what to do when the child node has no child records. By default, an empty child node does not affect access to the parent record. If you want to restrict access to the parent record, configure node filters to deny or allow access to the parent record.

Node Filter Rules

When you create a node filter for a business entity, you control access to the parent record when the node has no child records. For example, if a customer record has no address associated with it, you might allow users with the DataSteward-Global role permission to view the parent record, but all other data steward roles cannot view the parent record.

Before you create node filters, decide how you want to manage rules. You can define rules in terms of who is denied access, who is allowed access, or both. Familiarize yourself with the behaviors of each approach and create an overall plan. For maintenance purposes, it can be more straightforward to use only one type of rule.

Important: Avoid creating an excessive number of filters. The more filters, the longer it takes to process search requests. If you have performance issues after adding filters, consider reducing the number of filters.

Deny rules only

Deny rules are joined by a logical OR. If you define only deny rules for a node filter, the following behaviors apply:

- When a user role is not assigned to any deny rules, the user can see any parent record.
- When a user role is assigned to all deny rules, the user cannot see any parent record.
- When a user role is assigned to one deny rule, and there is no child record for the node, the user cannot see the parent record.
- When a user role is assigned to multiple deny rules, and at least one child record exists that satisfies the rule, the user cannot see the parent record.
- When a user role is not assigned to any deny rules, but one or more deny rules have the **Apply this rule** option selected, the user cannot see the parent record if a record satisfies any of these rules.

Allow rules only

Allow rules are joined by a logical AND. If you define only allow rules for a node filter, the following behaviors apply:

- When a user role is not assigned to any allow rules, the user cannot see any parent record.
- When a user role is assigned to all allow rules, the user can see any parent record.
- When a user role is assigned to one allow rule, and there is no child record for the node, the user can see the parent record.
- When a user role is assigned to multiple allow rules, and **all** the rules are satisfied, the user can see the parent record.
- When a user role is assigned to multiple allow rules, and any of the rules are not satisfied, the user cannot see the parent record.
- When a user role is not assigned to any allow rules, but one or more allow rules have the **Apply this rule** option selected, the user can see the record if all the rules are satisfied.

Both deny and allow rules

If you define both deny rules and allow rules for a node filter, the following behaviors apply:

- If a user role is not assigned to any deny rules or allow rules, the user cannot see any parent records.
- If a user role is assigned to one or more deny and allow rules, and a record satisfies any of the deny rules, the allow rules are ignored and the user cannot see the parent record.
- If a deny rule is not satisfied, the allow rules are processed as described in the **Allow rules only** section.

Multiple node filters exist for the same business entity

When a user role is assigned to more than one node filter, the following behaviors apply:

- If a deny rule is satisfied in any of the node filters, the user cannot see the parent record.
- If a deny rule is not satisfied among the node filters, the allow rules are processed as described in the **Allow rules only** section.

Adding Node Filters

To allow or deny access to the parent record when the node has no child records, create node filters.

Tip: To make your configuration easier to understand and maintain, avoid mixing allow and deny rules. If you decide to create both types of rules, and the rules result in a conflict for a user role, the deny rule wins.

1. Click **Business Entity** > **Modeling**, and then select **Business Entities**.
2. Select the business entity.
3. In the tree view panel, select **Node Filters**, and then click **Create**.

   A **[New nodeFilter]** option appears under **Node Filters**, and a form opens in the properties panel.
4. In the properties panel, enter a name for the filter.
5. Select a node.
   a. Beside the **Node** field, click the **Browse** button.

      A list of nodes opens in a dialog box.
   b. Select the node, and then click **Select**.

      The selected node appears in the **Node** field.

6. To define the filter in terms of who *cannot* access the parent record when the node record is empty, create deny rules.
    a. Beside the **Deny Access to <parent record>** heading, click the **Add** icon.
    b. From the **User Role** list, select the user roles, and then click the **Add** icon.
    c. To make this rule a default for any unassigned user roles in the **Deny** section, click **Apply this rule to the user roles that are not specified in any Deny rule**.
    d. Repeat the steps to configure more filter rules.
7. To define the filter in terms of who *can* access the parent record when the node record is empty, create allow rules.
    a. Beside the **Allow Access to <parent record>** heading, click the **Add** icon.
    b. From the **User Role** list, select the user roles, and then click the **Add** icon.
    c. To make this rule a default for any unassigned user roles in the **Allow** section, click **Apply this rule to the user roles that are not specified in any Allow rule**.
    d. Repeat the steps to configure more filter rules.
8. Click **Apply**.

    The node filter is saved but not published to the MDM Hub.

Role Privileges for Nodes Filters

Ensure that the user role has the necessary permissions on the objects and other resources.

The following table lists the minimum privileges that are required for the ORS resources to create a record:

ORS Resource	Resource Privileges
Object	Update
Base Object	Create
Lookup base objects	Read
Cleanse Functions used in transformations	Execute

The following table lists the minimum privileges that are required for the ORS resources to read a record:

ORS Resource	Resource Privileges
Base object	Read
Lookup base objects	Read

The following table lists the minimum privileges that are required for the ORS resources to update a record:

ORS Resource	Resource Privileges
Object	Update
Base object	Update

ORS Resource	Resource Privileges
Lookup base objects	Read
Cleanse Functions used in transformations	Execute

The following table lists the minimum privileges that are required for the ORS resources to delete a record:

ORS Resource	Resource Privileges
Object	Update
Base object	Delete
XREF object	Delete

The following table lists the minimum privileges that are required for the ORS resources to merge records:

ORS Resource	Resource Privileges
Base object (with merge task)	Update
Base object (without merge task)	Merge

The following table lists the minimum privileges that are required for the ORS resources to unmerge a record:

ORS Resource	Resource Privileges
One or more nodes	Merge

Security and the Data Director User Interface

Security and data filters affect the Data Director user interface.

The following table summarizes how the permission to perform an action affects the behavior of the user interface:

Business Entity Action	Data Director Behavior
Create records	In the **New** menu, you can see only the types of business entities for which you have permission to create records.
Read records	You can view the records for the business entities that you have permission to read. Other records are inaccessible.
Update records	In a record view, the **Edit** button is enabled when you have permission to edit business entity records. Otherwise, the **Edit** button is not visible.

Business Entity Action	Data Director Behavior
Delete records	The **Delete** icon is enabled when you have permission to delete business entity records.
Search for records	When searching for business entities by using search, search filters and search results contain only the business entity types that you have permission to view.
Merge and unmerge records	In the **Matching Records** view, the **Merge** button and the check boxes at the top of each record are enabled when you have permission to merge and unmerge records. If you add records to the view, the search filters and search results contain records only for the business entities that you have permission to view.

CHAPTER 13

Configuring the Content Security Policy

This chapter includes the following topics:

- Content Security Policy Overview, 127
- Content Security Policy Defaults, 127
- Configuring the Content Security Policy, 128

Content Security Policy Overview

Content Security Policy (CSP) is a standard that prevents code injection attacks, such as cross-site scripting. Multidomain MDM has content security policy rules to allow approved origins of content that a browser can load to display website content. The content security policy rules are applied across the system.

For example, to allow content from Twitter to display on your website, you can set a content security policy rule to allow content originating from `platform.twitter.com`. If you don't set the content security policy rules to allow content originating from Twitter, the linked content appears as a broken link.

Content Security Policy Defaults

When you install Multidomain MDM, default rules are applied that allow content required for Multidomain MDM to run.

Multidomain MDM has the following default rules:

```
default-src 'self'; script-src 'self', 'unsafe-eval'; style-src 'self', 'unsafe-inline',
'unsafe-eval', "use.fontawesome.com", "maxcdn.bootstrapcdn.com", "fonts.googleapis.com",
"netdna.bootstrapcdn.com"; font-src 'self' data:, "fonts.gstatic.com",
"use.fontawesome.com", "maxcdn.bootstrapcdn.com", "fonts.googleapis.com",
"netdna.bootstrapcdn.com"; img-src 'self' data:; frame-src 'self' blob:;
```

The default rules applied during installation can't be changed or deleted. Any additional rules applied during upgrade or after installation can be modified or deleted.

Note: When you upgrade, the upgrade process checks for custom user interface components in the Operational Reference Store (ORS) databases. The following rules are added if any custom user interface components exist:

```
script-src *; font-src *; style-src *; frame-src *; image-src *; default-src *;
```

Customize these rules after upgrading Multidomain MDM to secure your system and prevent code injection attacks.

To optimize the content security policy for your organization, configure rules to allow only the approved origins for required content.

Configuring the Content Security Policy

You can configure the content security policy to declare approved origins of content that a browser can load to display website content.

1. Click **Configuration** > **General Settings**, and then select **Content Security Policy**.
2. Click **Content Security Policy**.

   The tree view panel and the content security policy panel display the content security policy options.
3. In the **Content Security Policy** panel, click **Add Content Security Policy Tag**.
4. Select a policy tag.
5. Enter a value for the policy tag.

   Based on the selected policy tag, you can set standard values, such as `'self'; 'unsafe-inline'; 'none'`, or you can add URLs to allow or block content. Use the **Add** button to add multiple values for a policy tag. Use the **-** button to delete values.
6. To apply the updated content security policy to your system, click **Apply**.
7. Click **Publish**.
8. Review the changes and click **Confirm**.

CHAPTER 14

Integrating Data as a Service

This chapter includes the following topics:

- Overview, 129
- WSDL File for a DaaS Service, 130
- Transformations, 130
- SOAP Services, 131
- DaaS Providers, 131
- DaaS Providers Component, 132
- Integrating a DaaS Provider, 132

Overview

You can enrich and augment your business entity data with data from third-party data providers. You can integrate with Data as a Service (DaaS) providers to access data that is reliable, accurate, and complete. You can request a DaaS provider for relevant business entity information, such as revenues, parent-child or corporate linkages, detailed company profiles, and D-U-N-S numbers. You can use the data from the DaaS providers to create and update business entity records.

You can enable or disable the access to the DaaS providers. You can configure access to the DaaS services based on the user roles and the operations that a user performs. For example, you might enable a service only when a user creates a record. You can specify the business entities that you want to enrich with the DaaS data. You can select the business entity fields that you want to populate with the data from the DaaS provider. You can request data from the DaaS provider, compare the search results, and select the records that you want to convert and merge into a business entity record.

To integrate with a DaaS service, you must first register with the DaaS provider. In the Provisioning tool, you must create the required configurations, create a DaaS component, and add the component to a Data Director record view layout. When you open the record view for a business entity record, the DaaS providers appear in the DaaS panel.

WSDL File for a DaaS Service

The DaaS services are web services that support the Simple Object Access Protocol (SOAP) for the requests and responses from client applications. To use a DaaS service, you must upload the Web Services Description Language (WSDL) file for the service. A WSDL file describes a web service.

A WSDL file contains the following information:

- XML descriptions of the available web services.
- Location of the web service.
- Methods that the service uses.
- Parameters that the service takes.
- Formats of the SOAP requests and responses.

Transformations

When you integrate with a DaaS provider for data enrichment, you must match some of the business entity information you have with the information from the DaaS provider. Send a request to the DaaS web service with some minimum required information about a business entity. Based on the information, the DaaS provider sends a response that you can use to enrich the business entity data.

Use the SOAP protocol for requests and responses. The request to and the response from the DaaS service is an XML structure. A business entity service accepts some business entity data as input and maps it to fields in an XML document. The service calls the DaaS web service with this information and receives the response in the form of an XML document. The service converts the XML document to the business entity fields and saves the information as business entity data. You must configure transformations and map the business entity fields to the XML request fields and the XML response fields to the business entity fields .

Configure the following transformations for DaaS:

- **Business entity to XML**. Mapping required to send a request to a DaaS service.
- **XML to business entity**. Mapping required to read and convert a response from a DaaS service.

Transformation of Business Entity to XML

You can configure the transformation of a business entity to an XML format. When you send a request to a DaaS service, you must send the request in XML format. You transform the data in the business entity into an XML document. You must map the business entity fields to the XML fields that the DaaS service requires in the request.

Note: Before you configure the transformation, ensure that you upload the WSDL for the DaaS service that you plan to use. The WSDL file specifies the parameters that the service requires.

Transformation of XML to Business Entity

You can configure the transformation of an XML document to a business entity. The DaaS service sends the response in XML format. Parse the XML response and map the fields in the XML document to the business entity fields.

Note: Before you configure the transformation, ensure that you upload the WSDL for the DaaS service you plan to use.

The corporate linkage service from Duns & Bradstreet (D&B) returns the parent and the related entities of a requested organization. If you want to import the linkage information, you must develop a custom application that can use the linkage service.

You configure a transformation of an XML document to business entity and expose it as a service. When you expose an XML document to business entity transformation as a service, the process creates the DaaS Import and DaaS Update business entity services. Use the DaaS Import and DaaS Update services to import business entity data and update changes to business entity data.

For more information about how to use the linkage service, see the Supporting Corporate Linkage Service chapter in the *Multidomain MDM Business Entity Services Guide*.

SOAP Services

The data enrichment DaaS services are a suite of web services that support the SOAP protocol. You must integrate with the DaaS services and make SOAP calls to access the information to enrich your business entity data.

You must register the external SOAP services. If a service requires authentication information for calls, provide the information in the SOAP header when you configure the SOAP service.

DaaS Providers

Communication with a DaaS provider is through SOAP requests and responses. Send a request to the DaaS provider with the minimum required information about a business entity. The DaaS provider responds with information that you can use to expand the business entity view and enrich the business entity data.

You must configure a DaaS provider as an extension in the Provisioning tool. A DaaS provider configuration combines a web service call with business entity to XML format and XML format to business entity transformations. You must specify the metadata, such as required fields and the type of service, such as READ or SEARCH.

For example, if you specify the name of an organization, the DaaS service searches for the name and returns a list of records with matching names. If you specify a unique value, such as the D-U-N-S number, the DaaS service performs a direct match, and returns the details of the organization. You can configure multiple DaaS providers.

Note: Every DaaS provider must have at least one required field. Specify the required fields when you configure a DaaS provider.

Communication with a DaaS service is through the DaaS component that you add to a record view layout.

DaaS Providers Component

The DaaS Providers component is a component that you can add to a record view layout. The component lists the DaaS providers that you configure for the record that is open in a record view.

You can use the DaaS Providers component to request a DaaS provider for data. You can search for records, compare the results, and select the records that you want to convert and merge into a business entity record. Create the DaaS Providers component in the Components Editor of the Provisioning tool. After you create the component, the component appears in the Layout Designer. Use the Layout Designer to create a record view layout and add the component to the layout.

The following image shows a sample DaaS Providers component configuration:



Integrating a DaaS Provider

To integrate a DaaS provider, create the required configurations and DaaS Provider component. Design a record view layout and add the DaaS Provider component in it.

Perform the following tasks to integrate a DaaS Provider with the MDM Hub:

1. Upload the WSDL for the DaaS service.
2. To transform a business entity to XML format, map the business entity fields to the input for the SOAP service request.
3. Transform the XML response from the SOAP service and map the XML elements to the fields in the business entity.
4. Register the SOAP service.
5. Configure a DaaS provider by bringing together the WSDL, SOAP service, and transformations, and then publish the configuration to the MDM Hub.
6. Create a DaaS Providers UI component.
7. Design a record view layout, add the DaaS Providers component to the layout, and publish the configuration to the MDM Hub.

Uploading a WSDL File

Upload the WSDL file for the DaaS web service you want to use. The WSDL file describes the operations along with the format and data types of the requests and responses.

1. Click **Business Entity** > **Extensions**, and then select **WSDL**.
2. Click **Create**.

3. In the properties panel, specify the following properties:

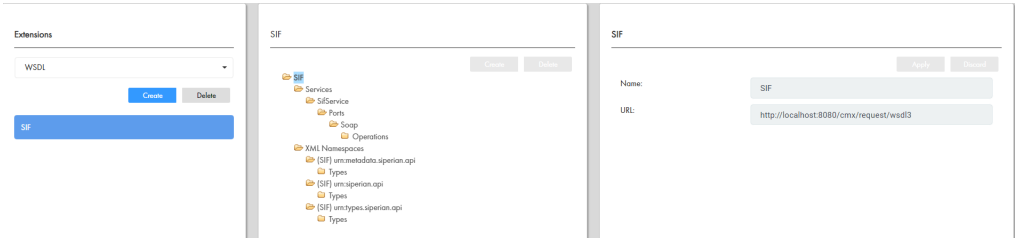
Property	Description
Name	Name for the WSDL file.

URL	URL of the WSDL file for the web service. You can refer to the following sample URLs:
	- For email verification and phone number validation:
	`http://ws.strikeiron.com/OnDemandEntityID1?wsdl`
	- For Services Integration Framework (SIF) APIs:
	`http://localhost:8080/cmx/request/wsdl3`

4. Click **Apply**.

   The details of the WSDL appear in the tree view panel.

   The following image shows a WSDL file for a DaaS service:



Transforming a Business Entity to XML Format

A request to the web service requires inputs in XML format. You must define the mappings for the service request. To transform a business entity into an XML format, map the business entity fields to the input fields of a service request.

1. Click **Business Entity** > **Transformations**, and then select **Business Entity to XML**.
2. Click **Create**.
3. In the properties panel, specify the following properties:

Property	Description
Name	Name for the transformation.
URL	Identifies the namespace where the SOAP operation is defined.
Business Entity	Business entity for which you request the service.
Type	Select the namespace.
Target	Request element. Click the **Browse** button and select the root element.

4. Click **Apply**.
5. To configure the transformation for the root node, select **Transformations** in the tree view panel, and then click **Create**.

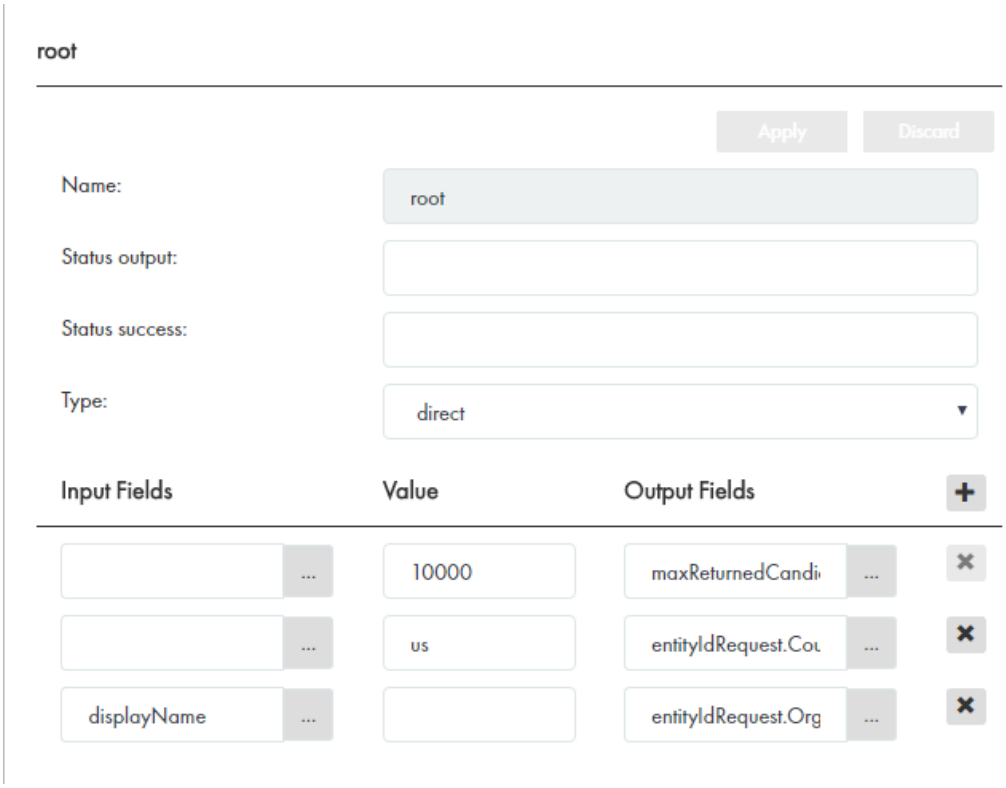
6. In the properties panel, specify the following properties:

Property	Description
Name	Name for the transformation.
Status Output	Name of the cleanse output parameter. Used in cleanse transformations. Not applicable to transformations for DaaS.
Status Success	A regular expression. When the cleanse operation successfully completes, it returns a value in the cleanse output parameter that matches the regular expression. Not applicable to transformations for DaaS.
Type	Type of transformation to map data from a business entity field to the XML field. Direct transformations keep the data unchanged.

7. Map the business entity fields to the fields in the request:
   a. In the **Input Fields** column, click the **Browse** button.
   b. Select the business entity field that you want to map, and then click **Select**.
   c. In the **Output Fields** column, click the **Browse** button.
   d. Select the field in the request, and then click **Select**.

      Note: If you specify a field, the value is taken from the Service Data Object (SDO), from the business entity or the XML data. For some output fields, you might specify constant values, and need not map the values to any input fields. For example, you can specify the maximum number of candidates that the service can return or the country code. You do not map such values to any input field. Specify the value in the **Value** field and map it to an Output Field.
   e. To map additional information, click the **Add** icon, and then follow Steps `a` to `d`.
   f. Click **Apply**.

   The following image shows a sample mapping of fields for the root entity:



Transforming XML Data to Business Entity

To use the DaaS services, you must define the mappings for the response that the service returns. Map the fields in the response to the business entity fields. Specify which fields from the XML output of the service you would like to store in your business entity.

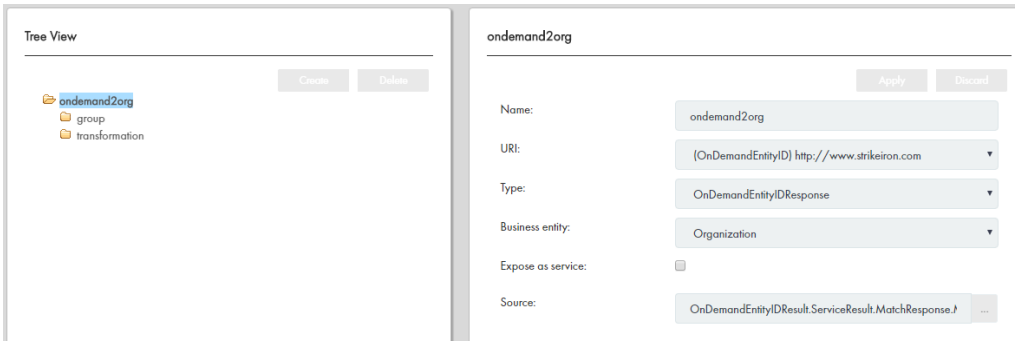
1. Click **Business Entity** > **Transformations**, and then select **XML to Business Entity**.
2. Click **Create**.
3. In the properties panel, specify the following properties:

Property	Description
Name	Name for the transformation.
URL	Identifies the namespace where the SOAP operation is defined.
Type	Select the namespace.
Business Entity	Business entity you want to transform.

Property	Description
Expose As Service	If set to true, the transformation is exposed as two business entity services, Import and Update. These services accept XML data, transform it into business entity data, and save it as a business entity record.
Source	Element in the response object. Click the **Browse** button to search for the element.

4. Click **Apply**.

   The following image shows a sample XML data to business entity transformation:

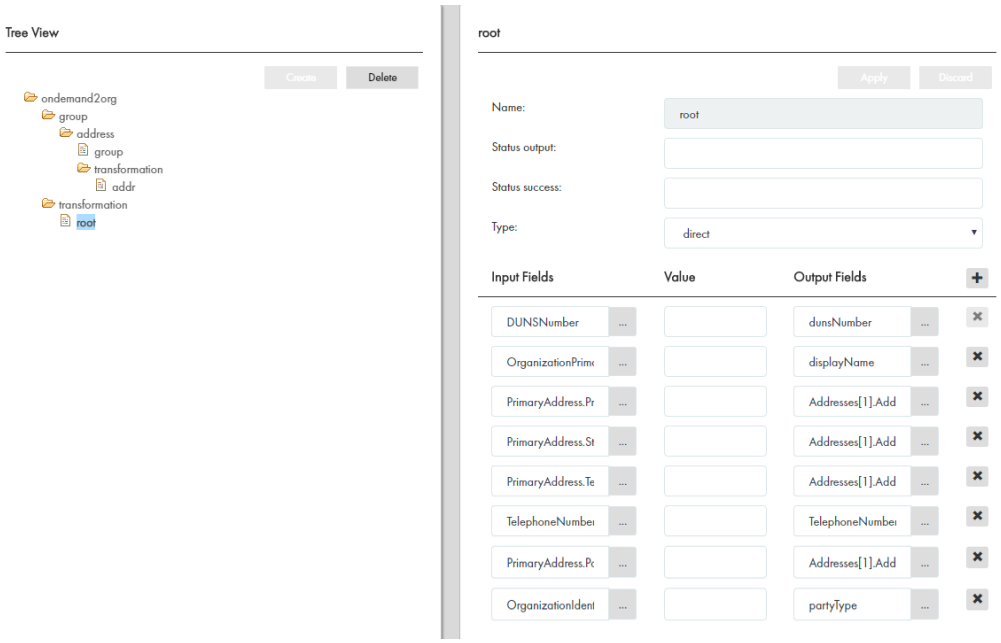


5. To configure the transformation for the root node, select **Transformations** in the tree view panel, and then click **Create**.
6. In the properties panel, specify the following properties:

Property	Description
Name	Name for the transformation. For example, transformation for root.
Status Output	Name of the cleanse output parameter. Used in cleanse transformations. Not applicable to transformations for DaaS.
Status Success	A regular expression. When the cleanse operation successfully completes, it returns a value in the cleanse output parameter that matches the regular expression. Not applicable to transformations for DaaS.
Type	Type of transformation to directly map data from a business entity field to the XML. Direct transformations keep the data unchanged.

7. Map the XML elements in the service response to the business entity fields:
   a. In the **Input Fields** column, click the **Browse** button.
   b. Select the field, and then click **Select**.
   c. In the **Output Fields** column, click the **Browse** button.
   d. Select the business entity field to which you want to map, and then click **Select**.
   e. To map additional information, click the **Add** icon, and then follow steps `a` to `d`.
   f. Click **Apply**.

   The following image shows a sample mapping of fields for the root node:



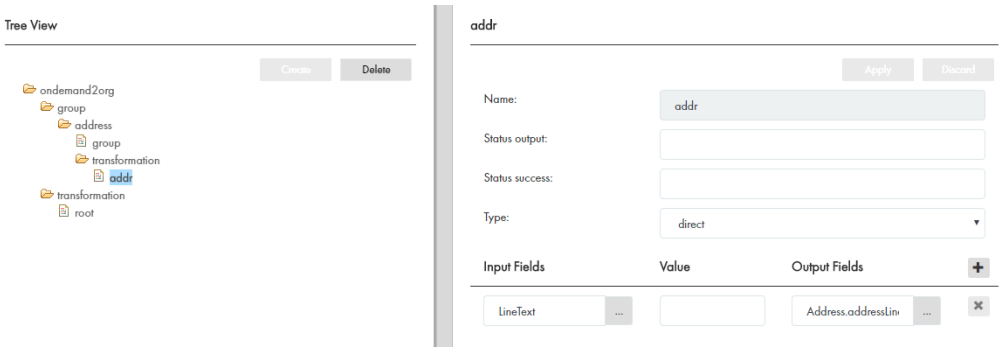
8. To configure the transformation for child nodes of the business entity, add the node to a group, and then configure the transformations:
   a. Select the **Groups** folder, and then click **Create**.
   b. In the properties panel, enter a name for the node as you want it to appear in the group folder.
   c. In the **Source** field, select a field, and in the **Target** field, select the child node in the business entity. Repeat for each child node.
   d. Click **Apply**.
   e. Expand a **child node**, select **Transformations**, then click **Create**.
   f. In the properties panel, enter a name for the transformation.
   g. From the **Type** list, select **Direct**.
   h. In the **Input Fields** column, click the **Browse** button.
   i. Select the field, and then click **Select**.
   j. In the **Output Fields** column, click the **Browse** button.
   k. Select the business entity field to which you want to map, and then click **Select**.

      Note: You can specify Service Data Object (SDO) XML path directly in a `field` attribute. When a field contains `/`, it indicates the XPath of the SDO. If you use a cleanse transformation before you populate the business entity with the XML data, you might require some parent XML elements in the child node mappings. However, parent XML elements are not available for selection in the transformation of child nodes. You must manually type the path to the required parent XML element in the **Field Name**. Use the Service Data Object (SDO) XPath expression to access a parent element, where `/` represents the root element. For example, when you configure the transformation of the address child node, you might concatenate the `dunsNumber` element with the `city` element. Because the `dunsNumber` element is a parent element, it is not available for selection. To access the `dunsNumber` element, use the path `/OrderCompanyProfileResult[1]/ServiceResult[1]/OrderProductResponseDetail[1]/InquiryDetail[1]/DUNSNumber[1]`
   l. Repeat Steps `h` to `k` for other fields.

m. Click **Apply**.

9. Repeat Step 9 for each child node.

   The following image shows a sample transformation and mapping of fields for a child node:



Registering a SOAP Service

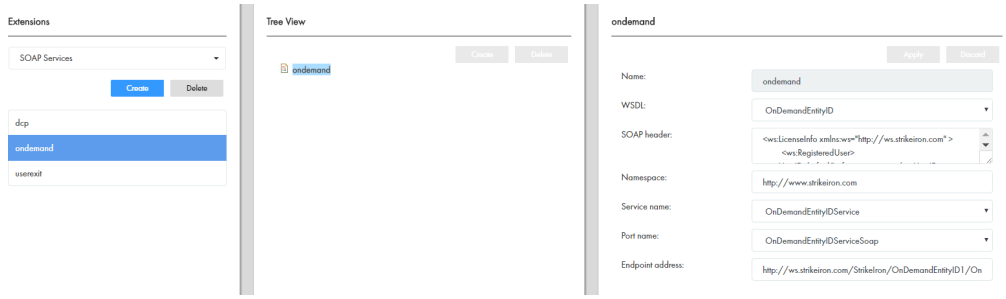
To make SOAP calls to the DaaS services, you must first register the required SOAP service. After you upload a WSDL file, register the SOAP service and the operations that the WSDL file describes.

1. Click **Business Entity** > **Extensions**, and then select **SOAP Services**.
2. Click **Create**.
3. In the properties panel, specify the following properties:

Property	Description
Name	Name of the SOAP service.
WSDL	URL of the WSDL file for the SOAP service.
SOAP Header	Additional application-specific information, such as authentication. For example, the service might require that each call to an operation has a valid user name and password in the SOAP header.
Namespace	Namespace associated with the SOAP service.
Service	Name of the service.
Port	Port name through which a client application can access the SOAP service.
Endpoint	URL or address where a client application can access the SOAP service.

4. Click **Apply**.

   The following image shows a sample SOAP service configuration:



Configuring a DaaS Provider

Configure a DaaS provider for making service calls to enrich the business entity data.

1. Click **Business Entity** > **Extensions**, and then select **DaaS Providers**.
2. Click **Create**.
3. In the properties panel, specify the following properties:

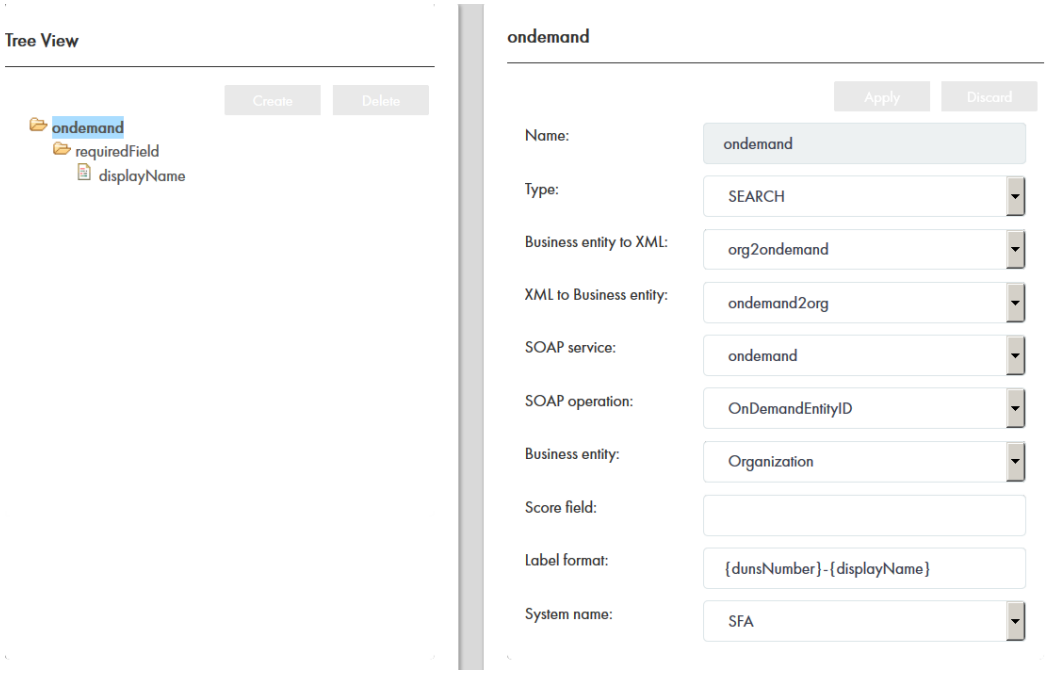
Properties	Description
Name	Name for the DaaS provider.
Type	Type of service that the provider uses, whether READ or SEARCH.
BE to XML	Business entity to XML transformation that you want to use.
XML to BE	XML to business entity transformation that you want to use.
SOAP Service	SOAP service that you want to use.
SOAP Operation	SOAP operation that you want to perform.
Business Entity	Business entity for which you want to request data from the DaaS provider.
Score Field	Additional information for each row in the search results. You map a field of the incoming XML data to this field.
Label Format	Format that will appear in the search result. Not applicable to DaaS.
Source System	Source system name, such as Sales Force Automation system (SFA).

4. Click **Apply**.
5. In the tree view panel, select **Required Fields**, and then click **Create**.

   Note: Each DaaS provider must have at least one required field. If you do not specify the required field or fields, the DaaS provider is disabled in the DaaS Providers component in the Data Director record view.
6. In the properties panel, select the required field.

   When you make a service call, the DaaS provider performs a search based on the value in the required field.
7. Click **Apply**.

The following image shows a sample DaaS provider configuration:



Creating a DaaS Providers Component

Create a DaaS Providers component in the Component Editor. After you create the DaaS Providers component, the component appears in the Layout Designer.

1. Click **Configuration** > **Component Editor**, and then select **DaaS Providers**.
2. Click **Create**.
3. In the properties panel, enter a name for the component, and then click **Apply**.

   The DaaS Providers component that you created appears in the tree view panel.
4. In the DaaS Providers component tree, click **daasProviders**, and then click **Create**.

   The DaaS Provider properties appear in the properties panel.
5. In the **Name** field, enter a name for the DaaS provider.
6. From the **Search** list, select the DaaS provider that you configured for a search operation.
7. From the **Read** list, select the DaaS provider that you configured for a read operation.
8. Click **Apply**.

The following image shows a sample DaaS Provider configuration:



Designing a Record View Layout with DaaS

You design a record view layout to display a record with the DaaS Provider. You can design the record view in the Layout Designer and add the DaaS Provider component to it. You can request the DaaS provider for data for the record that is open in the record view.

Related Topics:

- "Designing a Business Entity View Layout" on page 190

Deleting Transformations

You can delete transformations. If you delete a transformation on which components or extensions are based, the components and extensions are deleted.

Deleting WSDL files, SOAP Services, and DaaS Providers

You can delete the extensions, such as WSDL file, SOAP service, and DaaS provider.

If you delete an extension on which components or extensions are based, the components and extensions are deleted.

CHAPTER 15

Configuring External Calls

This chapter includes the following topics:

- Overview, 142
- Configuring External Calls, 143

Overview

External providers provide web services for cleansing, analysis, and transformation of business entity data. You can use the external web services for custom validation, such as checking if the address field is empty when you add a business entity. You can use the external web services to apply custom logic when you transform business entity data. For example, when you merge two business entity records, you can merge addresses, but you cannot merge telephone numbers.

You can configure calls to the external web services for certain steps in the business entity execution logic. You must configure the business entities and the events for which you want to call an external service. You must configure external calls as extensions to business entity services. Use the XML Schema Definition (XSD) for the external web services to get the request and response types. An external web service operation must use the request and response types for the input and output elements. Based on the logic you implement, requests go to the external services for analysis or transformation of business entity data.

The XSD file is in the following location: `<MDM Hub installation directory>\hub\ server\lib\mdm-spi.jar`

Use the sample WSDL file in the Resource Kit to understand the external services, operations, methods, and the data types that the service methods exchange. The sample `custom-logic-service.wsdl` file for the external web services is in the following location: `<MDM Hub installation directory>\hub\resourcekit \samples\BESExternalCall\source\resources\webapp\WEB-INF\wsdl\`

You must develop and deploy the external web services. Use the Ant build file, `build.xml` in the folder `<MDM Hub installation directory>\hub\resourcekit\samples\BESExternalCall\` to build the `bes-external-call.ear` file. You must deploy the EAR file on the application server and use the Provisioning tool to configure the calls to the web services.

Configuring External Calls

Configure the calls to the external services for custom logic and validation of business entity data.

To configure external calls, perform the following tasks:

1. Use the XML Schema Definition (XSD) to get the request and response types for external calls. Use the request and response types as the input and output elements for an external web service operation. Refer to the `bes-external-call.xsd` file in the `mdm-spi.jar` file.
2. Develop and deploy the external web service.
3. Upload the WSDL file for the web service.
4. Register the SOAP service.
5. Configure an external call and bind the WSDL, the SOAP service, and the SOAP operation. Publish your configuration to the MDM Hub.

For more information about WSDL files and SOAP calls to the web service, see the Integrating Data as a Service chapter.

For more information about the steps in the business entity services for which you can use external web services, see the *Multidomain MDM Business Entity Services Guide*.

Uploading a WSDL File

You can upload a WSDL file for the web service that you want to use.

1. Click **Business Entity** > **Extensions**, and then select **WSDL**.
2. Click **Create**.
3. In the properties panel, specify the following properties:

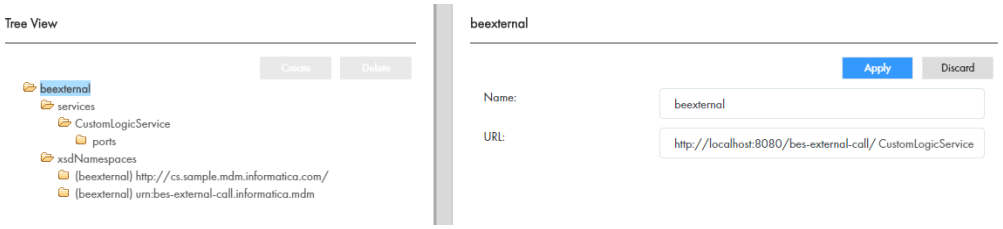
Property	Description
Name	Name for the WSDL file.

URL	URL of the WSDL file for the web service. You can refer to the following sample URLs:
	- For email verification and phone number validation:
	`http://ws.strikeiron.com/OnDemandEntityID1?wsdl`
	- For Services Integration Framework (SIF) APIs:
	`http://localhost:8080/cmx/request/wsdl3`

4. Click **Apply**.

   The details of the WSDL appear in the tree view panel.

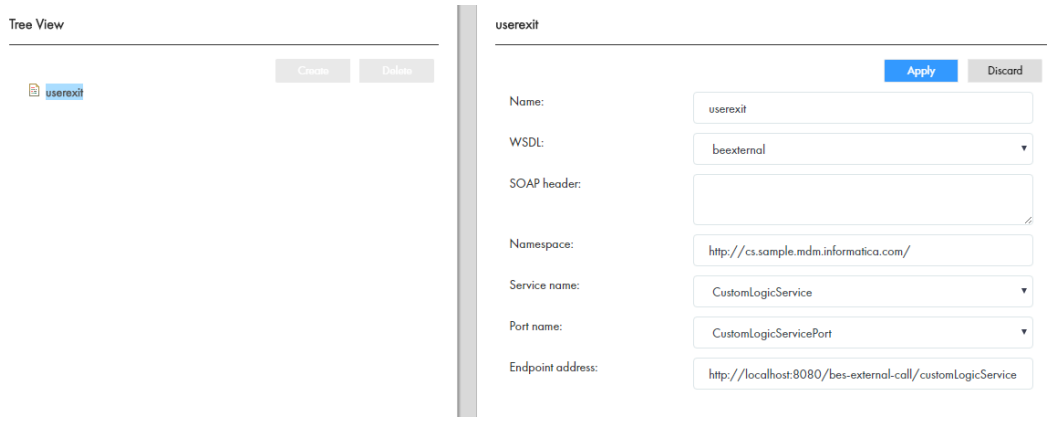
   The following image shows an uploaded WSDL file:



Registering a SOAP Service

Register the SOAP service to make SOAP calls to the external providers to access the web services for custom validation and logic to process the business entity data.

The following image shows a sample SOAP service configuration:



Related Topics:

- "Registering a SOAP Service" on page 138

Configuring an External Call

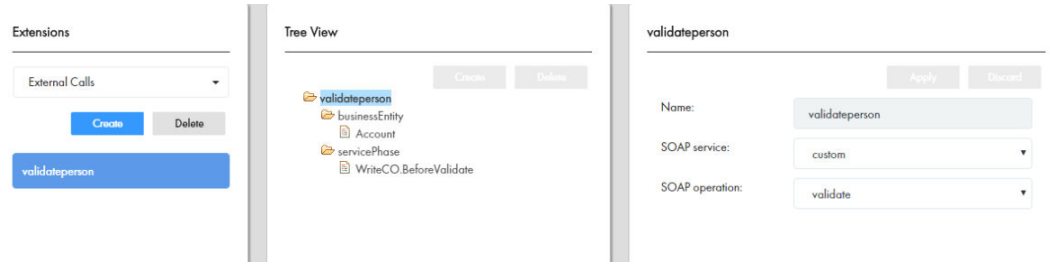
External calls are extensions to the business entity services. You can bind the WSDL, SOAP Service, and SOAP operation and configure an external call.

1. Click **Business Entity** > **Extensions**, and then select **External Calls**.
2. Click **Create**.
3. In the properties panel, specify the following properties:

Properties	Description
Name	Name for the external call.
SOAP Service	SOAP service that you want to use.
SOAP Operation	SOAP operation that you want to perform.

4. Click **Apply**.
5. Publish your changes.

The following image shows a sample external call configuration:



CHAPTER 16

Designing the Data Director User Interface

This chapter includes the following topics:

- Designing the Data Director User Interface Overview, 145
- User Interface Design Examples, 146
- User Interface Design Prerequisites, 146
- User Interface Layouts, 146
- Primary User Interface Components, 154
- Secondary User Interface Components, 161
- Custom User Interface Components , 172
- How to Design the Data Director User Interface, 175

Designing the Data Director User Interface Overview

You can design the Data Director user interface to add, edit, and manage records and to manage tasks. To design the user interface, create an application and user interface components, and then assemble the components into layouts.

When you design the Data Director user interface, you design layouts for the **Home** page, additional pages, and record views. If the user interface requirements change, you can update the layouts that you created.

To design the user interface, use the Application Editor, the Component Editor, and the Layout Designer in the Provisioning tool. The user interface can contain primary, secondary, and custom components. You can create custom components in the Component Editor. After you create the custom components, use the Layout Designer to select a layout template, and then drag and assemble the components that you want in the layout.

User Interface Design Examples

You have a request to design a **Home** page, an additional page that links to an external web application, and some custom record views for Account Managers.

To design a layout for a **Home** page, an additional page, and a record view, perform the following steps:

1. Gather user requirements.
2. Visualize the design.
3. Create the components.
4. Select a layout.
5. Add components to the layout.
6. Publish the page or view.
7. Get feedback from your users and adjust the page or view as necessary.

User Interface Design Prerequisites

Before you design the Data Director user interface, perform the following prerequisite tasks:

1. Configure the MDM Hub.
2. Create user roles that need access to the user interface layouts. You create user roles in the MDM Hub Console.
3. Create a source system with which you want to associate the application. You create source systems in the MDM Hub Console.
4. Create a Data Director application.
5. Define the business entity models.
6. If you want to use the user interface to manage business entity relationships, create the relationship types that you want to manage.
7. If you want to design the user interface to display a view of a business entity, create business entity views.

User Interface Layouts

The Data Director user interface includes a **Home** page, additional pages, and record views. Design **Home** pages and additional pages that meet the needs of your end users. For the record views, you can either enable the default record views or design custom record views.

When users access a Data Director application, the **Home** page appears. The **Home** page can contain a task inbox, charts, and reports. Also, users can select any additional pages that are created for them from the navigation bar. The additional pages can contain primary, secondary, and custom components such as external links.

Record views help users add, view, and manage records and review tasks. The default record views use preconfigured layouts that consist of a single panel. Custom record views can consist of multiple panels with primary, secondary, and custom components.

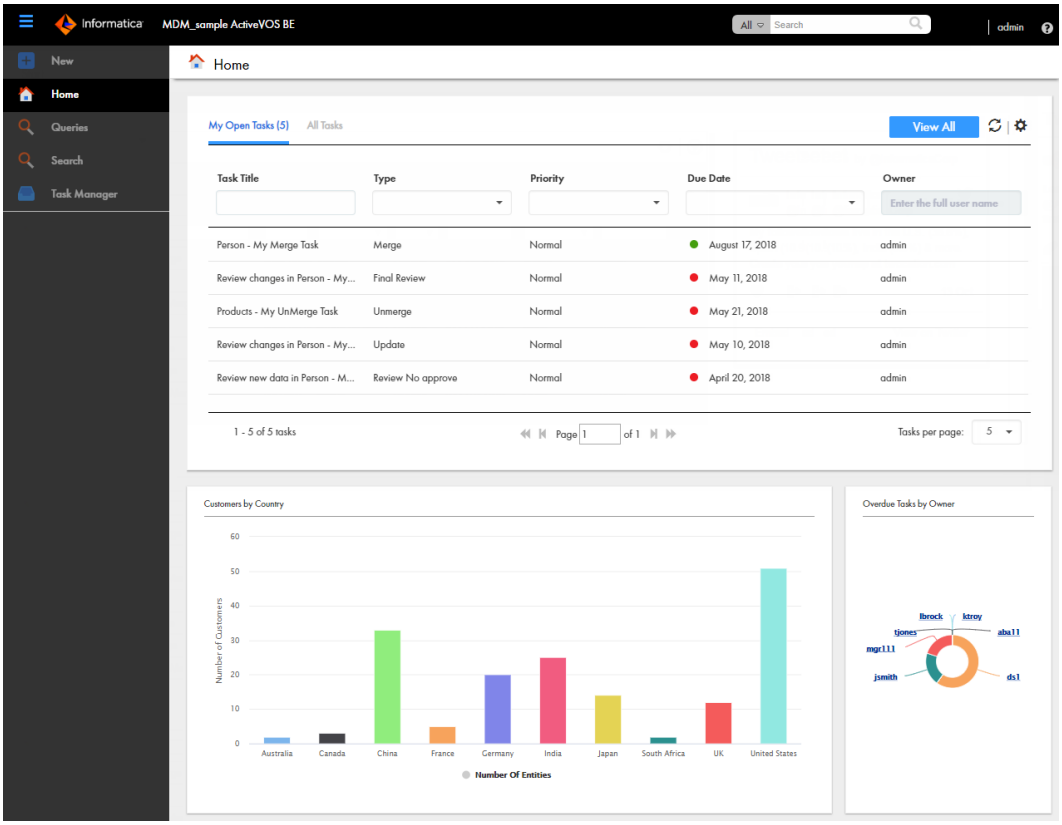
Note: If you upgraded from a version prior to 10.2, layouts might not appear properly in the Provisioning tool. Verify that the layouts appear properly in the application. If required, you can edit the layouts in the Provisioning tool.

Home Page Layouts

You can design multiple **Home** page layouts. You can create layouts that are applicable to multiple user roles or applicable to unique user roles. A **Home** page appears when you log in to Data Director. You can design **Home** page layouts to contain a task inbox and custom resources such as charts and social media feeds.

When you design a **Home** page, select a template that can display the components that you want. For example, to display the task inbox and charts in the **Home** page, the template must have a panel that is big enough to display the components.

The following image shows a **Home** page in Data Director with the Chart component and the Task Inbox component:



Related Topics:

- "Designing a Home Page Layout" on page 187

Additional Page Layouts

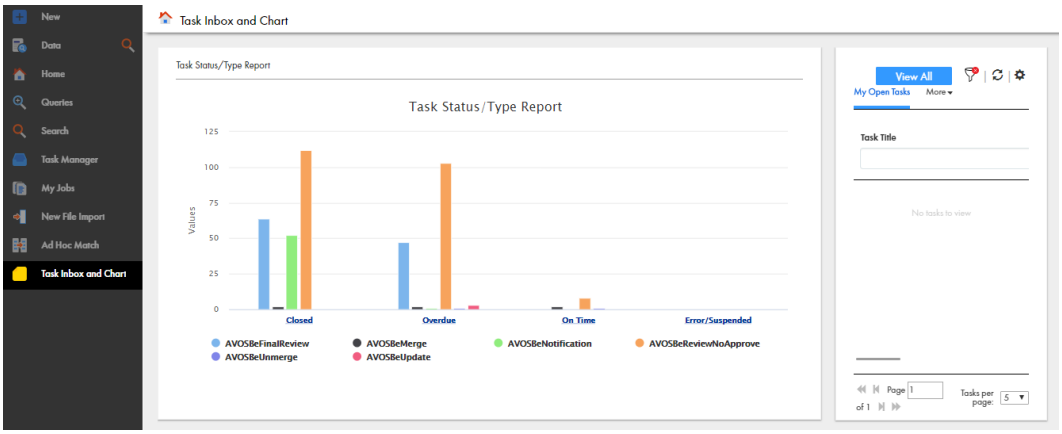
Additional pages can provide access to information that users need in addition to the **Home** page. You can design multiple additional page layouts that are applicable to multiple user roles or applicable to unique user roles.

An additional page layout can contain primary, secondary, and custom components. For example, you can design an additional page layout with a component that connects to an external web application like the Informatica Data Quality dashboard.

When you design an additional page layout, you select a template that can fully display the components that you want. For example, to display a Chart component, the template must have a panel that is big enough to display the chart fully.

When you design an additional page layout, the Provisioning tool creates a custom tab to which the additional page layout is associated. In Data Director, users access the additional page by clicking the custom tab in the navigation bar.

The following image shows an additional page in Data Director with the Chart component and the Task Inbox component:



Record View Layouts

Record views appear in an entity tab when users create or manage records or when users manage tasks in the Task Manager. You can design multiple record view layouts. You can add primary components to layouts to manage records. Add secondary or custom components to display the information related to records, such as matched records and social media feeds.

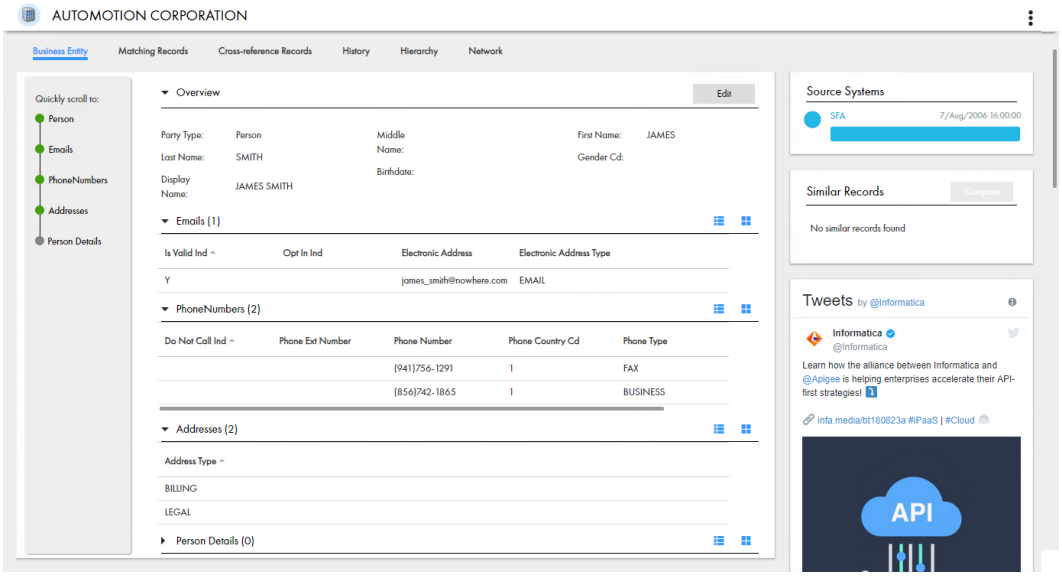
You can configure actions that users can perform using the record view, such as creating a record or reviewing tasks.

You can design Record View and Review Task, Merge Task, and Unmerge Task layouts. The layout type determines the primary components that you can add to the layout and the actions that users can perform. For example, if you design an Unmerge Task layout, only the primary components that help users to unmerge records are available.

When you design a record view, enable the default record views. Users can manage records using the default record views while you design the custom record views. If enabled, the names of the default record views appear as tabs. After you design custom record views, you can hide the default record views.

When you design a record view, select a template that can display the components that you want. For example, to display data in a record, the template must have a panel that is big enough to display the record details. When users log in to Data Director, layouts that are applicable to their user roles appear in the application.

The following image shows a record view with the Record Details component in the main panel and secondary components that display related information in the sidebar:



Tip: If you do not want to design record view layouts, enable the default record views. The default record views contain a primary component, but do not contain any secondary or custom components.

Record View and Review Task Layout Type

The Record View and Review Task layout type is the most flexible type of record view layout that you can design. You can use the same layout to manage both records and tasks. Or, you can create separate layouts for records and review tasks.

Record View Layout Type

You can design custom record views to create or view records. You can also design views to merge duplicate records, unmerge merged records, view record history, or view hierarchical relationships between records.

To design custom record views that can replace the default record views, select the same purpose and use the same primary component as the default record views. Then add the secondary and custom components that you want.

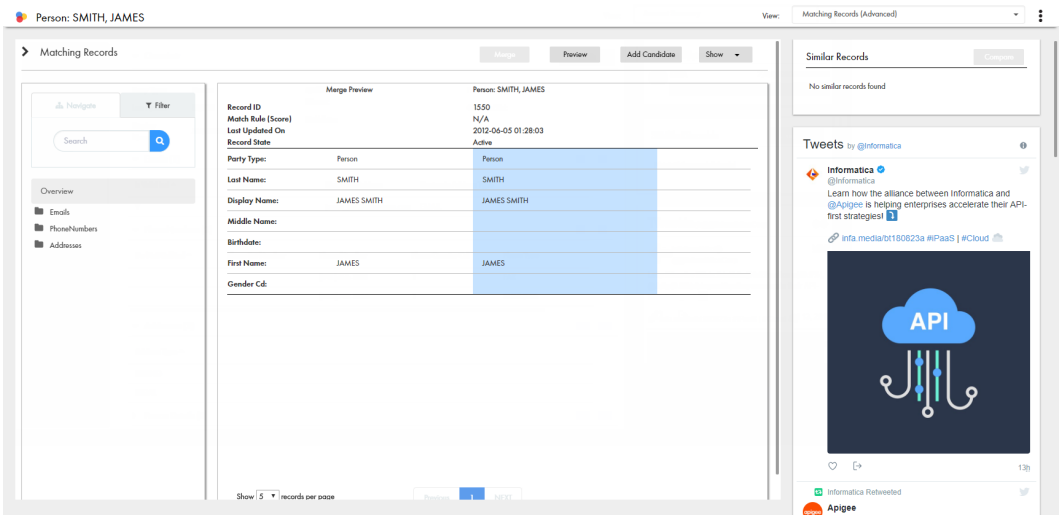
The following table lists the default record views and the selections to make on the **Purpose** page and **Design** page:

Default Record View	Purpose Page	Design Page
Business Entity view	Create records View or edit records	Record Details component
Matching Records view	View or edit records	Matching Records (Advanced) or Matching Records component

Default Record View	Purpose Page	Design Page
Cross-reference Records view	View or edit records	Cross-reference Records component
History view	View or edit records	History component
Hierarchy view	View or edit records	Hierarchy Manager component

Tip: To use the same layout for the Business Entity view and Review Tasks, select all options on the **Purpose** page of the Layout Designer.

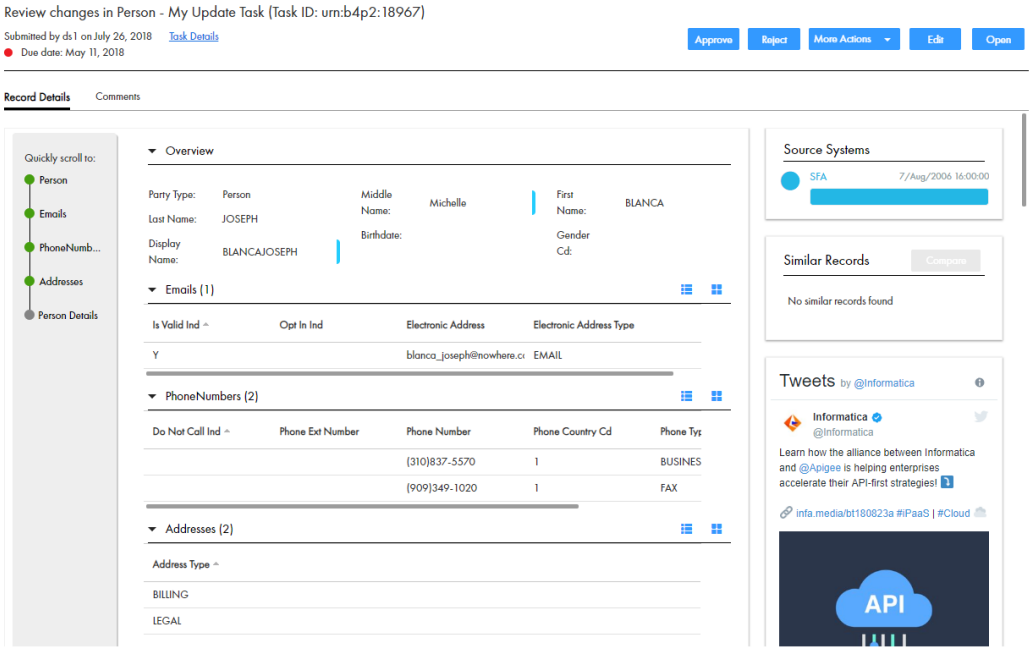
The following image shows an example of a custom Matching Records view layout:



Review Task Layout Type

You can design a record view layout to manage review tasks in the Task Manager, such as reviewing new records or edited records. A typical review task layout contains the Record Details component and any secondary or custom components that you want.

The following image shows an example of a custom review task layout:



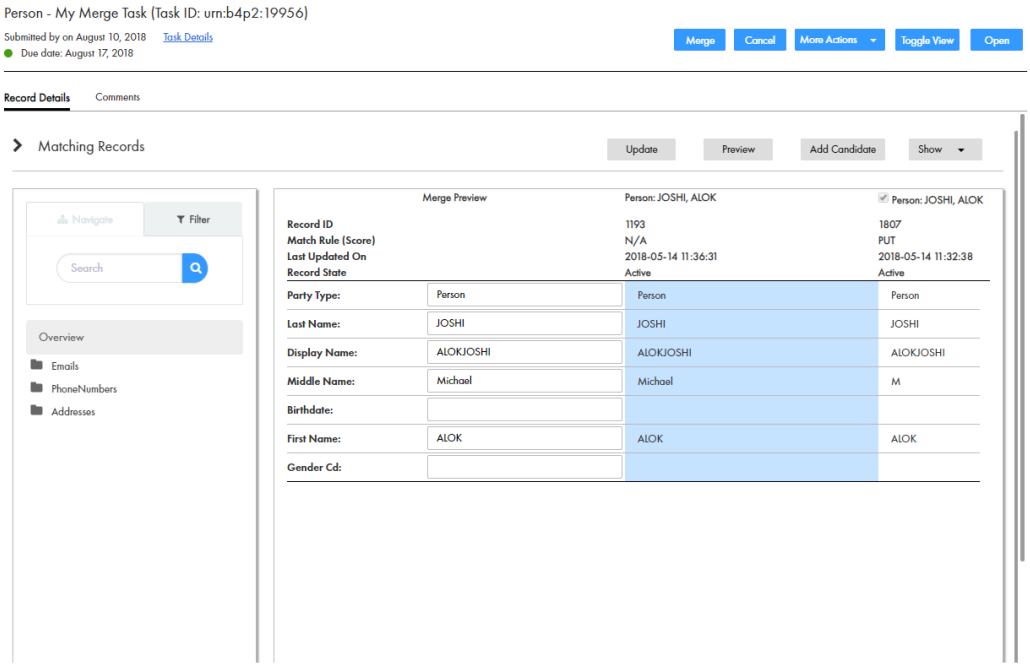
Related Topics:

- "Designing a Business Entity View Layout" on page 190
- "Designing a Cross-reference Records View Layout" on page 196
- "Designing a Matching Records View Layout" on page 194
- "Designing a History View Layout" on page 197
- "Designing a Review Task Layout" on page 199

Merge Task Layout Type

You can design a record view layout with the Merge Task layout type. Use Merge Task layouts to manage merge tasks in the Task Manager. A typical merge task layout contains the Matching Records (Advanced) component or the Matching Records component and any secondary or custom components that you want.

The following image shows an example merge task layout in the Task Manager:



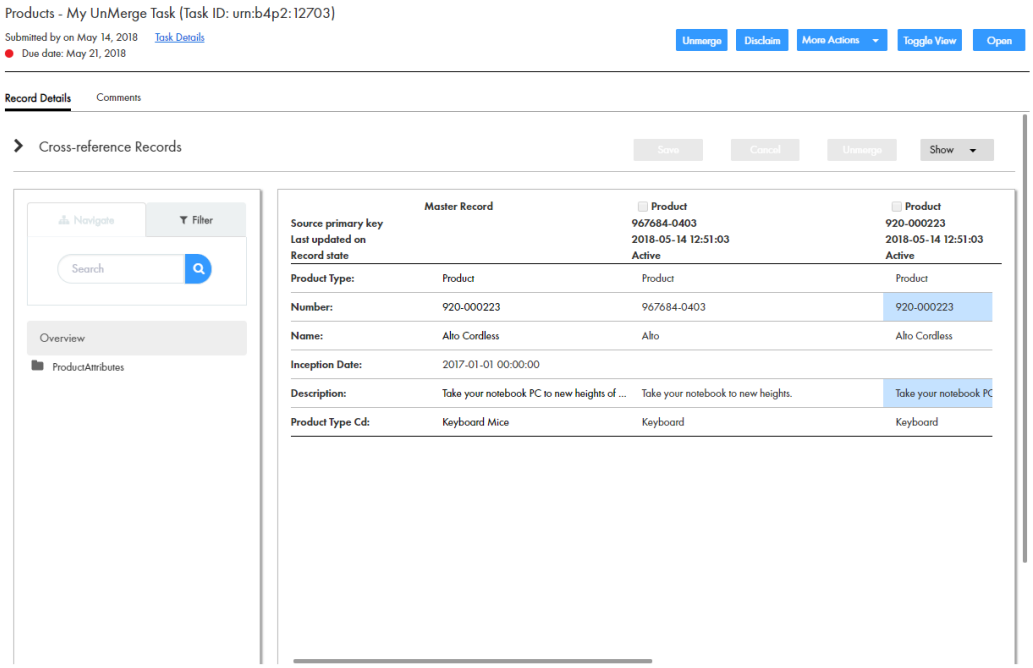
Related Topics:

- "Designing a Merge Task Layout" on page 201

Unmerge Task Layout Type

You can design a record view layout with the Unmerge Task layout type. Use Unmerge Task layouts to manage unmerge tasks in the Task Manager. A typical unmerge task layout contains the Cross-reference Records component and any secondary or custom components that you want.

The following image shows an example unmerge task layout in the Task Manager:



Related Topics:

- "Designing an Unmerge Task Layout" on page 203

Default Record Views

The default record views use preconfigured layouts that contain a primary component, but do not contain any secondary or custom components. The default record views appear as tabs in open records in Data Director.

You might want the default record views to appear when you begin designing layouts so that users can view and manage records. After you design custom record views, you can hide the default record views to reduce confusion for your end users.

The following set of default record views can be hidden or shown as a group:

Business Entity view

Displays the details of a record in a form layout. End users can view or edit records based on their user role permissions. This record view layout contains the Record Details component.

Cross-reference Records view

Displays the source data that contributed to the master record and unmerge records. This record view layout contains the Cross-reference Records component.

History view

Displays past changes to a record. This record view layout contains the History component.

Matching Records view

Displays a merge comparison panel to find and add potential duplicate records, select values you want to override or promote, view the merge preview, and merge the open record with the duplicate records. This record view layout contains the Matching Records (Advanced) component.

Related Topics:

- "Hiding the Default Record Views" on page 212

Primary User Interface Components

You can add primary components to layouts. Primary components help users manage records.

The following table describes the primary components that appear when designing **Home** page layouts in the Layout Designer:

Primary Component	Description
Task Inbox	Displays task notifications that the review process generates.
Dashboard Reports	Displays a Jaspersoft report. Use the component if you already have Jaspersoft reports configured for subject areas in Data Director.

The following table describes the primary components that appear when designing record view layouts in the Layout Designer:

Primary Component	Description
Record Details	Displays the data in a record. Use the component to create, view, and edit records.
Matching Records (Advanced)	Displays an advanced merge comparison panel with an updated look and feel. Use the component to find duplicate records and merge the open record with the duplicate records. You can select root or child values in the records to override or promote, and preview the merged record.
Matching Records	Displays a merge comparison panel. Use the component to find duplicate records and merge the open record with the duplicate records. You can select root values in the record to override or promote and preview the merged record.
Cross-reference Records	Displays the source data that contributed to the master record. Use the component to unmerge merged records.
History	Displays past changes to a record. Use the component to investigate data changes.

Note: When designing record view layouts, the layout type you select determines the primary components available.

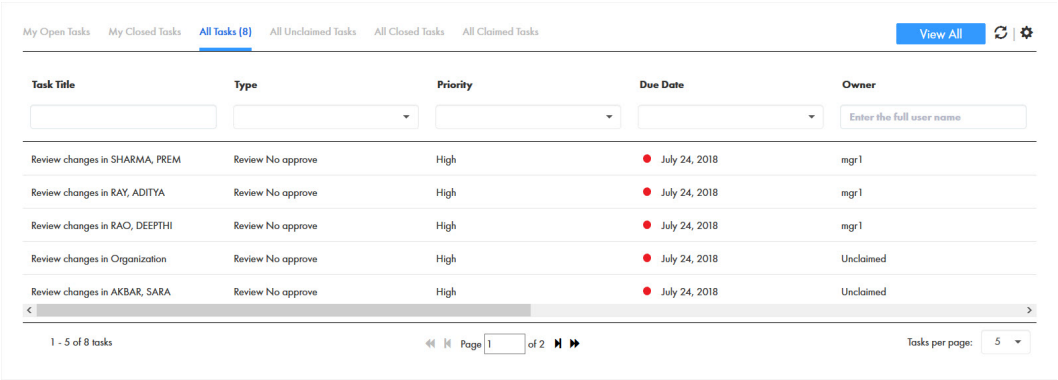
Related Topics:

- "Configuring the Task Inbox Component" on page 176

Task Inbox

The task inbox component is a primary component that you can configure and add to a **Home** page layout. The component displays task notifications that the review process generates. You can use the task inbox to organize, claim, and resolve tasks.

The following image shows a sample task inbox in a **Home** page:



You can configure the fields and filters that the task inbox component displays for the users.

The following table describes the fields that the task inbox component can display as columns:

Field	Description
Task Title	A link that displays a short description of the review task. To review the record and take action on the task, click the task title. The task opens in the Task Manager and displays the associated record in the review panel. If your role permits it, you can edit the record from this view before taking action on the task.
Type	Name of a people activity within the process.
Priority	Priority of the task. The task priority can be Low, Normal, or High. When generating tasks, the review process assigns the same priority to all tasks of the same type.
Due Date	Date the task is due. The review process calculates a default due date based on the task creation date plus a predefined time period. The due date of each task is prefixed with one of the following colored dots: - Green. Indicates that the task is not overdue. - Red. Indicates that the task is overdue. - Amber. Indicates that the task will be overdue in the next 24 hours.
Owner	Name of the signed-in user.
ID	Instance ID of the review process that generated the task.

Field	Description
Status	Status of the task can be Open or Closed. When the task status is Open, you can view the record associated with the task.
Creator	Name of the user who created the task.
Creation Date	Date on which the user created the task.

The following table describes the filters that the task inbox can display:

Filter	Description
My Open Tasks	Shows your open tasks.
My Closed Tasks	Shows the tasks that you closed.
All Tasks	Shows all the closed, claimed, and unclaimed tasks that are assigned to your user group.
All Unclaimed Tasks	Shows all the unclaimed tasks that are assigned to your user group.
All Closed Tasks	Shows all the tasks that the users in your user group closed.
All Claimed Tasks	Shows all the tasks that the users in your user group claimed.

Dashboard Reports

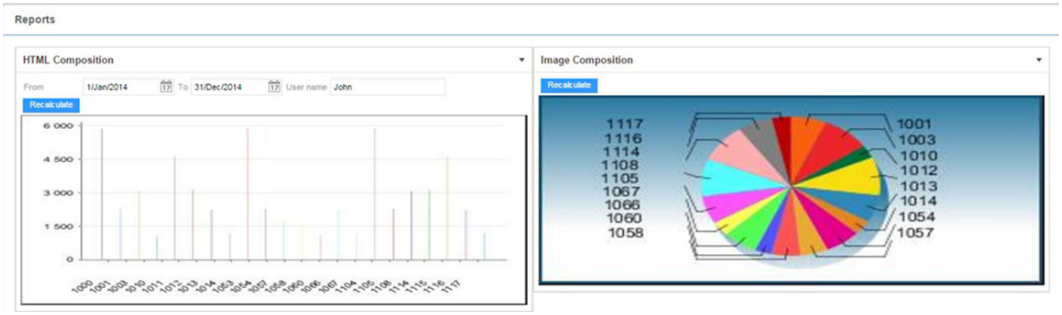
The Dashboard Reports component is a primary component that you can add to a **Home** page layout. If you use the business intelligence tool Jaspersoft to develop and analyze reports, use this component to add a Jaspersoft report to the layout.

Note: This component is for Jaspersoft reports only. To display charts using the Highcharts framework, create a Chart component. To display charts from other business intelligence tools, create a custom component.

Use the Dashboard Reports component if you already have Jaspersoft reports configured for subject areas in Data Director. The component acts as an inline frame for all reports configured in the IDD configuration file.

Note: If you want to display the Dashboard Reports component in a **Home** page, use Template 1 to design the **Home** page layout.

The following image shows a sample Dashboard Reports component in a **Home** page:

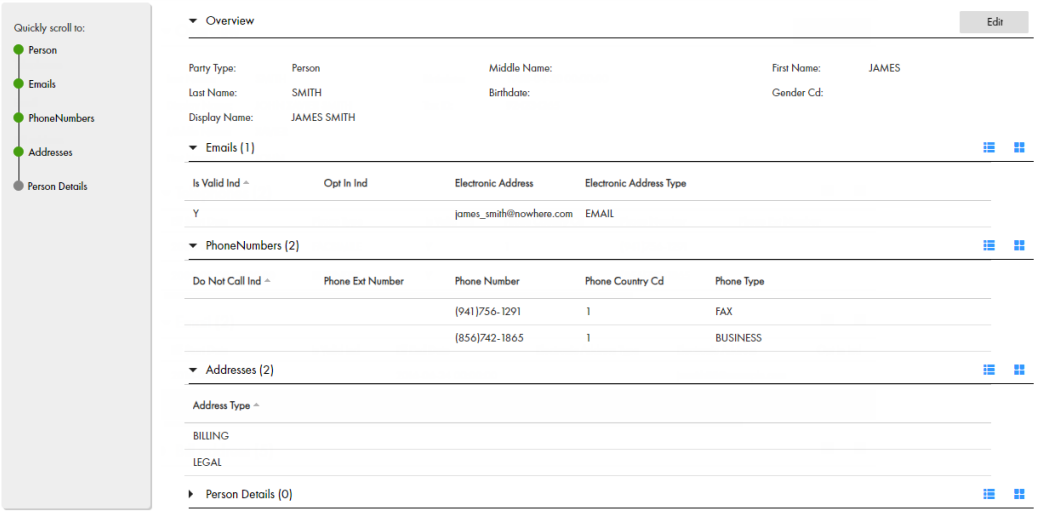


Record Details

The Record Details component is a primary component that displays data in a business entity record. Add the Record Details component to a custom Business Entity view layout or review task layout.

The Record Details component appears in the Layout Designer after a business entity model is created in the Modeling page. When you design a layout, you can configure the Record Details component to display the information you want, such as telephone numbers, emails, and addresses. If you display multiple child records that contain large amounts of information, you might want to enable the business entity menu to help users navigate among the child records.

The following image shows a sample Record Details component with the business entity menu in Data Director:



Best Practices for Record Details Component

Informatica recommends that you use only one Record Details component in a Business Entity view layout.

However, if you must use multiple Record Details components in a Business Entity view layout, consider the following implications:

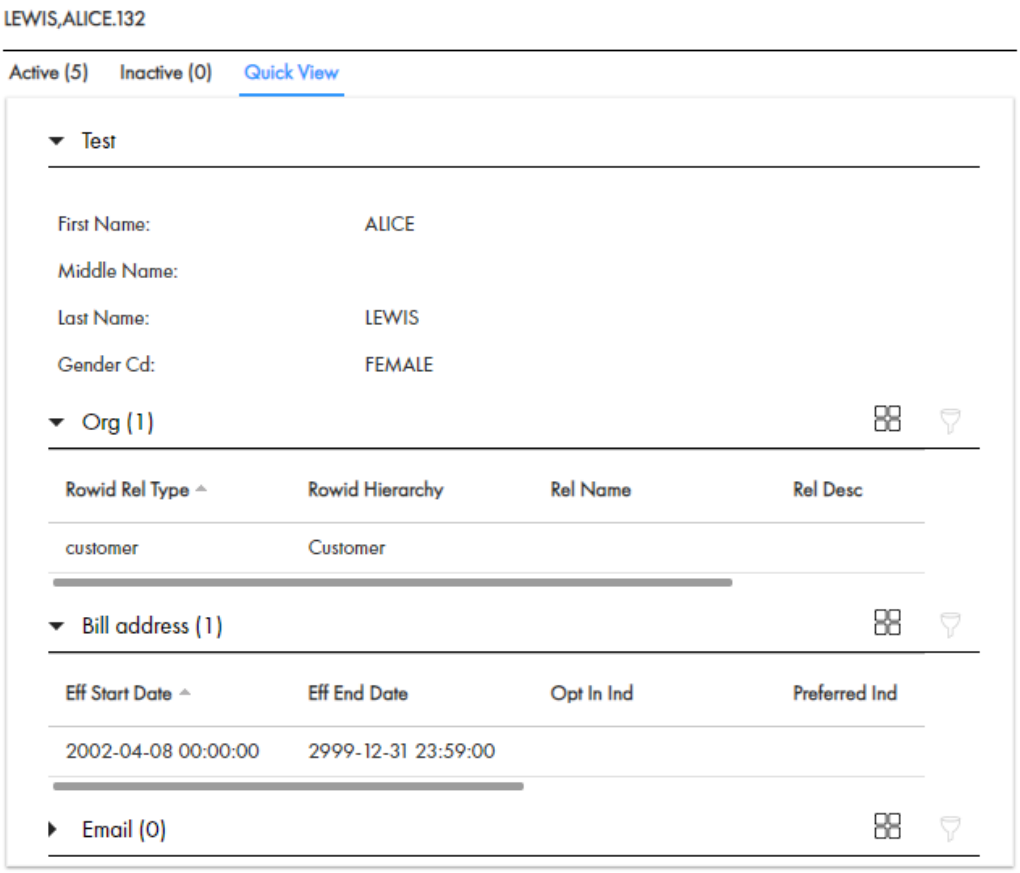
- If the set of fields for which you configure a validation or data cleansing rule are split across Record Details components, the validation and cleanse operations fail. Ensure that you configure validation and data cleansing rules for a set of fields that are within a single component.
- Users need to save changes to fields within one component before making changes to fields in another component. Otherwise, changes made to the fields in the component will be lost.

Record Details Component in the Quick View Tabs

You can create a Record Details component that appears in the **Quick View** tab of the **Network** view or the **Hierarchy** view. Record Details components that appear in the **Quick View** tabs provide a summary of the details of a record.

When you design a record view layout, you choose whether you want to design the **Quick View** tab for the **Network** view or **Hierarchy** view.

The following image shows a sample **Quick View** tab in the **Network** view:



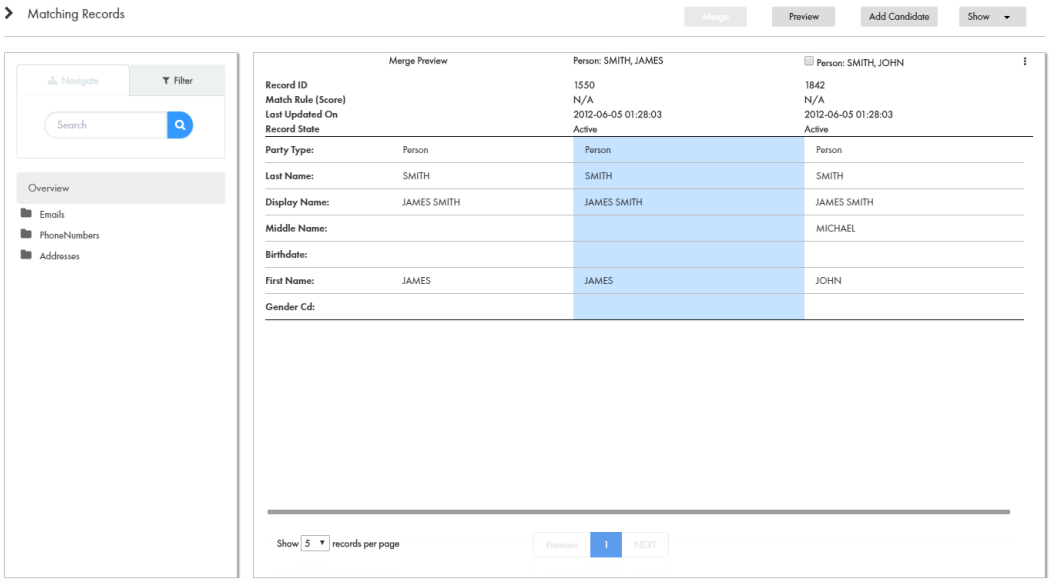
Matching Records (Advanced)

The Matching Records component with advanced features is a primary component that displays a merge comparison panel. Add the Matching Records (Advanced) component to a custom Matching Records view layout or merge task layout.

Data Director users use the Matching Records (Advanced) component to add potential duplicate records or override or promote values to contribute to the master record. Then preview the master record and merge the open record with the duplicate records.

The advanced features allow users to view, override, and promote values at the root and child levels of the record. To only allow users to promote values in the root levels, use the Matching Record component.

The following image shows a sample Matching Records (Advanced) component in Data Director:



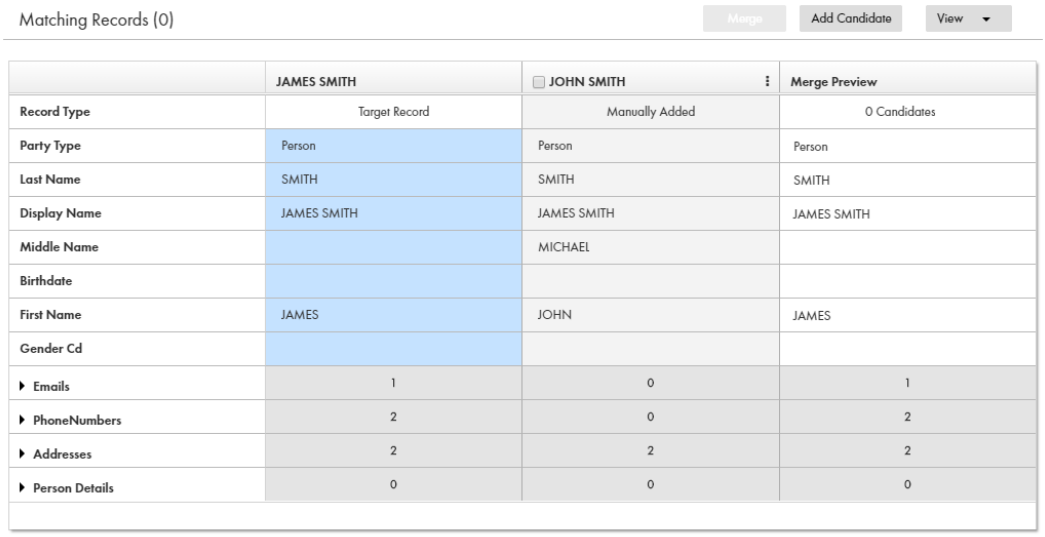
Matching Records

The Matching Records component is a primary component that displays a merge comparison panel. Add the Matching Records component to a custom Matching Records view layout or merge task layout.

Data Director users use the Matching Records component to add duplicate records and override or promote values to contribute to the master record. Then preview the master record and merge the open record with the duplicate records.

Data Director users use the component to override and promote values at the root level of the record. To allow users to view, override, and promote values in the child levels, use the Matching Records (Advanced) component.

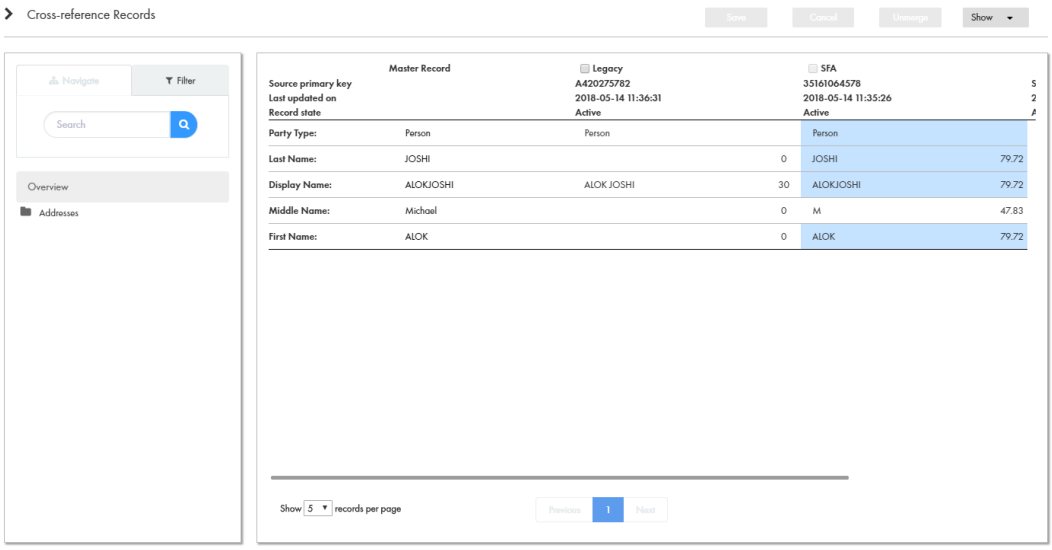
The following image shows a sample Matching Records component in Data Director:



Cross-reference Records

The Cross-reference Records component is a primary component that displays the source data that contributes to the master record. Add the component to a custom Cross-reference Records view layout or unmerge task layout. Data Director users use the component to adjust how data is consolidated and to unmerge merged records.

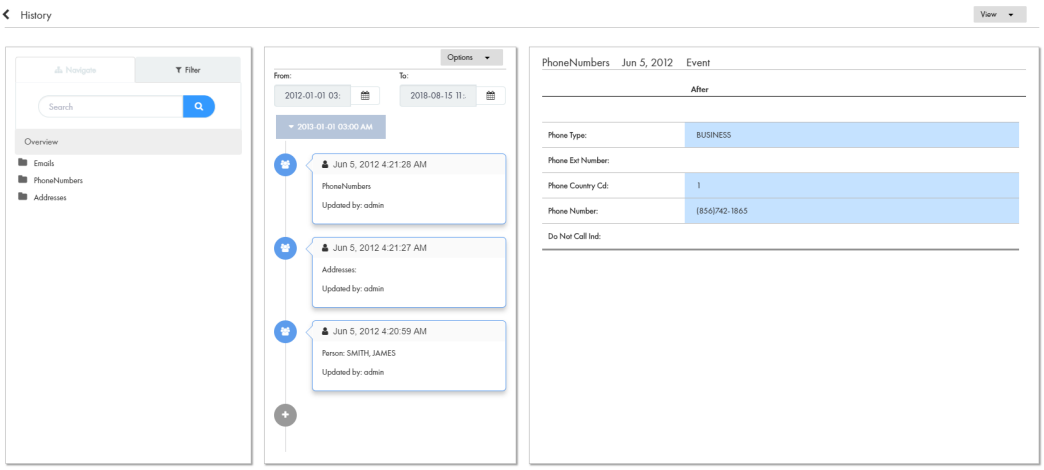
The following image shows a sample Cross-reference Records component in a record view:



History

The History component is a primary component that displays past changes to a record. Add the History component to a custom History view.

The following image shows a sample History component in a record view:

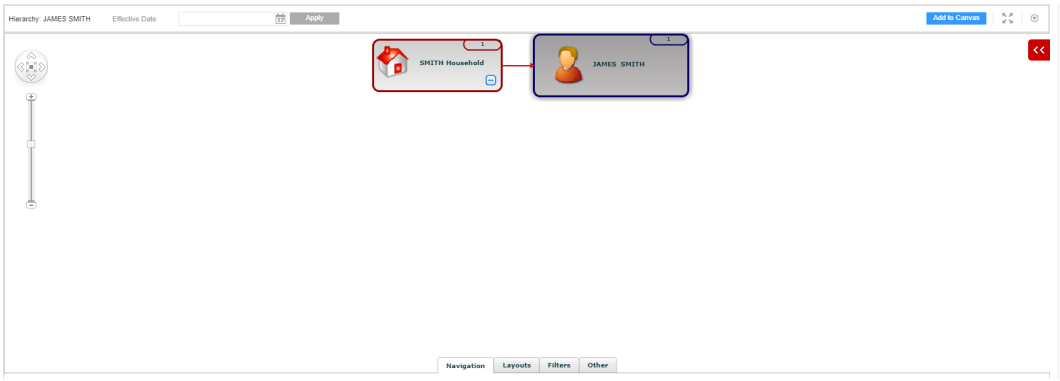


Hierarchy Manager

The Hierarchy Manager component is a primary component that displays a diagram that shows how a record is related to other records. Add the Hierarchy Manager component to a custom Hierarchy view.

To display related records, ensure that relationships are defined for the business entity model on which you base the record view layout. You create business entity models and define relationships between business entities in the Modeling page.

The following image shows a sample Hierarchy Manager component in a record view:



Secondary User Interface Components

You can add secondary components to layouts. Some secondary components require configuration to use in the Layout Designer. To configure the secondary components, use the Component Editor.

The following table describes the secondary components that appear in the Layout Designer:

Secondary Component	Description	Requires Configuration
Chart	Displays data about records, tasks, or any other types of information you collect about your master data. You create charts from datamart tables that include aggregated ActiveVOS data, business entities data, or custom data that you create.	Yes
Matched Records	Displays the records that match the selected record.	No
Source Systems	Displays the source systems that contributes to the selected record.	No

Secondary Component	Description	Requires Configuration
Related Records	Displays the records that are related to the selected record. Based on the configuration, the Data Director users can use the component to add, edit, or delete relationships.	Yes
Related Tasks	Displays all tasks that are related to the selected record. Only tasks created in the BE-AVOS adapter appear.	No
Similar Records	Allows users to search for similar records before they create one. Displays records that are similar to the record that you create.	Yes
Hierarchy Widget	Displays all the related records of a record in a hierarchical format. You can also open the record details of any record from the hierarchy.	Yes

Chart

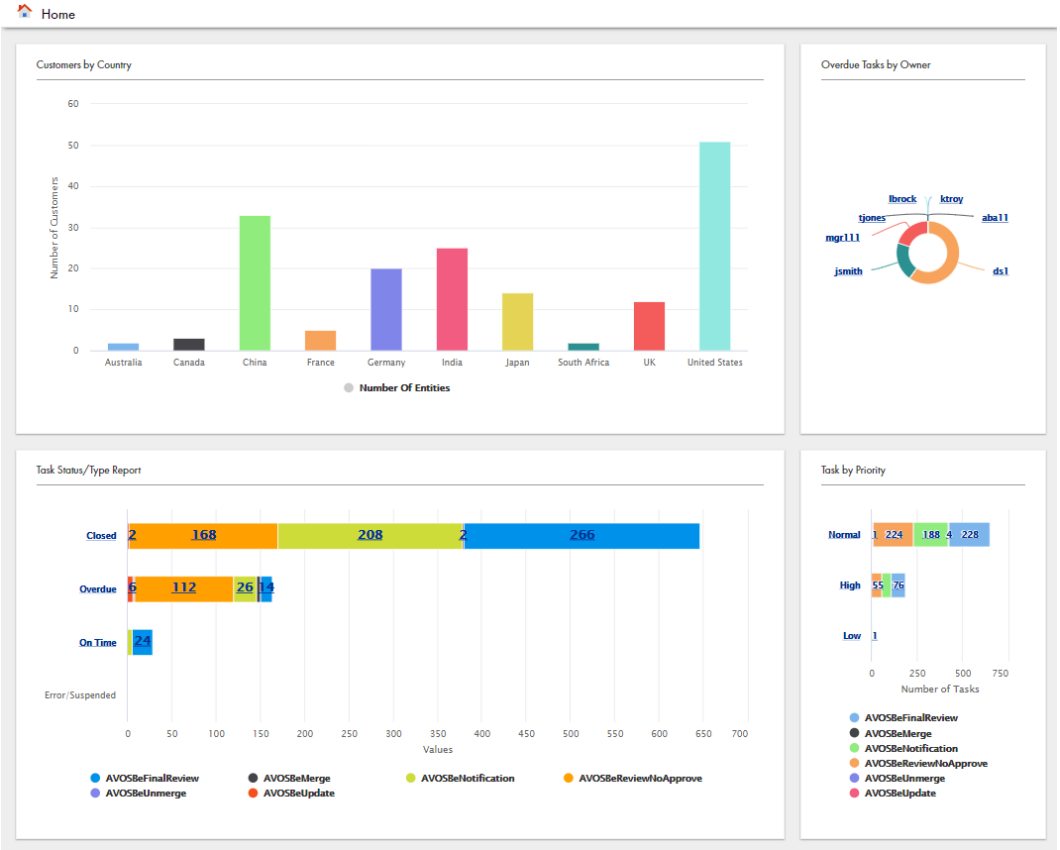
The Chart component is a secondary component that you can add to a **Home** page layout or additional page layout. The component displays report data in a chart, such as a bar graph or pie graph.

You populate charts with data collected from reports. You can filter and group the report data used in charts. You can also customize the appearance of charts.

Chart components display a single layer of data. To show another layer of data granularity, you can link data in a chart to a drilldown chart. To show the referenced records or tasks, you can link data in a chart to an interface in Data Director. You can link to the **Query** tab, **Search** tab, or Task Manager.

Before you add the Chart component to a **Home** page layout or additional page layout, create the component in the Component Editor. After you create the component, the component appears in the Layout Designer.

The following image shows a **Home** page in Data Director with Chart components:



Note: To populate charts with report data, you must register the reports that you want to use to collect data. For more information, see the *Multidomain MDM Business Entity Services Guide*.

Source Data

You choose the source data that you want to use in the chart. The source data comes from reports that collect information about your master data. You can also filter or group the source data used in charts.

Note: Before you can generate reports or populate charts, you must configure the data mart database connection. Then you can populate the data mart with the report data. For more information, see the *Multidomain MDM Resource Kit Guide*.

Reports

You can use the out of the box reports to collect information about your records and tasks. The out of the box reports are available by default and collect information about predefined criteria. If you need to collect other types of information about your master data, you can create custom reports.

For example, you might use a report to collect information about the number of tasks assigned to each user. Or, you might use a report to collect information about the number of records added for each business entity type.

Note: You must register the out of the box reports and custom reports that you want to use. For more information, see the *Multidomain MDM Business Entity Services Guide*.

The following table describes the out of the box reports:

Report ID	Report	Description
MDM.RPT.1	Tasks Overview by Status	Displays the total number of tasks and breaks down the tasks by status. You can drill down to users assigned for each task status.
MDM.RPT.2	Task Overview by Priority	Displays the total number of tasks and breaks down the tasks by priority. You can drill down to users assigned for each task priority.
MDM.RPT.4	Business Entities Added by Year	Displays the number of records added for each business entity type and breaks down the records by year.
MDM.RPT.5	Customer Onboarding Time	Displays how long it took to qualify records.
MDM.RPT.6	Business Entities by Source Systems	Displays the total number of records for each business entity type that comes from each source system. The detailed report breaks down the records for each pair of business entity type and source system by year.
MDM.RPT.7	Assigned Tasks by Users	Displays the total number of tasks assigned to each user.
MDM.RPT.8	Open Tasks by User Roles	Displays the total number of tasks that are open by user roles.
MDM.RPT.10	Closed Tasks by Users	Displays the total number of tasks that are closed by each user.

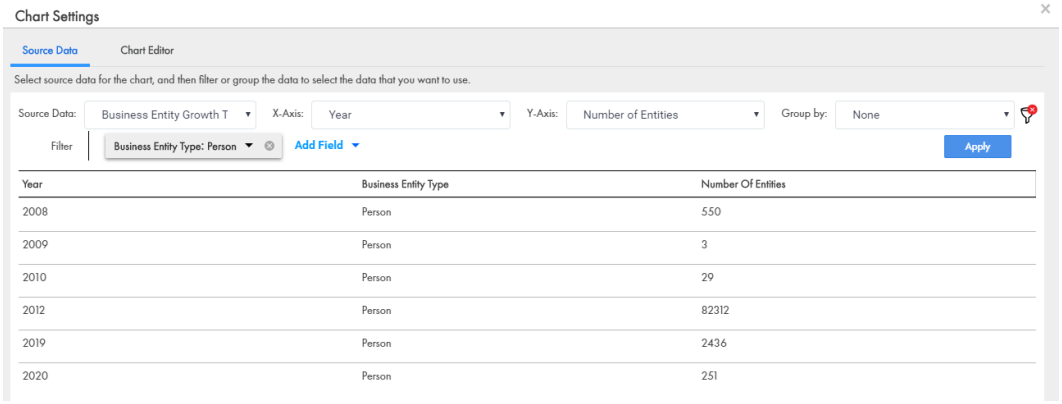
Filters

To narrow the focus of a chart, you can filter the source data used in the chart. You can filter the source data by data dimensions collected in the report.

For example, you might populate the chart with data collected by the out of the box Business Entities Added report. The report collects information about the number of records added for each business entity type and breaks down the records by year. The chart shows business entity growth trends for all business entity types. You can filter the source data to only include records created for the Person business entity type. Then the chart shows business entity growth trends for the Person business entity type.

You can also use filters to ensure that new data is included in the chart. For example, you might populate the chart with data collected by the Business Entities Added report and filter the data to only included records created for the Person business entity type. Then when users create new records for the Person business entity type, the chart displays the new data in the chart.

The following image shows a chart that displays the number of records created for the Person business entity type and breakdowns the records by year:



Group by

To segment data in a data series into groups, you can group the source data displayed in a chart by a criteria in the underlying data. The grouped data appears side by side or stacked. You might want to use the group by criteria for bar graphs.

For example, you might create a chart that shows tasks by status and then grouped by task type. The chart displays the tasks by status, with the data about each status grouped by task type.

The following image shows a chart with grouped data:

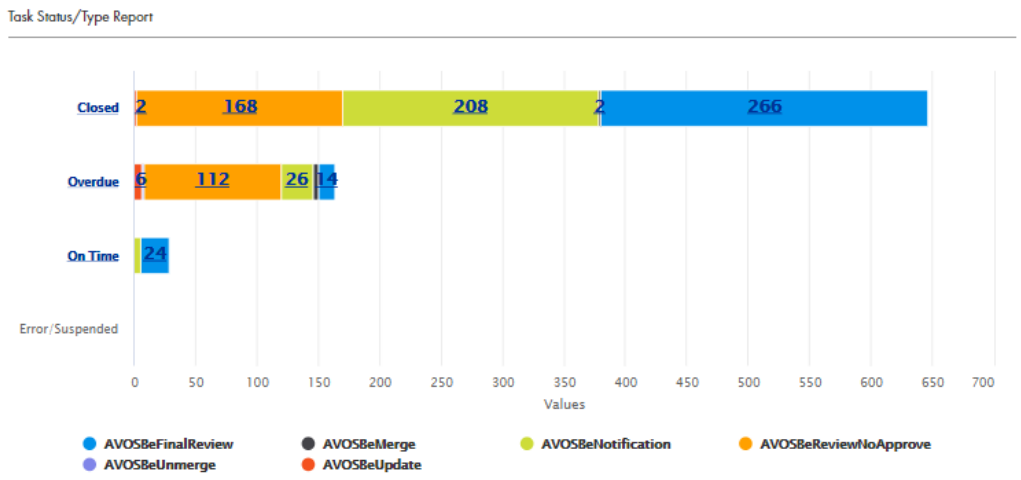
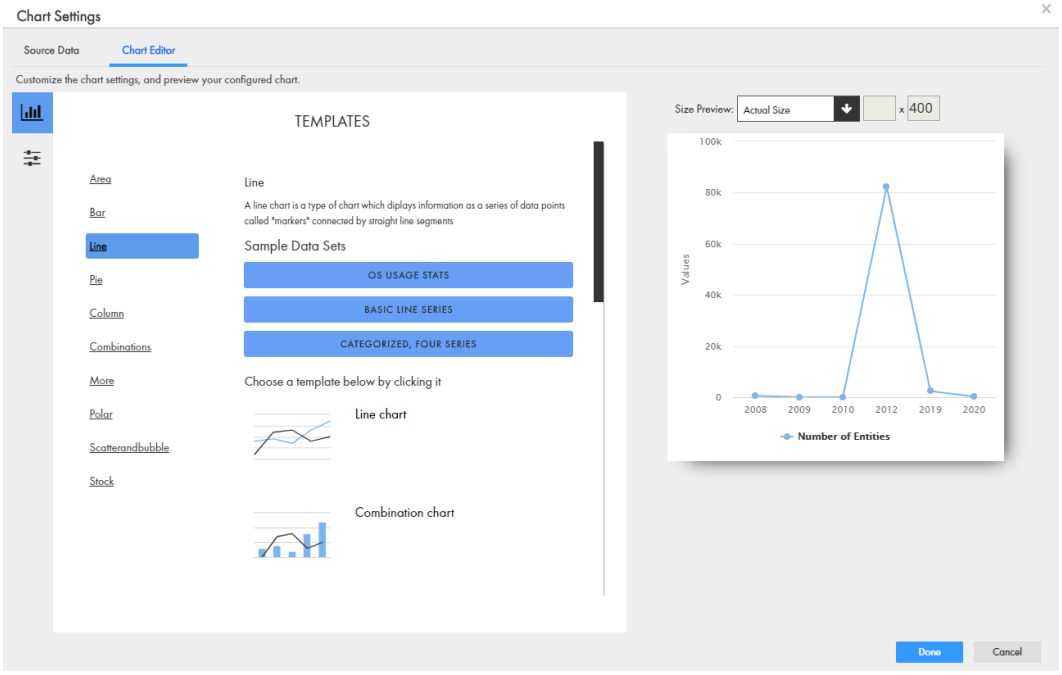


Chart Editor

You can customize the look and feel of your charts. You can choose the chart template, colors, and titles. You can also preview your chart to see how the source data might appear based on your configurations.

The following image shows the **Chart Editor** tab in the **Chart Settings** dialog box:



Linked Charts

When you configure the Chart component, you create a root chart. The root chart shows a single layer of data. You can link data in the root chart to a drilldown chart to show another layer of data granularity. Or, you can link data in a chart to another interface in Data Director, such as the **Queries** tab, **Search** tab, or Task Manager.

When you link data in a root chart to a drilldown chart, you can continue to link drilldown charts to other drilldown charts. Or, you can link a drilldown chart to another interface. If you link a root chart or drilldown chart to an interface, you cannot create a link to another chart or another interface.

When users select data in a chart that is linked to a drilldown chart, they drill down to a drilldown chart that provides more information about the selected data. When users select data in a chart that is linked to an interface, they navigate to a linked interface, such as the **Queries** tab, **Search** tab, or Task Manager. The configured interface opens with predefined search criteria or filters. For example, the Task Manager might open with predefined filters that show the open tasks for the Person business entity type.

The following table describes the charts and interfaces you can link your chart to:

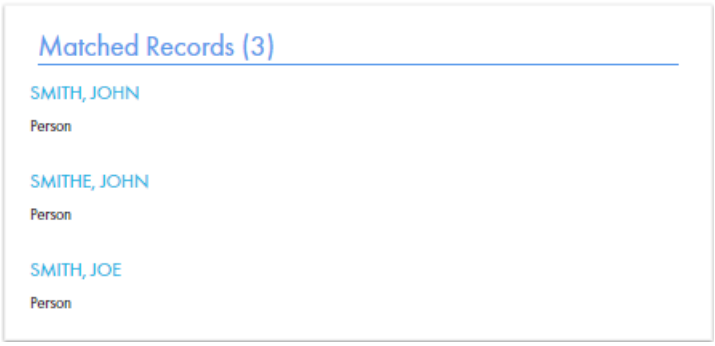
Link to	Description	Example
Drilldown Chart	Provides another layer of data granularity about the selected data.	You might have a chart that displays open tasks by task type. When users select the data segment for Automerge tasks, they drill down to see the total number of Automerge tasks assigned to each user in a drilldown chart.
Query tab	Opens the **Query** tab with predefined query parameters and shows the relevant records.	You might have a chart that displays business entities added. When users select the data segment for Person business entity types, they navigate to the **Query** tab. The **Query** tab shows records for the Person business entity type.

Link to	Description	Example
Search tab	Opens the **Search** tab with predefined search criteria and shows the relevant records.	You might have a chart that displays business entities added. When users select the data segment for Household business entity types added in 2019, they navigate to the **Search** tab. The **Search** tab shows records added in 2019 for the Household business entity type.
Task Manager	Opens the Task Manager with the predefined quick filter tab and filter criteria and shows the relevant tasks.	You might have a chart that displays assigned tasks by priority. When users select the data segment for high priority tasks, they navigate to the Task Manager. The Task Manager shows the assigned high priority tasks.

Matched Records

The Matched Records component is a secondary component that you can add to a record view layout. The component lists the records that are potential matches of the record that is open in the record view. Each matched record in the component is a link that you can open in a record view.

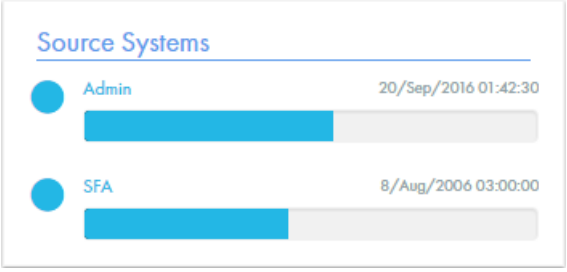
The following image shows a sample Matched Records component that appears for John E. Smith in an record view:



Source Systems

The Source Systems component is a secondary component that you can add to a record view layout. The component displays a horizontal bar graph of the source systems that contributed to the record. The source systems appear in order from the largest to the smallest contributor of cross-reference records.

The following image shows a sample Source Systems component that appears in a record view:



Related Records

The Related Records component is a secondary component that you can add to a record view layout. The component lists records that are directly related to the record that is open in a record view. Each related record in the component is a link that you can open in a record view.

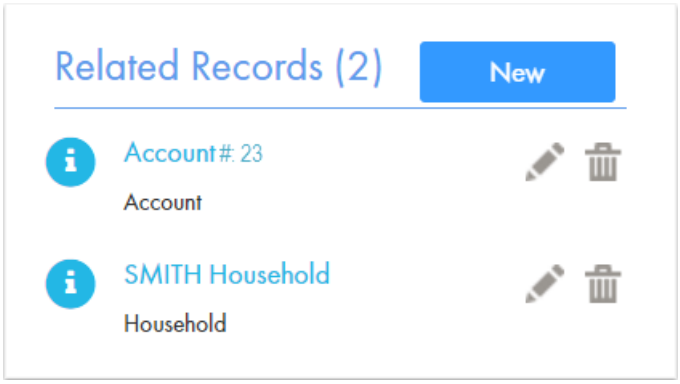
Note: The Related Records component shows only the records that the user has permission to view. To be able to add, edit, and delete relationships, the user must have these permissions for the types of business entities that they want to relate. To set permissions, you use the Security Access Manager in the MDM Console. For more information, see the *Multidomain MDM Configuration Guide*.

You can use the Related Records component in the view mode or the edit mode. In the view mode, users can only view the records. In the edit mode, users can add, edit, and delete relationships for records that appear in the Related Records component.

Before you add the Related Records component to a record view layout, create the component in the Component Editor. After you create the component, the component appears in the Layout Designer.

To display related records, ensure that relationships are defined for the business entity model on which you base the record view. You can create business entity models and define relationships between business entities in the Modeling page.

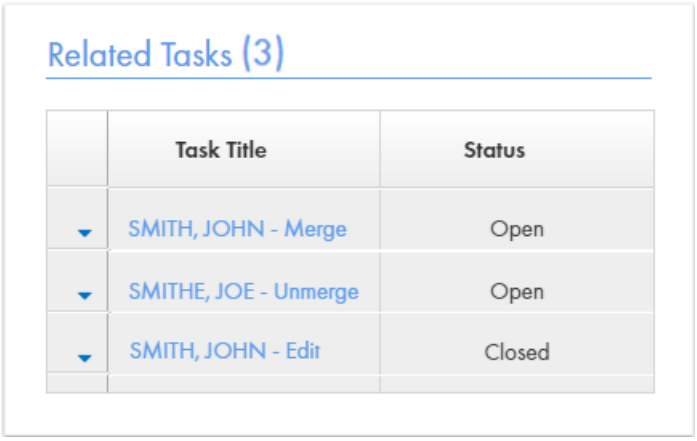
The following image shows a sample Related Records component in the edit mode with records related to a person record in a record view:



Related Tasks

The Related Tasks component is a secondary component that you can add to a record view layout. The component lists all the tasks that are related to the record that is open in the record view. Each task in the component is a link that you can open to view the task details.

The following image shows a sample Related Tasks component that appears in a record view:



Similar Records

The Similar Records component is a secondary component that you can add to a record view layout. When you create a record, the component displays records that are similar to record that the you create. You can review the records that appear in the component to ensure that the records you create are not duplicates.

In the Similar Records component, the records are displayed as a link with an information icon. To preview a record, you click the information icon. To open a record, click the record link.

Before you can add the Similar Records component to a record view layout, create the component in the Component Editor. After you create the component, the component appears in the Layout Designer.

You can configure the Similar Records component to use one of the following search options:

Search

Use to find and display similar records by using search. Ensure that search is configured in the MDM Hub environment. Also, you must configure the search to include only the business entity fields that are identified as searchable. For more information about configuring search, see the *Multidomain MDM Configuration Guide*. For more information about searchable fields, see the *Multidomain MDM Provisioning Tool Guide*.

Queries

Use to find and display similar records by using queries. Queries are based on match rules. Ensure that the match rule sets that you want to use for querying similar records are configured in the MDM Hub environment. For more information about queries and match rule sets, see the *Multidomain MDM Configuration Guide*.

You can configure the Similar Records component for a business entity or a business entity view.

When you configure the Similar Records component for a business entity view, consider the following guidelines:

- Base the search on fields that are part of the business entity view.

- Do not base the search on child record fields, such as Address and Email, that are transformed to appear as the parent record fields.

The following table describes the XML elements that you can use to configure the Similar Records component:

Element	Description
searchableFields	Specifies one or more fields on which you want to base the search. The `searchableFields` element is the parent of the `field name` element.
field name	Specifies the name of a field on which you want to base the search for similar records. The `field name` element is a child of the `searchableFields` element. You can configure multiple `field name` elements.
searchType	Specifies the type of search to perform. The `searchType` element can contain the following child elements: - `smartSearch` - `searchMatch`
smartSearch	Specifies that you want to use search to find similar records.
searchMatch	Specifies that you want to use queries to find similar records. The `searchMatch` element can contain the following child elements: - `fuzzy` - `matchRuleSet`
fuzzy	Specifies whether you want to perform a fuzzy search. To perform a fuzzy search, set to `true`. The `fuzzy` element is a child of the `searchMatch` element. If you do not add this element, exact search is performed.
matchRuleSet	Specifies the name of the match rule set to use for finding similar records. The `matchRuleSet` element is a child of the `searchMatch` element.
label	Specifies the label format for the search field values. To configure the label format, use the `existsFormat` attribute.
column	Specifies a single column to use in the label format. To configure the column for the label, use the `columnUid` attribute, which is the unique identifier of the column. The `column` element is a child of the label element. You can specify more than one column for the label.

The following image shows the Similar Records component with three records that are similar to the record for John Smith:



Related Topics:

- "Creating the Similar Records Component" on page 180

Similar Records with Search Example

You want to configure the Similar Records component with search. You want to base the search for similar records on the searchable fields first name and last name. You specify `smartSearch` as the `searchType`.

To configure the Similar Records component with search, you use the following code:

```
<config>
    <searchableFields>
        <field name="firstName"/>
        <field name="lastName"/>
       </searchableFields>

                <searchType>
                <smartSearch>
                </smartSearch>
                </searchType>

    <label existsFormat="{1} {2} {3}">
        <column columnUid="firstName" />
        <column columnUid="middleName"/>
        <column columnUid="lastName" />
    </label>
</config>
```

Similar Records with Queries Example

You want to configure the Similar Records component with queries. You want to base the search for similar records on the searchable fields first name and last name. You specify `searchMatch` as the `searchType`.

To configure the Similar Records component with queries, you use the following code:

```
<config>
    <searchableFields>
        <field name="firstName"/>
        <field name="lastName"/>
       </searchableFields>

        <searchType>
                 <searchMatch>
                <fuzzy>true</fuzzy>
                <matchRuleSet>WS</matchRuleSet>
                 </searchMatch>
        </searchType>

    <label existsFormat="{1} {2} {3}">
        <column columnUid="firstName" />
        <column columnUid="middleName"/>
        <column columnUid="lastName" />
    </label>
</config>
```

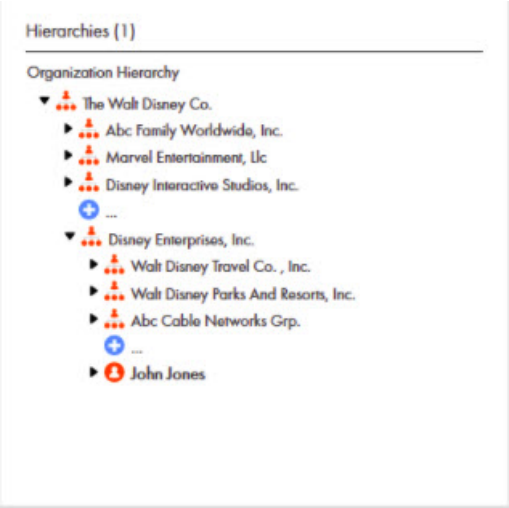
In the example, a fuzzy search is configured. If you want to configure an exact search, remove the following element from the code: `<fuzzy>true</fuzzy>`

Hierarchy Widget

The Hierarchy Widget is a secondary component that you can add to a layout. The widget displays all the related records of a record in a hierarchical format.

You can expand each node in the hierarchy to view the list of related records. For example, to view the list of related records for the Walt Disney Company, expand the Walt Disney Company record and a list of related records appears.

You can also open any record in the hierarchy to view the details of the record. The following image shows a sample Hierarchy Widget that appears in the Dashboard layout:



Custom User Interface Components

You can add custom components to layouts, such as charts and social media feeds. You can configure custom components that you need in the Component Editor. After you configure the custom components, the components appear in the Layout Designer.

The following table describes the custom components that can appear in the Layout Designer when designing **Home** pages, additional pages, or record views:

Custom Component	Description
External Links	Displays custom external resources such as a Twitter feed and web page.

Important: Make sure that the component names that you configure are not the same as any business entity name. For example, if you have a Person business entity in your environment, do not name any custom component Person.

External Links

The External Links component is a custom component that you can add to **Home** page layouts, additional page layouts, and record view layouts. You can create multiple External Links components. The component can display external web pages and RSS feeds.

You can display external resources, such as a Bing search page or Twitter feeds, related to records. If you use Jaspersoft, which is a business intelligence tool to develop and analyze reports, you can display the reports as external resources.

Before you add the External Links component to a layout, create the components in the Component Editor. When you create the External Links component, use an inline frame to embed an external web page or use HTML code and Javascript. After you create the component, the component appears in the Layout Designer.

Related Topics:

- "Creating the External Links Component" on page 181

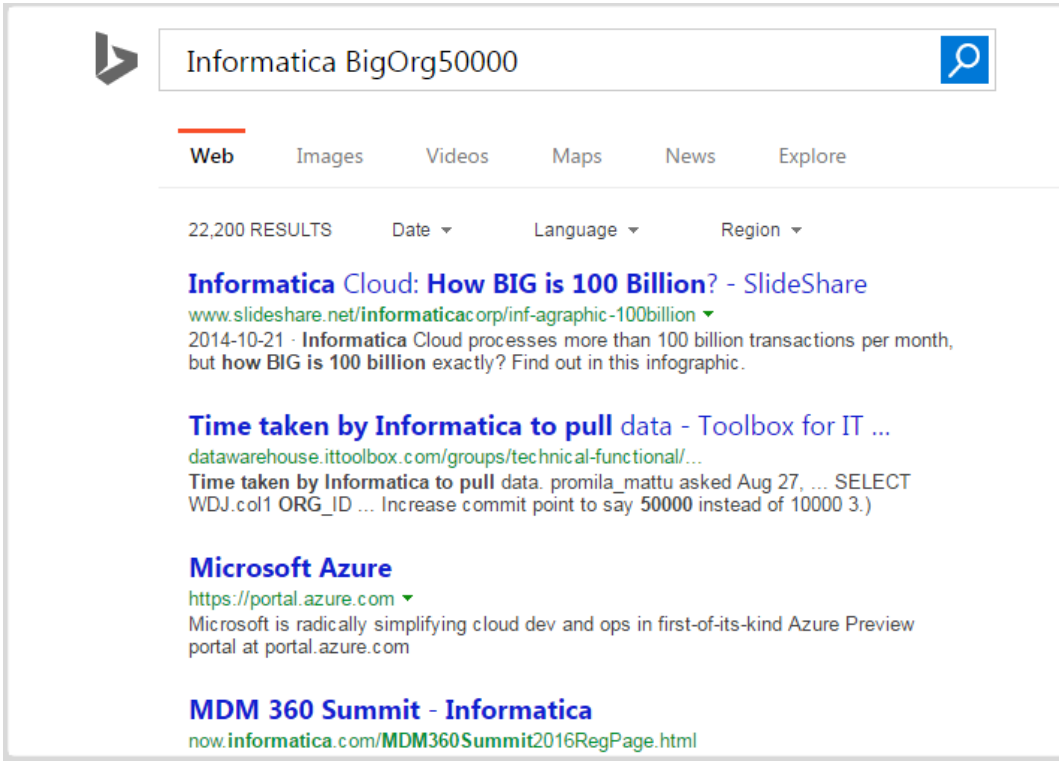
Web Page Embedding Example

You want to configure an inline frame to display a Bing search page that populates with the display name from the business entity.

The following sample code shows the inline frame configuration that you need for the Bing search page to appear in an Entity view:

```
<style>
  #box {
        height: 500px;
        overflow: auto;
       }
</style>
<div id="box">
        <iframe src="http://www.bing.com/search?q=Informatica+{{displayName}}"
            allowtransparency="true" frameborder="0" scrolling="no"
            style="border:none;overflow:auto;height:100%;width:100%" >
        </iframe>
</div>
```

The following image shows the Bing search page that appears in the Entity view:



RSS Feed Configuration Example

You want to display the Twitter feed that belongs to a business entity. You use HTML and Javascript to configure how the Twitter feed appears.

The following sample code shows the HTML and Javascript code that you need for the Twitter feed to appear in an Entity view:

```
<div>
    <a target='_blank' class="twitter-timeline" href="https://twitter.com/{{taxID}}"
       data-screen-name="{{taxID}}" data-tweet-limit="5" data-show-replies="false"
       data-widget-id="473181088752222208">Tweets by {{displayName}}
                </a>
    <script>!function(d,s,id){
              var widgetElem = d.getElementById(id);
              if (widgetElem) {
                  var widgetParent = widgetElem.parentNode;
                  widgetParent.removeChild(widgetElem);
              }
              var js,
              fjs=d.getElementsByTagName(s)[0],
              p=/^http:/.test(d.location)?'http':'https';
              js=d.createElement(s);
              js.id=id;
              js.src=p+"://platform.twitter.com/widgets.js";
              fjs.parentNode.insertBefore(js,fjs);
            }(document,"script","twitter-wjs");
    </script>
</div>
```

The following image shows the Twitter feed that appears in the Entity view:



For more information about the Twitter API code in the example, see the Twitter documentation regarding embedded timelines.

How to Design the Data Director User Interface

Design and update Data Director user interface layouts for the **Home** page, additional page, and record views.

To design and update layouts, perform the following tasks:

1. Create custom components if you need custom components to display data related to records.
2. Design a layout, add components to the layout, and then publish the changes.
3. If the layout needs further changes to meet requirements, you can update the layout and publish the changes.

Creating a Data Director Application

The user interface layout that you want to design must be associated with a Data Director application. To create a Data Director application, use the Application Editor.

1. From the **Database** list, select the database to which you want to associate your configurations.
2. Click **Configuration** > **Application Editor**.

   The **Applications** page appears.
3. Click **Create**.
4. In the properties panel, specify the following properties:

Property	Description
Name	Name of the application that appears in the Applications panel.
Display Name	Name of the application that appears in Data Director.
Source System	Source system with which you want to associate the application.

Property	Description
Session Timeout (minutes)	Time in minutes to wait before an idle Data Director session can time out.
Display Default Record Views	Enables the default record views in Data Director. The default record views appear as tabs in opened records.
Enable Draft	Enables the **Drafts** menu option in the Data Director navigation bar. Specifies whether Data Director users can add, edit, and submit business entity records in the draft state.
Display attributes in row order	Indicates whether to display the attributes that you add to a record details component in row order. You can add the attributes in the Record Details Component Designer dialog box. By default, Data Director displays the attributes in column order.
Smart search	Enables the **Search** menu option in the Data Director navigation bar.
Queries	Enables the **Query** menu option in the Data Director navigation bar.
Queries dialog in application	This option appears when you enable both the **Smart search** and the **Queries** options. Enable the **Queries dialog in application** option to allow the use of queries to search for records within a record view, such as **Matching Records** or while using a component, such as **Related Records**. **Note:** Both the **Query** and the **Search** menu options appear in the Data Director navigation bar.

5. Click **Apply**.

   The application that you created appears in the tree view panel and in the **Applications** panel.

6. Publish the changes to the MDM Hub.

   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.

   b. Review the changes or publish without a review.

      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Configuring the Task Inbox Component

Create the task inbox component in the Component Editor. After you create the task inbox component, the component appears in the Layout Designer. You can configure the fields that you want to display as columns and the predefined filters for the task inbox.

1. Click **Configuration** > **Component Editor**.

   The **Component Editor** appears.

2. From the component type list, select **Task Inbox**, and click **Create**.

   The properties panel displays **Basic** and **Advanced** tabs. To configure the task inbox in the user interface, use the **Basic** tab. To configure the task inbox with the XML code, use the **Advanced** tab. See "XML Code to Configure a Task Inbox" on page 178.

3. On the **Basic** tab, enter a name for the task inbox component.

4. To configure the fields for the task inbox component, in the **Display Fields** field, click **Edit**.

   The **Table Settings** dialog box appears.

   - To add or remove a field, select the field and drag it to the required section.
   - To set the order for a field, move the field up or down.
5. Click **OK**.
6. To configure the filters for the task inbox component, in the **Display Filters** field, click **Edit**.

   The **Table Settings** dialog box appears.

   - To add or remove a filter, select the filter and drag it to the required section.
   - To set the order for a filter, move the filter up or down.
7. Click **OK**.
8. Click **Apply**.
9. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

The task inbox component that you create and publish appears in the **Component Editor** panel.

Related Topics:

- "Primary User Interface Components" on page 154

XML Code to Configure a Task Inbox

On the **Advanced** tab of the task inbox configuration page, use the XML code to configure the task inbox.

The following table lists the elements that you can use in the XML code to configure the task inbox:

Element	Description
displayFields	Name of the field that you want to display in the task inbox. Use one of the following fields: - TASK_TITLE. A short description of the review task. - TYPE. Name of a people activity within the process. - PRIORITY. Priority of the task. - DUE_DATE. Date the task is due. - OWNER. Name of the signed-in user. - TASK_ID. Instance ID of the review process that generated the task. - STATUS. Status of the task. - CREATOR. Name of the user who created the task. - CREATION_DATE. Date on which the user created the task.
displayFilters	Filter that you want to display in the task inbox. Use one of the following filters: - MY_OPEN_TASKS. Shows your open tasks. - MY_CLOSED_TASKS. Shows the tasks that you closed. - ALL_TASKS. Shows all the closed, claimed, and unclaimed tasks that are assigned to your user group. - ALL_UNCLAIMED_TASKS. Shows all the unclaimed tasks that are assigned to your user group. - ALL_CLOSED_TASKS. Shows all the tasks that the users in your user group closed. - ALL_CLAIMED_TASKS. Shows all the tasks that the users in your user group claimed.
excludeTaskTypes	Task type that you do not want to view in the task inbox.

The following sample XML code includes all the fields and filters and excludes the draft and merge tasks in the task inbox:

```
<dataViewLayout xmlns="http://www.informatica.com/mdm/TaskInbox">
   <taskInboxConfig>
      <displayFields>TASK_TITLE</displayFields>
      <displayFields>TYPE</displayFields>
      <displayFields>PRIORITY</displayFields>
      <displayFields>DUE_DATE</displayFields>
      <displayFields>OWNER</displayFields>
      <displayFields>TASK_ID</displayFields>
      <displayFields>STATUS</displayFields>
      <displayFields>CREATOR</displayFields>
      <displayFields>CREATION_DATE</displayFields>
      <displayFilters>MY_OPEN_TASKS</displayFilters>
      <displayFilters>MY_CLOSED_TASKS</displayFilters>
      <displayFilters>ALL_TASKS</displayFilters>
      <displayFilters>ALL_UNCLAIMED_TASKS</displayFilters>
      <displayFilters>ALL_CLOSED_TASKS</displayFilters>
      <displayFilters>ALL_CLAIMED_TASKS</displayFilters>
      <excludeTaskTypes>AVOSBeDraftState</excludeTaskTypes>
      <excludeTaskTypes>AVOSBeMerge</excludeTaskTypes>
   </taskInboxConfig>
</dataViewLayout>
```

Creating the Related Records Component

Create the Related Records component in the Component Editor. After you create the Related Records component, the component appears in the Layout Designer.

1. Click **Configuration** > **Component Editor**.

   The **Component Editor** appears.

2. Create a Related Records component.
   a. From the component type list, select **Related Records**, and click **Create**.
   b. In the properties panel, specify the following Related Records component properties:

Property	Description
Name	Name of the Related Records component that appears in the Component Editor panel.
Label	Label of the Related Records component that appears in the component list for the component in the Layout Designer workspace.
Honor Direction	Specifies whether the Related Records component honors the direction of the relationship between records. When enabled, the component displays related records that are children of the selected record.
Read Only	Specifies whether users can edit related records from the Related Records component. To configure the component as a non-editable component, enable the property.

   c. Click **Apply**.

      The Related Records component that you created appears in the **Component Editor** panel and in the tree view panel.

3. Create filters for the business entities that appear in the Related Records component.
   a. In the tree view panel, click **Filter**, and then click **Create**.

      The filter properties appear in the properties panel.
   b. In the **Name** field, enter a name for the filter.
   c. From the **Object Type** list, choose one of the following filter types:

Filter Type	Description
Relationship	Filter based on a relationship type.
Business Entities	Filter based on a business entity.

      If you select Relationship, the relationship names appear in the **Object** list. If you select Business Entity, the business entity names appear in the **Object** list.

   d. Select an item from the **Value** list, and click **Apply**.

   You can create multiple filters for the business entities in the Related Records component.

4. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

The Related Records component that you create and publish appears in the **Component Editor** panel.

Creating the Similar Records Component

Create the Similar Records component in the Component Editor. After you create the Similar Records component, the component appears in the Layout Designer.

1. Click **Configuration** > **Component Editor**.

   The **Component Editor** appears.
2. From the Component type list, select **Similar Records**, and click **Create**.

   The property fields appear in the properties panel.
3. Enter a name for the Similar Records component.
4. In the **XML** field, enter the following XML configuration that includes a list of fields to search for similar records:

```
<config>
    <searchableFields>
        <field name="<field name 1>"/>
        <field name="<field name 2>"/>
                                ...
        <field name="<field name n>"/>
    </searchableFields>

                <searchType>
                                        ...
                </searchType>

    <label existsFormat="{1} {2} {3}">
        <column columnUid="<field name 1>" />
        <column columnUid="<field name 2>"/>
        <column columnUid="<field name 3>" />
    </label>
</config>
```

   Where `<field name>` is the name of the field on which you want to base the search for similar records. You can base the search for similar records on field names in the business entity model that are configured as searchable. Also, you need to specify the format in which the search field values must appear.
5. Click **Apply**.

   The Similar Records component that you created appears in the **Component Editor** panel and in the tree view panel.
6. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.

b. Review the changes or publish without a review.
   - To publish without a review, click **Publish**.
   - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Related Topics:

- " Similar Records" on page 169

Creating the External Links Component

Create the External Links components in the Component Editor. After you create the External Links components, the components appear in the Layout Designer.

1. Click **Configuration** > **Component Editor**.

   The **Component Editor** appears.
2. From the Component type list, select **External Links**, and click **Create**.

   The property fields appear in the properties panel.
3. Specify the following component properties:

Property	Description
Name	Name of the External Links component that appears in the Component Editor panel.
Code	Code to display the external resource. You can use an inline frame to embed an external web page, or use HTML code and Javascript. You must use the HTML tags that are allowed within the <body> element. **Note:** If the external resource for which you want to create a component does not have a predefined height, specify the height of the component in the code.

4. Click **Apply**.

   The External Links component that you created appears in the **Component Editor** panel and in the tree view panel.
5. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Related Topics:

- "External Links" on page 173

Creating the Chart Component

Create the Chart component in the Component Editor. You can configure the source data that you want to use in the chart and the appearance of the chart. After you create the Chart component, the component appears in the Layout Designer.

Before you begin, register the report that you want to use in the chart. For more information, see the *Multidomain MDM Business Entity Services Guide*.

1. Click **Configuration** > **Component Editor**.

   The **Component Editor** appears.
2. From the component type list, select **Chart**, and then click **Create**.

   A chart node appears in the tree view panel.
3. In the properties panel, enter a name for the chart in the **Chart Name** field.
4. In the **Display Name** field, enter a display name for the chart.
5. To configure the chart, in the **Source Data** field, select **Select**.

   The **Chart Settings** dialog box appears.
6. Configure the source data that you want to use.
   a. From the **Source Data** list, select the report with data that you want to use in the chart.

      Note: Your registered reports appear in the **Source Data** list. For more information about registering reports, see the *Multidomain MDM Business Entity Services Guide*.

      The **X-Axis** and **Y-Axis** lists display the axes defined for the report.

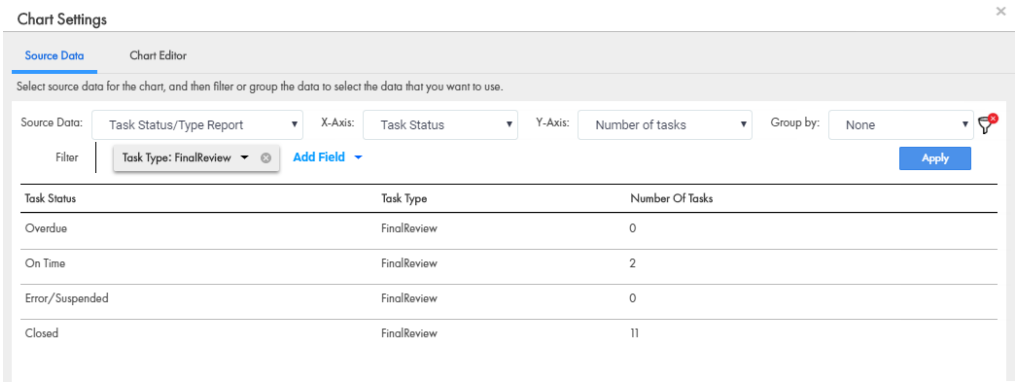
   b. To filter the report data that you want to use in the chart, click the **Filter** Icon, and then add filters.

      The table filters the report data based on the selected filters.

   c. To group the report data by a criteria in the underlying data, in the **Group by** list, select a criteria.

      The table groups the report data based on the selected group by criteria.

   The following image shows the **Source Data** tab with filtered report data:



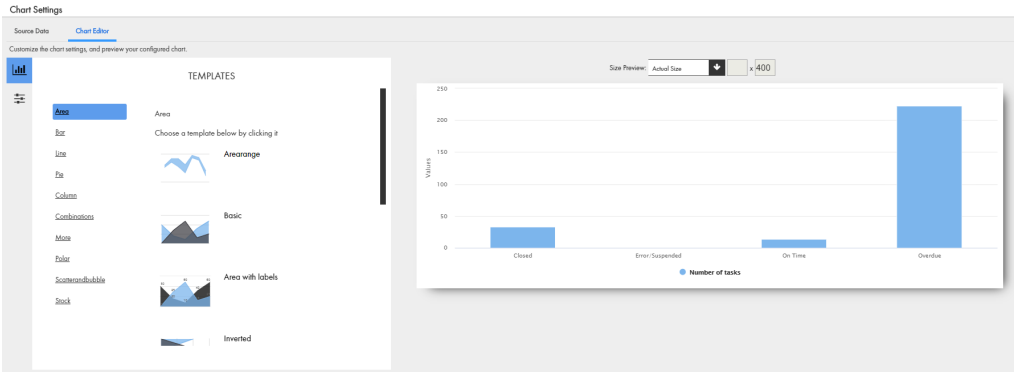
   The **Source Data** tab displays the report data you want to use in the chart.
7. Configure the chart template and customize the chart.
   a. Click **Chart Editor**.

b. Select the chart template that you want to use.

c. Click the **Customize Chart** tab, and then customize the chart.
For example, you might want to configure the chart titles and appearance.

d. Preview the chart to ensure it appears as expected.

The following image shows the **Chart Editor** tab with a chart preview:



The **Chart Editor** tab displays a preview of the chart.

8. Click **Done**.

   In the properties panel, the fields display your selections.

9. Click **Apply**.

10. Publish the changes to the MDM Hub.

    a. Click **Publish**.

       A confirmation dialog box appears that prompts you to publish or review the changes.

    b. Review the changes or publish without a review.

       - To publish without a review, click **Publish**.
       - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

The Chart component that you created and published appears in the **Component Editor** panel. You can add the Chart component to a **Home** page layout or additional page layout.

You can link the chart to a drilldown chart or to another interface in Data Director.

For information about linking to the **Query** tab, see "Linking a Chart to the Query Tab" on page 185.

For information about linking to the **Search** tab, see "Linking a Chart to the Search Tab" on page 186.

For information about linking to the Task Manager, see "Linking a Chart to the Task Manger" on page 186.

Linking a Chart to a Drilldown Chart

Link a chart to a drilldown chart. When Data Director users select data in the chart, they drill down to view another layer of data granularity in the drilldown chart.

Before you begin, create the **Chart** component. For more information, see the *Multidomain MDM Business Entity Services Guide*.

1. Click **Configuration** > **Component Editor**.

   The **Component Editor** appears.

2. From the component type list, select **Chart**, and then select the Chart component you want to configure.

The Chart node appears in the tree view panel.

3. Link the chart to a drilldown chart and then configure the drilldown chart.
   a. In the tree view panel, click the **Add** icon, and then select the drilldown chart to which you want to link to the chart.

      Note: The available drilldown charts depend on your report configuration. For more information, see the *Multidomain MDM Business Entity Services Guide*.

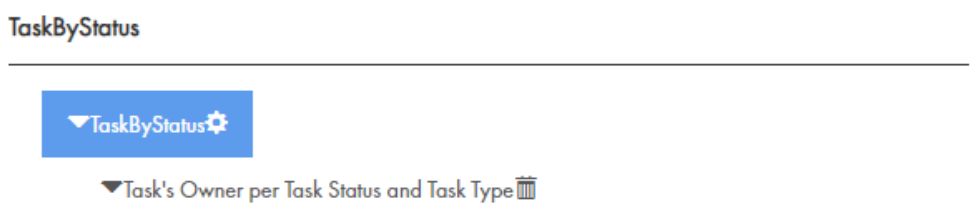
      The drilldown chart node appears in the tree view panel.

   b. Optionally, configure the source data that you want to use. In the **Source Data** field, click the source data link, and then configure the source data.
      For example, you might want to configure the axes, group by criteria, or filters.

   c. Optionally, customize the chart. Click **Chart Editor**, and then customize the chart.
      For example, you might want to configure the chart template or the appearance of the chart.

   d. Click **Done**.

   The following image shows an example of a chart linked to a drilldown chart:



   The drilldown chart node appears in the tree view panel.

4. Optionally, to link the drilldown chart you configured to a drilldown chart, repeat step 3.

   The drilldown chart node appears in the tree view panel.

5. Click **Apply**.
6. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.

   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

You updated the Chart component and linked the chart to a drilldown chart. You can add the Chart component to a **Home** page layout or additional page layout.

You can link the chart to a drilldown chart or to another interface in Data Director.

For information about linking to the **Query** tab, see "Linking a Chart to the Query Tab" on page 185.

For information about linking to the **Search** tab, see "Linking a Chart to the Search Tab" on page 186.

For information about linking to the Task Manager, see "Linking a Chart to the Task Manger" on page 186.

Linking a Chart to the Query Tab

Link a chart to the **Query** tab, and configure the search criteria for the query. When Data Director users select data in the chart, they navigate to the **Query** tab. The **Query** tab displays query results based on the search criteria.

Before you begin, create the Chart component.

1. Click **Configuration** > **Component Editor**.

   The **Component Editor** appears.
2. From the component type list, select **Chart**, and then select the Chart component you want to configure.

   The Chart node appears in the tree view panel.
3. In the tree view panel, click the **Add** icon, and then select **Query**.

   The Query node appears in the tree view panel.
4. In the properties panel, click the **Add** icon.

   The **New Query** dialog box appears.
5. Configure the search criteria for the query.
   a. From the **Business Entity** list, select the business entity within which you want to perform the search.
   b. Select a query type.

Query Type	Description
Basic	Uses simple text comparisons to find records that contain the query values.
Extended	Uses matching logic and match-enabled attributes to find records that contain the query values.

   c. If you selected the extended query type, from the **Match Rule Set** list, select a match rule set.
   d. Select the fields that you want to include as the search criteria for the query.
   e. In the **Selected Fields** section, drag to rearrange the sequence of the fields.
   f. To make the query case sensitive, enable the **Case sensitive** option.
   g. To sort search results, from the **Sort By** list, select a field name by which you want the search results sorted.
   h. Select the sort order of the query to be ascending or descending.
   i. From the **Business Entity** list, select the business entity within which you want to perform the search.
   j. Click **OK**.
6. Click **Apply**.
7. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.

- To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

You updated the Chart component and linked the chart to the **Query** tab. You can add the Chart component to a **Home** page layout or additional page layout.

Linking a Chart to the Search Tab

Link a chart to the **Search** tab, and configure the search criteria. When Data Director users select data in the chart, they navigate to the **Search** tab. The **Search** tab displays search results based on the search criteria.

Before you begin, create the Chart component.

1. Click **Configuration** > **Component Editor**.

   The **Component Editor** appears.
2. From the component type list, select **Chart**, and then select the Chart component you want to configure.

   The Chart node appears in the tree view panel.
3. In the tree view panel, click the **Add** icon, and then select **Search**.

   The Search node appears in the tree view panel.
4. Configure the search criteria.
   a. In the **Business Entity** field, select the business entity type that you want to use as the search criteria.
   b. In the **Search Input** field, select the underlying data in the report that you want to use as the search criteria.

   For example, if you have a Customers Distributed by City report, you might select the Customer business entity and the City Names search input.
5. Click **Apply**.
6. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

You updated the Chart component and linked the chart to the **Search** tab. You can add the Chart component to a **Home** page layout or additional page layout.

Linking a Chart to the Task Manger

Link a chart to the Task Manager, and configure the filter criteria. When Data Director users select data in the chart, they navigate to the **Task Manager** tab. The **Task Manager** tab displays tasks based on the configured filter criteria.

Before you begin, create the Chart component.

1. Click **Configuration** > **Component Editor**.

   The **Component Editor** appears.
2. From the component type list, select **Chart**, and then select the Chart component you want to configure.

   The Chart node appears in the tree view panel.

3. In the tree view panel, click the **Add** icon, and then select **Task Manager**.

   The Task Manager node appears in the tree view panel.

4. Configure the filter criteria for the Task Manager.
   a. In the **Task Manager Tab** field, select the quick filter tab that you want to open in the Task Manager.
   b. Optionally, to filter the tasks using a search term, in the **Search Criteria** field, enter the search term.
   c. Optionally, to filter the tasks using custom filters, in the **Filters** section, click the **Add** icon, and then configure filters.

   For example, you might select the **All Tasks** tab, enter John as the search team, and add the Task Type filter.

5. Click **Apply**.
6. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.

   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

You updated the Chart component and linked the chart to the Task Manager. You can add the Chart component to a **Home** page layout or additional page layout.

Designing a Home Page Layout

You can design multiple **Home** pages for Data Director. You can configure which roles can access the **Home** pages. For example, you might design a **Home** page layout for Account Managers that displays the task inbox and charts.

Note: If you do not design a Home page, a Data Director application displays the legacy start page.

1. Click **Configuration** > **Layout Designer**.

   The **Layout Designer** appears.

2. Click **Create** > **Home Page Layout**.

   The **Properties** page appears.

3. Specify the following properties:

Property	Description
View Name	Label for the option.
Layout Name	Label for the layout that you design. After you design and publish the layout, the layout name appears in the **Layout Categories** panel.
Layout ID	Label used for generating a system ID.
Description	Optional. A meaningful description to identify the layout.

4. Click **Next**.

   The **User Roles** page appears.

5. Select the user roles that can access the **Home** page.

   The user roles that appear on the **User Roles** page are configured in the MDM Hub.

6. Click **Next**.

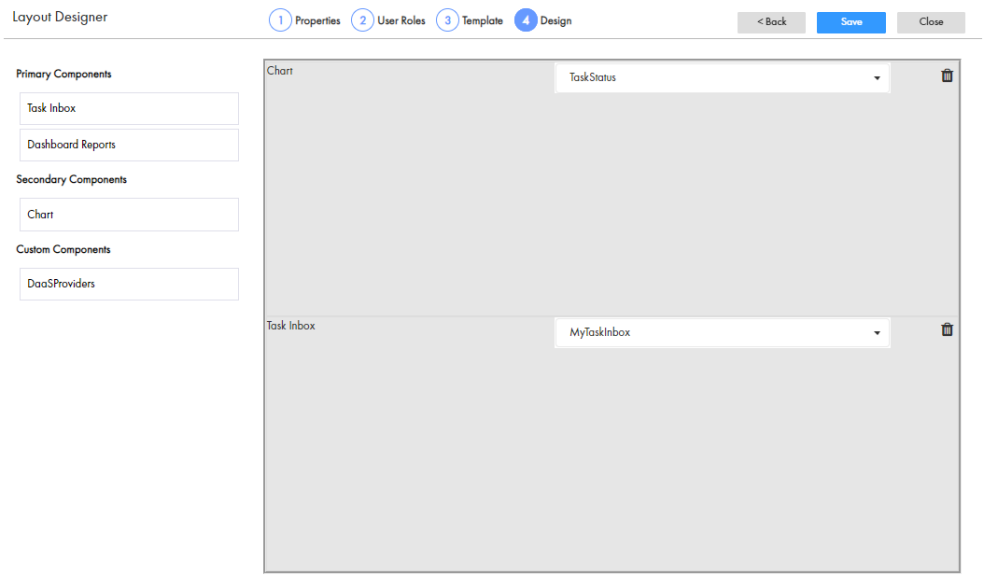
   The **Template** page appears.

7. Select a template that supports how users will interact with the **Home** page, and click **Next**.

   The **Design** page appears.

8. Drag the components that you want in the **Home** page layout into the workspace.
   For example, drag the **Task Inbox** component and the **Chart** component into the workspace.

   The following image shows the **Design** page with the **Task Inbox** component and the **Chart** component in the workspace:



9. Click **Save**.

   The changes are saved to the temporary workspace.

10. Publish the changes to the MDM Hub.

    a. Click **Publish**.

       A confirmation dialog box appears that prompts you to publish or review the changes.

    b. Review the changes or publish without a review.

       - To publish without a review, click **Publish**.
       - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

The layout appears in **Layout Categories** panel. When users log in to Data Director, the **Home** page designed for their user role appears.

Related Topics:

- "Home Page Layouts" on page 147

Designing an Additional Page Layout

You can design multiple pages for Data Director in the Layout Designer. You can configure which roles can access the additional pages. For example, you might design an additional page layout for a customer that displays the task inbox and an external search.

1. Click **Configuration** > **Layout Designer**.

   The **Layout Designer** appears.

2. Click **Create** > **Additional Page Layout**.

   The **Properties** page appears.

3. Specify the following properties:

Property	Description
Tab Name	Label for the page tab that you want to create and associate to the additional page. In Data Director, the custom tab appears in the navigation bar. Users click the custom tab to view the associated additional page.
Layout ID	Label used for generating a system ID.
Description	Optional. A meaningful description to identify the layout.
Layout Name	Label for the layout that you design. After you design and publish the layout, the layout name appears in the Layout Categories panel.

4. Select the Data Director application to which you want to associate the layout.
5. Click **Next**.

   The **User Roles** page appears.

6. Select the user roles that can access the additional page.

   The user roles that appear on the **User Roles** page are configured in the MDM Hub.

7. Click **Next**.

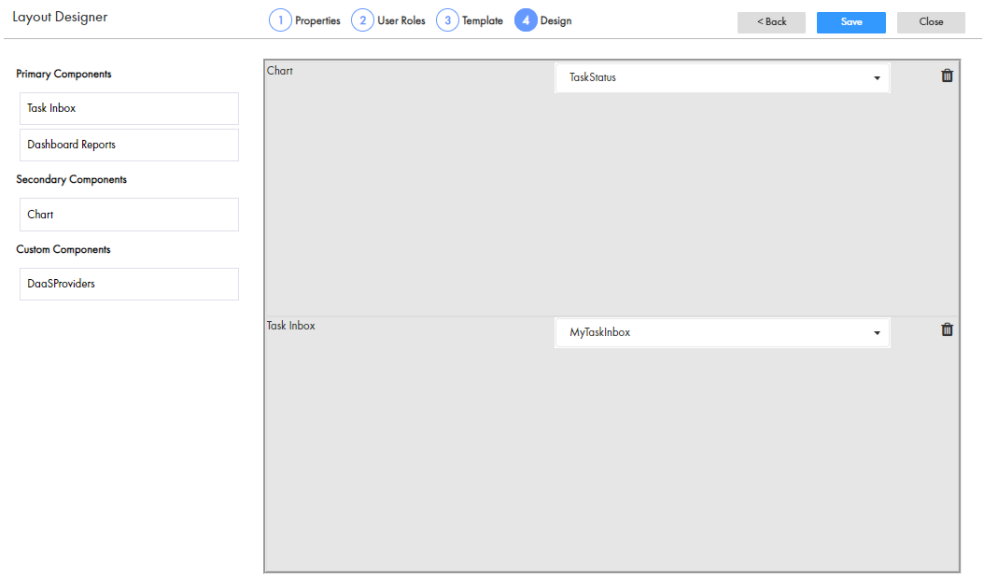
   The **Template** page appears.

8. Select a template that supports how users will interact with the additional page, and click **Next**.

   The **Design** page appears.

9. Drag the components that you want in the additional page layout into the workspace.
   For example, drag the **Task Inbox** component and the **Chart** component into the workspace.

The following image shows the **Design** page with the **Task Inbox** component and the **Chart** component in the workspace:



10. Click **Save**.

    The changes are saved to the temporary workspace.

11. Publish the changes to the MDM Hub.

    a. Click **Publish**.

       A confirmation dialog box appears that prompts you to publish or review the changes.

    b. Review the changes or publish without a review.

       - To publish without a review, click **Publish**.
       - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

The layout appears in Layout Categories panel. In Data Director, additional pages appear in the associated custom tab in the navigation bar.

Designing a Business Entity View Layout

You can design a custom record view layout to replace the default Business Entity view. Add the Record Details components, and add any secondary or custom components that you want.

1. Click **Configuration** > **Layout Designer**.

   The **Layout Designer** appears.

2. Click **Create** > **Record View Layout**.

   The **Properties** page appears.

3. From the **Layout Type** list, select **Record View and Review Task**.

4. Define the properties of the layout.
   a. Specify the following properties:

Property	Description
View Name	Label for the option. The view name appears as a tab in an opened record in Data Director.
Layout Name	Label for the layout that you design. After you design and publish the layout, the label appears in the **Layout Categories** panel in the Layout Designer.
Layout ID	Label used for generating a system ID.
Description	Optional. A meaningful description to identify the layout.
List Option Icon	Optional. Icon for the list option. The list option icon appears in the **View** list in Data Director.
List Order	Optional. Placement of the view name in the list of tabs in the opened record in Data Director.

   b. Choose the business entity or business entity view for which you want to design the layout.

      For example, choose **Person**.

5. Click **Next**.

   The **User Roles** page appears.

6. Select the user roles that can access the record view layout, and click **Next**.

   Tip: To create user roles, use the MDM Hub Console.

   The **Purpose** page appears.

7. Select the actions that users can perform using the record view layout.
   - To create records, select **Create records**.
   - To view or edit records, select **View or edit records**.
   - To create, view, or edit records, select **Create records** and **View or edit records**.

   Tip: To use the same layout for the Business Entity view and Review Tasks, select all options on the **Purpose** page of the Layout Designer.

8. Click **Next**.

   The **Template** page appears.

9. Select a template that supports how users will interact with the record view layout, and click **Next**.

   Tip: Primary components require more horizontal space than other components. When designing record view layouts, select a template with a panel that is big enough to display a primary component.

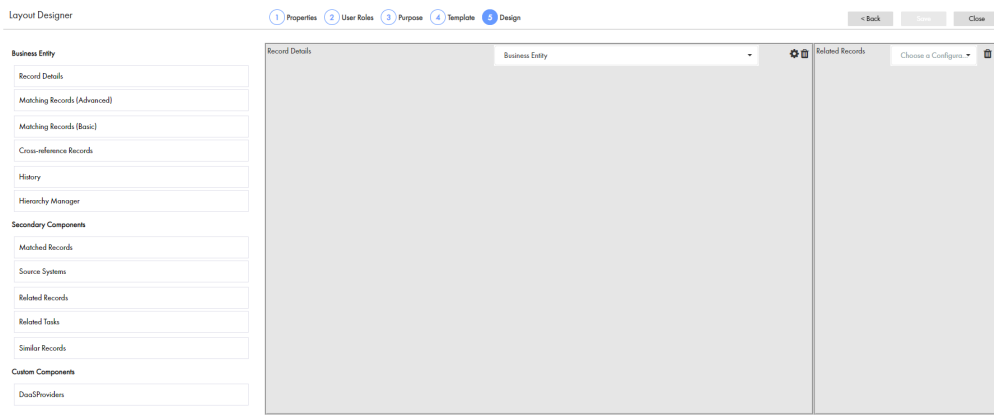
   The **Design** page appears.

10. Drag components into the workspace.
    a. Drag the **Record Details** component into the workspace.

       Note: Drag only one primary component into the workspace.

    b. To display related information, drag secondary or custom components into the workspace.

The following image shows a **Design** page with the Record Details component and the Related Records component in the workspace:



11. If you want selected fields of the **Record Details** component to appear in the record view, configure the component.
    a. From the component configuration list, select a configuration.

       You can select a default or a custom configuration.
       The following default configurations are available:

       - default-full. Configuration of the business entity that includes the root node and all the child nodes.
       - default-root. Configuration of the root node of the business entity.

    b. Click the **Configure** icon of the component in the workspace.

       The **Record Details Component Designer** appears.

    c. In the **Configuration name** field, enter a configuration name.

       Important: Ensure that the configuration name is not the same as any business entity name. For example, if you have a Person business entity in your environment, do not name the configuration Person.

    d. If you want to configure the amount of sections in which you can add attributes, in the **Sections** list, select a number.

       Note: When the layout contains multiple sections, some field values in the record details might not be visible.

    e. If you want to use a vertical menu to access the business entity details, enable **Navigation pills**.

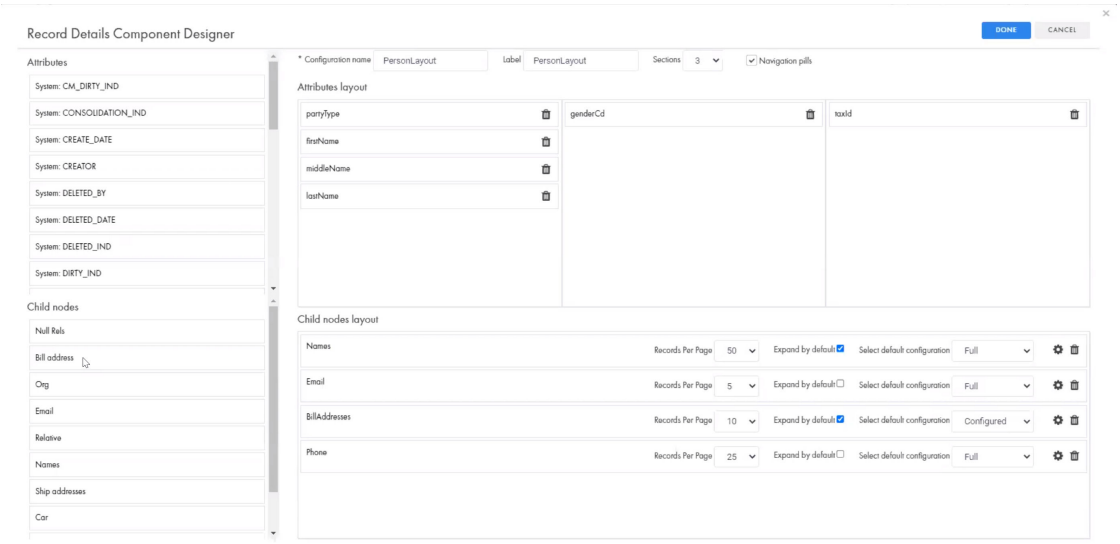
    f. From the **Attributes** section, drag the business entity attributes that you want to display in a record view into the **Attributes layout** section.

       For example, drag the First Name, Middle Name, Last Name, Birth Date, and Tax ID into the **Attributes layout** section.

    g. From the **Child nodes** section, drag the child business entities that you want to display in a record view to the **Child nodes layout** section.

       For example, drag Bill address, Ship address, Telephones, and Email into the **Child nodes layout** section.

The following image shows the **Record Details Component Designer** page with the layout of attributes and child nodes defined:



h. To customize the display of the child nodes in the **Child nodes layout** section, configure the following view options:

Records Per Page:

Select the maximum number of records that you want to display in the page. Default is 5.

Expand by default:

Choose to show an expanded view of the child records.

Select default configuration:

From the list, select one of the following values:

- `Configured`. Displays the root node without the child nodes. Default is `Configured`.
- `Full`. Displays the root node and the child nodes.

i. If you want selected fields of the child business entities to appear in the record view, click the **Configure** icon next to the child entity.

The **Record Details Component Designer** appears, where you can design the child entity.

j. Click **Done**.

The **Design** page appears.

12. Click **Save**.

The changes are saved to the temporary workspace.

13. Publish the changes to the MDM Hub.

a. Click **Publish**.

A confirmation dialog box appears that prompts you to publish or review the changes.

b. Review the changes or publish without a review.

- To publish without a review, click **Publish**.
- To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

The layout appears in the **Layout Categories** panel. When users create or open records in Data Director, the record view layouts designed for their user role appear in tabs in the record.

Related Topics:

- "Record View and Review Task Layout Type" on page 149

Designing a Matching Records View Layout

You can design a record view layout to replace the default Matching Records view. Add the Matching Records (Advanced) component or the Matching Records component, and add any secondary or custom components that you want.

1. Click **Configuration** > **Layout Designer**.

   The **Layout Designer** appears.
2. Click **Create** > **Record View Layout**.

   The **Properties** page appears.
3. From the **Layout Type** list, select **Record View and Review Task**.
4. Define the properties of the layout.
   a. Specify the following properties:

Property	Description
View Name	Label for the option. The view name appears as a tab in an opened record in Data Director.
Layout Name	Label for the layout that you design. After you design and publish the layout, the label appears in the **Layout Categories** panel in the Layout Designer.
Layout ID	Label used for generating a system ID.
Description	Optional. A meaningful description to identify the layout.
List Option Icon	Optional. Icon for the list option. The list option icon appears in the **View** list in Data Director.
List Order	Optional. Placement of the view name in the list of tabs in the opened record in Data Director.

   b. Choose the business entity or business entity view for which you want to design the layout.

      For example, choose **Person**.
5. Click **Next**.

   The **User Roles** page appears.
6. Select the user roles that can access the record view layout, and click **Next**.

   Tip: To create user roles, use the MDM Hub Console.

   The **Purpose** page appears.
7. Select the **View or edit records** option, and click **Next**.

   The **Template** page appears.
8. Select a template that supports how users will interact with the record view layout, and click **Next**.

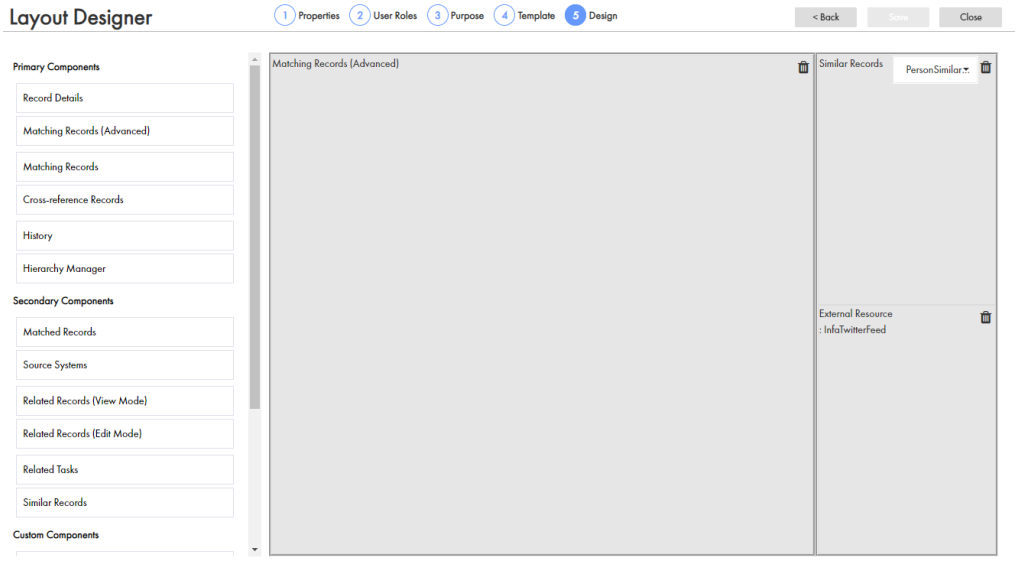
   Tip: Primary components require more horizontal space than other components. When designing record view layouts, select a template with a panel that is big enough to display a primary component.

The **Design** page appears.

9. Drag one primary component into the workspace.
   - To merge records using the merge comparison panel with advanced features, drag the Matching Records (Advanced) component into the workspace.
     Use the Matching Records (Advanced) component to override and promote root and child values in records.
   - To merge records using the merge comparison panel, drag the Matching Records component into the workspace.
     Use the Matching Records component to override root values in records.

   Note: Drag only one primary component into the workspace.

   The following image shows the **Design** page with the Matching Records (Advanced) component, the Similar Records component, and the Twitter Feed component in the workspace:



10. Optionally, drag secondary or custom components into the workspace.
11. Click **Save**.

    The changes are saved to the temporary workspace.
12. Publish the changes to the MDM Hub.
    a. Click **Publish**.

       A confirmation dialog box appears that prompts you to publish or review the changes.
    b. Review the changes or publish without a review.
       - To publish without a review, click **Publish**.
       - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

The layout appears in **Layout Categories** panel. When users open records in Data Director, the record view layouts designed for their user role appear in tabs in the record.

Related Topics:

- "Record View and Review Task Layout Type" on page 149

Designing a Cross-reference Records View Layout

You can design a record view layout to replace the Cross-reference Records view. Add the Cross-reference Records component, and add any secondary or custom components that you want.

1. Click **Configuration** > **Layout Designer**.

   The **Layout Designer** appears.
2. Click **Create** > **Record View Layout**.

   The **Properties** page appears.
3. From the **Layout Type** list, select **Record View and Review Task**.
4. Define the properties of the layout.
   a. Specify the following properties:

Property	Description
View Name	Label for the option. The view name appears as a tab in an opened record in Data Director.
Layout Name	Label for the layout that you design. After you design and publish the layout, the label appears in the **Layout Categories** panel in the Layout Designer.
Layout ID	Label used for generating a system ID.
Description	Optional. A meaningful description to identify the layout.
List Option Icon	Optional. Icon for the list option. The list option icon appears in the **View** list in Data Director.
List Order	Optional. Placement of the view name in the list of tabs in the opened record in Data Director.

   b. Choose the business entity for which you want to design the layout.

      For example, choose **Person**.
5. Click **Next**.

   The **User Roles** page appears.
6. Select the user roles that can access the record view layout, and click **Next**.

   Tip: To create user roles, use the MDM Hub Console.

   The **Purpose** page appears.
7. Select the **View or edit records** option, and click **Next**.

   The **Template** page appears.
8. Select a template that supports how users will interact with the record view layout, and click **Next**.

   Tip: Primary components require more horizontal space than other components. When designing record view layouts, select a template with a panel that is big enough to display a primary component.

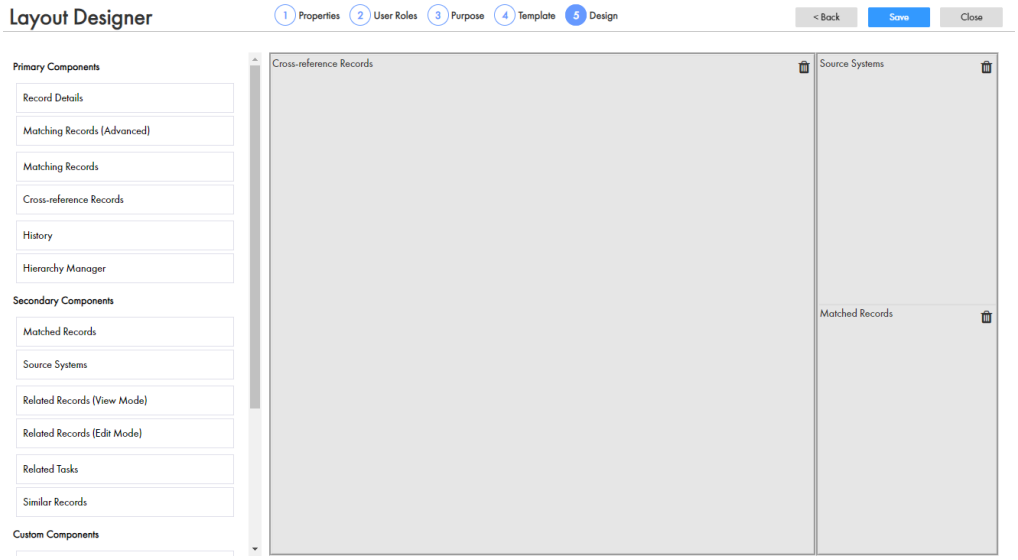
   The **Design** page appears.

9. Drag components into the workspace.
   a. Drag the Cross-reference Records component into the workspace.

      Note: Drag only one primary component into the workspace.

   b. To display related information, drag secondary or custom components into the workspace.

   The following image shows the **Design** page with the Cross-reference Records component, the Source Systems component, and the Matched Records component in the workspace:



10. Click **Save**.

    The changes are saved to the temporary workspace.

11. Publish the changes to the MDM Hub.
    a. Click **Publish**.

       A confirmation dialog box appears that prompts you to publish or review the changes.

    b. Review the changes or publish without a review.
       - To publish without a review, click **Publish**.
       - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

The layout appears in **Layout Categories** panel. When users open records in Data Director, the record view layouts designed for their user role appear in tabs in the record.

Related Topics:

- "Record View and Review Task Layout Type" on page 149

Designing a History View Layout

You can design a custom record view layout to replace the History view. Add the History component, and add any secondary or custom components that you want.

1. Click **Configuration** > **Layout Designer**.

   The **Layout Designer** appears.

2. Click **Create** > **Record View Layout**.

   The **Properties** page appears.
3. From the **Layout Type** list, select **Record View and Review Task**.
4. Define the properties of the layout.
   a. Specify the following properties:

Property	Description
View Name	Label for the option. The view name appears as a tab in an opened record in Data Director.
Layout Name	Label for the layout that you design. After you design and publish the layout, the label appears in the **Layout Categories** panel in the Layout Designer.
Layout ID	Label used for generating a system ID.
Description	Optional. A meaningful description to identify the layout.
List Option Icon	Optional. Icon for the list option. The list option icon appears in the **View** list in Data Director.
List Order	Optional. Placement of the view name in the list of tabs in the opened record in Data Director.

   b. Choose the business entity or business entity view for which you want to design the layout.

      For example, choose **Person**.
5. Click **Next**.

   The **User Roles** page appears.
6. Select the user roles that can access the record view layout, and click **Next**.

   Tip: To create user roles, use the MDM Hub Console.

   The **Purpose** page appears.
7. Select the **View or edit records** option, and click **Next**.

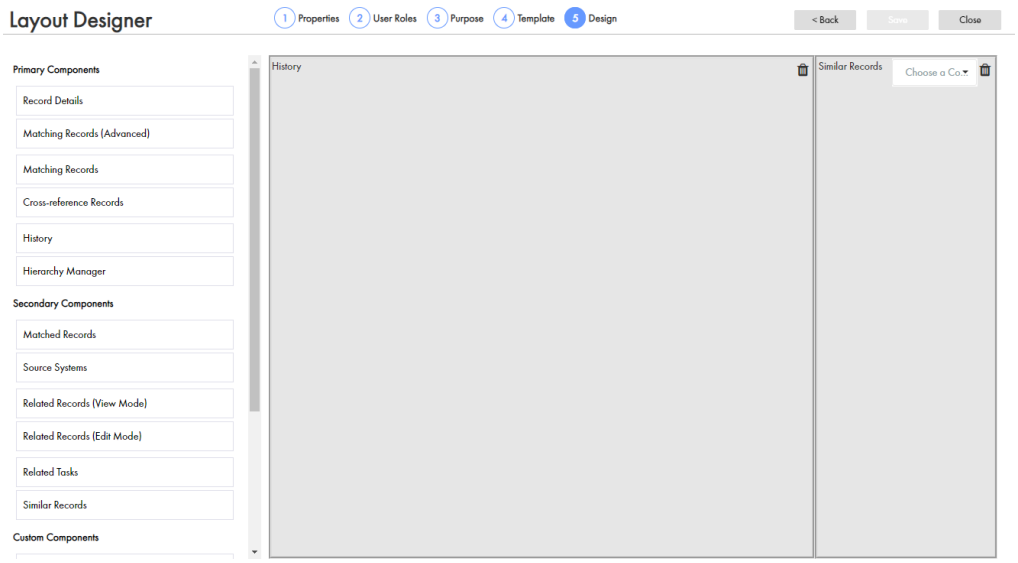
   The **Template** page appears.
8. Select a template that supports how users will interact with the record view layout, and click **Next**.

   Tip: Primary components require more horizontal space than other components. When designing record view layouts, select a template with a panel that is big enough to display a primary component.

   The **Design** page appears.
9. Drag components into the workspace.
   a. Drag the History component into the workspace.

      Note: Drag only one primary component into the workspace.
   b. To display related information, drag secondary or custom components into the workspace.

The following image shows the **Design** page with the History component and the Similar Records component in the workspace:



10. Click **Save**.

    The changes are saved to the temporary workspace.

11. Publish the changes to the MDM Hub.

    a. Click **Publish**.

       A confirmation dialog box appears that prompts you to publish or review the changes.

    b. Review the changes or publish without a review.

       - To publish without a review, click **Publish**.
       - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

The layout appears in **Layout Categories** panel. When users open records in Data Director, the record view layouts designed for their user role appear in tabs in the record.

Related Topics:

- "Record View and Review Task Layout Type" on page 149

Designing a Review Task Layout

You can design a custom record view layout to manage review tasks in the Task Manager. Add the Record Details component, and add any secondary or custom components that you want.

1. Click **Configuration** > **Layout Designer**.

   The **Layout Designer** appears.

2. Click **Create** > **Record View Layout**.

   The **Properties** page appears.

3. From the **Layout Type** list, select **Record View and Review Task**.

4. Define the properties of the layout.
   a. Specify the following properties:

Property	Description
View Name	Label for the option. The view name appears as a tab in an opened record in Data Director.
Layout Name	Label for the layout that you design. After you design and publish the layout, the label appears in the **Layout Categories** panel in the Layout Designer.
Layout ID	Label used for generating a system ID.
Description	Optional. A meaningful description to identify the layout.
List Option Icon	Optional. Icon for the list option. The list option icon appears in the **View** list in Data Director.
List Order	Optional. Placement of the view name in the list of tabs in the opened record in Data Director.

   b. Choose the business entity or business entity view for which you want to design the layout.

      For example, choose **Person**.

5. Click **Next**.

   The **User Roles** page appears.

6. Select the user roles that can access the record view layout, and click **Next**.

   Tip: To create user roles, use the MDM Hub Console.

   The **Purpose** page appears.

7. Select the **View or edit records** option, and click **Next**.

   The **Template** page appears.

8. Select a template that supports how users will interact with the record view layout, and click **Next**.

   Tip: Primary components require more horizontal space than other components. When designing record view layouts, select a template with a panel that is big enough to display a primary component.

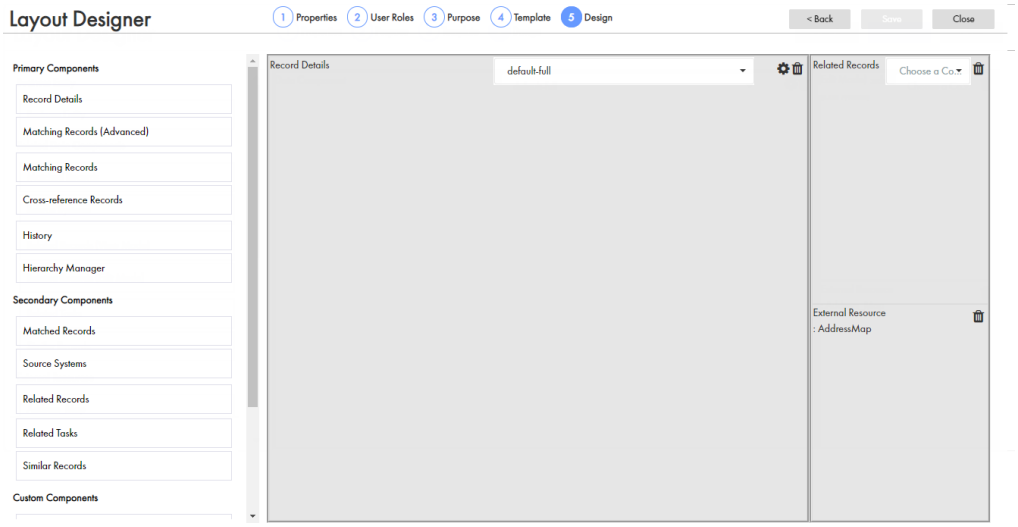
   The **Design** page appears.

9. Drag components into the workspace.
   a. Drag the Record Details component into the workspace.

      Note: Drag only one primary component into the workspace.

   b. To display related information, drag secondary or custom components into the workspace.

The following image shows the **Design** page with the Record Details component, the Related Records component, and the Address Map component in the workspace:



10. Click **Save**.

    The changes are saved to the temporary workspace.

11. Publish the changes to the MDM Hub.

    a. Click **Publish**.

       A confirmation dialog box appears that prompts you to publish or review the changes.

    b. Review the changes or publish without a review.

       - To publish without a review, click **Publish**.
       - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

The layout appears in **Layout Categories** panel. When users manage review tasks in the Task Manager, the record view layout designed for their user role appears in the review panel.

Designing a Merge Task Layout

You can design a custom record view layout to manage merge tasks in the Task Manager. Add the Matching Records (Advanced) component or the Matching Records component, and add any secondary or custom components that you want.

1. Click **Configuration** > **Layout Designer**.

   The **Layout Designer** appears.

2. Click **Create** > **Record View Layout**.

   The **Properties** page appears.

3. From the **Layout Type** list, select **Merge Task**.

4. Define the properties of the layout.
   a. Specify the following properties:

Property	Description
View Name	Label for the option. The view name appears as a tab in an opened record in Data Director.
Layout Name	Label for the layout that you design. After you design and publish the layout, the label appears in the **Layout Categories** panel in the Layout Designer.
Layout ID	Label used for generating a system ID.
Description	Optional. A meaningful description to identify the layout.
List Option Icon	Optional. Icon for the list option. The list option icon appears in the **View** list in Data Director.
List Order	Optional. Placement of the view name in the list of tabs in the opened record in Data Director.

   b. Choose the business entity or business entity view for which you want to design the layout.

      For example, choose **Person**.

5. Click **Next**.

   The **User Roles** page appears.

6. Select the user roles that can access the record view layout, and click **Next**.

   Tip: To create user roles, use the MDM Hub Console.

   The **Purpose** page appears.

7. Select any of the following actions that end users can perform using the record view:

   Merge tasks for business entity workflows.

   Merge tasks for subject area workflows.

8. Click **Next**.
9. Select a template that supports how users will interact with the record view layout, and click **Next**.

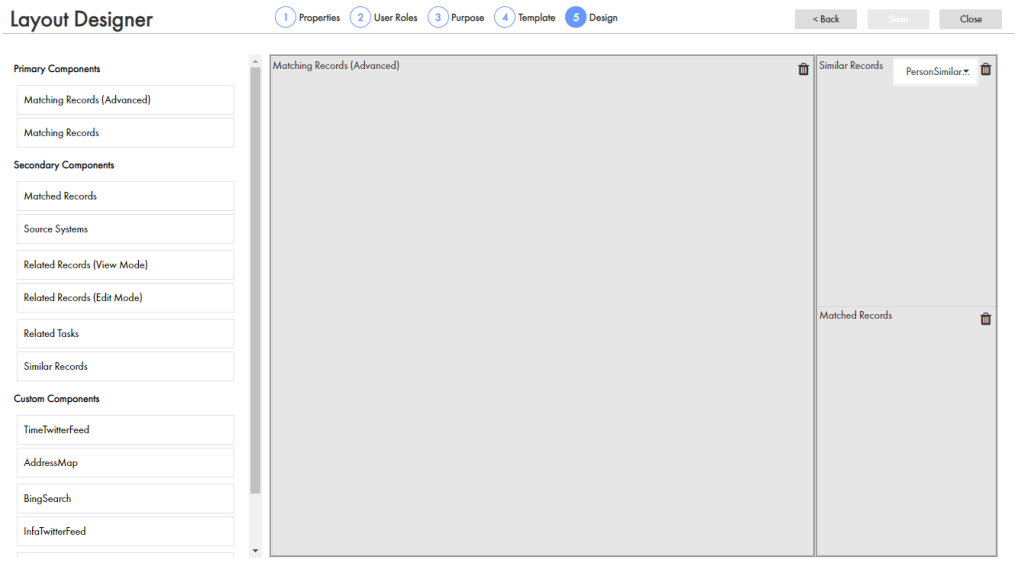
   Tip: Primary components require more horizontal space than other components. When designing record view layouts, select a template with a panel that is big enough to display a primary component.

   The **Design** page appears.

10. Drag one primary component into the workspace.
    - To manage merge tasks using the merge comparison panel with advanced features, drag the Matching Records (Advanced) component into the workspace.
      Use the Matching Records (Advanced) component to override and promote root and child values in records.
    - To manage merge tasks using the merge comparison panel, drag the Matching Records component into the workspace.
      Use the Matching Records component to override root values in records.

    Note: Drag only one primary component into the workspace.

The following image shows the **Design** page with the Matching Records (Advanced) component, the Similar Records component, and the Matched Records component in the workspace:



11. Optionally, drag secondary or custom components into the workspace.
12. Click **Save**.

    The changes are saved to the temporary workspace.
13. Publish the changes to the MDM Hub.
    a. Click **Publish**.

       A confirmation dialog box appears that prompts you to publish or review the changes.
    b. Review the changes or publish without a review.
       - To publish without a review, click **Publish**.
       - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

The layout appears in **Layout Categories** panel. When users manage merge tasks in the Task Manager, the record view layout designed for their user role appears in the review panel.

Related Topics:

- "Merge Task Layout Type" on page 152

Designing an Unmerge Task Layout

You can design a record view layout to manage unmerge tasks in the Task Manager. Add the Cross-reference Records component, and add any secondary or custom components that you want.

1. Click **Configuration** > **Layout Designer**.

   The **Layout Designer** appears.
2. Click **Create** > **Record View Layout**.

   The **Properties** page appears.
3. Click **Layout Type** > **Unmerge Task**.

4. Define the properties of the layout.
   a. Specify the following properties:

Property	Description
View Name	Label for the option. The view name appears as a tab in an opened record in Data Director.
Layout Name	Label for the layout that you design. After you design and publish the layout, the label appears in the **Layout Categories** panel in the Layout Designer.
Layout ID	Label used for generating a system ID.
Description	Optional. A meaningful description to identify the layout.
List Option Icon	Optional. Icon for the list option. The list option icon appears in the **View** list in Data Director.
List Order	Optional. Placement of the view name in the list of tabs in the opened record in Data Director.

   b. Choose the business entity for which you want to design the layout.

      For example, choose **Person**.

5. Click **Next**.

   The **User Roles** page appears.

6. Select the user roles that can access the record view layout, and click **Next**.

   Tip: To create user roles, use the MDM Hub Console.

   The **Purpose** page appears.

7. Select the **View or edit records** option, and click **Next**.

   The **Template** page appears.

8. Select a template that supports how users will interact with the record view layout, and click **Next**.

   Tip: Primary components require more horizontal space than other components. When designing record view layouts, select a template with a panel that is big enough to display a primary component.

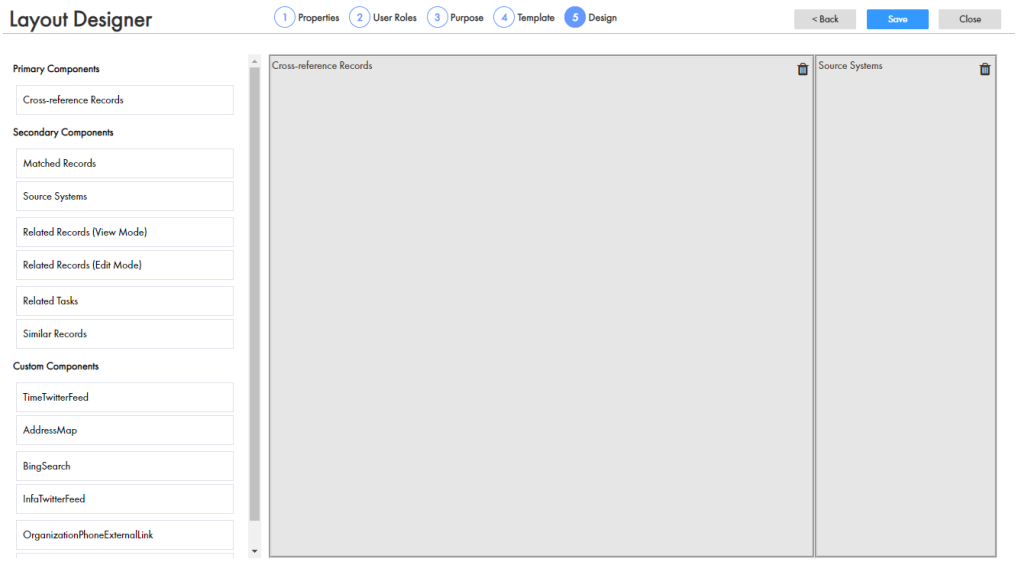
   The **Design** page appears.

9. Drag components into the workspace.
   a. Drag the Cross-reference Records component into the workspace.

      Note: Drag only one primary component into the workspace.

   b. To display related information, drag secondary or custom components into the workspace.

The following image shows the **Design** page with the Cross-reference Records component and the Source Systems component in the workspace:



10. Click **Save**.

    The changes are saved to the temporary workspace.

11. Publish the changes to the MDM Hub.

    a. Click **Publish**.

       A confirmation dialog box appears that prompts you to publish or review the changes.

    b. Review the changes or publish without a review.

       - To publish without a review, click **Publish**.
       - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

The layout appears in **Layout Categories** panel. When users manage unmerge tasks in the Task Manager, the record view layout designed for their user role appears in the review panel.

Related Topics:

- "Unmerge Task Layout Type" on page 153

Designing the Quick View Tab for the Hierarchy View

Design the **Quick View** tab to show the details of a record. In Data Director, users can view the **Quick View** tab in the **Hierarchy** view.

1. Click **Configuration** > **Layout Designer**.

   The **Layout Designer** appears.

2. Click **Create** > **Record View Layout**.

   The **Properties** page appears.

3. From the **Layout Type** list, select **Record View and Review Task**.

4. Define the properties of the layout.
   a. Specify the following properties:

Property	Description
View Name	Label for the Quick View.
Layout Name	Label for the layout that you design. After you design and publish the layout, the label appears in the **Layout Categories** panel in the Layout Designer.
Layout ID	Label used for generating a system ID.
Description	Optional. A meaningful description to identify the layout.

   b. Choose the business entity for which you want to design the layout.

      For example, choose **Person**.

5. Click **Next**.

   The **User Roles** page appears.
6. Select the user roles that can access the record view layout, and click **Next**.

   Tip: To create user roles, use the MDM Hub Console.

   The **Purpose** page appears.
7. Select the **View records in the Quick View tab of the Hierarchy View** check box.
8. Click **Next**.

   The **Template** page appears.
9. Select template 1, and then click **Next**.

   The **Design** page appears.
10. Drag **Record Details** into the workspace and configure the component.
    a. From the component configuration list, select a configuration.

       You can select a default or a custom configuration.
       The following default configurations are available:

       - default-full. Configuration of the business entity that includes the root node and all the child nodes.
       - default-root. Configuration of the root node of the business entity.

    b. Click the **Configure** icon of the component in the workspace.

       The **Record Details Component Designer** appears.

    c. In the **Configuration name** field, enter a configuration name.

       Important: Ensure that the configuration name is not the same as any business entity name. For example, if you have a Person business entity in your environment, do not name the configuration Person.

    d. From the **Section** list, select **1**.

    e. From the **Attributes** section, drag the business entity attributes that you want to display in a record view into the **Attributes layout** section.

For example, drag the First Name, Middle Name, Last Name, Birth Date, and Tax ID into the **Attributes layout** section.

f. From the **Child nodes** section, drag the child business entities that you want to display in a record view to the **Child nodes layout** section.

For example, drag Bill address, Ship address, Telephones, and Email into the **Child nodes layout** section.

g. Click **Done**.

The **Design** page appears.

11. Click **Save**.

The changes are saved to the temporary workspace.

12. Publish the changes to the MDM Hub.

a. Click **Publish**.

A confirmation dialog box appears that prompts you to publish or review the changes.

b. Review the changes or publish without a review.

- To publish without a review, click **Publish**.
- To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Designing the Quick View Tab for the Network View

Design the **Quick View** tab to show the details of a record. In Data Director, users can view the **Quick View** tab in the **Network** view.

1. Click **Configuration** > **Layout Designer**.

The **Layout Designer** appears.

2. Click **Create** > **Record View Layout**.

The **Properties** page appears.

3. From the **Layout Type** list, select **Record View and Review Task**.
4. Define the properties of the layout.

a. Specify the following properties:

Property	Description
View Name	Label for the Quick View.
Layout Name	Label for the layout that you design. After you design and publish the layout, the label appears in the **Layout Categories** panel in the Layout Designer.
Layout ID	Label used for generating a system ID.
Description	Optional. A meaningful description to identify the layout.

b. Choose the business entity for which you want to design the layout.

For example, choose **Person**.

5. Click **Next**.

The **User Roles** page appears.

6. Select the user roles that can access the record view layout, and click **Next**.

   Tip: To create user roles, use the MDM Hub Console.

   The **Purpose** page appears.

7. Select the **View records in the Quick View tab of the Network View** check box.
8. Click **Next**.

   The **Template** page appears.

9. Select template 1, and then click **Next**.

   The **Design** page appears.

10. Drag **Record Details** into the workspace and configure the component.
    a. From the component configuration list, select a configuration.

       You can select a default or a custom configuration.
       The following default configurations are available:

       - default-full. Configuration of the business entity that includes the root node and all the child nodes.
       - default-root. Configuration of the root node of the business entity.

    b. Click the **Configure** icon of the component in the workspace.

       The **Record Details Component Designer** appears.

    c. In the **Configuration name** field, enter a configuration name.

       Important: Ensure that the configuration name is not the same as any business entity name. For example, if you have a Person business entity in your environment, do not name the configuration Person.

    d. From the **Section** list, select **1**.
    e. From the **Attributes** section, drag the business entity attributes that you want to display in a record view into the **Attributes layout** section.

       For example, drag the First Name, Middle Name, Last Name, Birth Date, and Tax ID into the **Attributes layout** section.

    f. From the **Child nodes** section, drag the child business entities that you want to display in a record view to the **Child nodes layout** section.

       For example, drag Bill address, Ship address, Telephones, and Email into the **Child nodes layout** section.

    g. Click **Done**.

       The **Design** page appears.

11. Click **Save**.

    The changes are saved to the temporary workspace.

12. Publish the changes to the MDM Hub.
    a. Click **Publish**.

       A confirmation dialog box appears that prompts you to publish or review the changes.

    b. Review the changes or publish without a review.

       - To publish without a review, click **Publish**.

- To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Updating Home Page Layouts

You can update **Home** page layouts for Data Director in the Layout Designer. For example, you can update the **Home** page layout to display additional components.

Troubleshooting Tip: After you update a layout, the layout tab name is **Untitled Layout** in Data Director. The issue occurs with layouts created in a version prior to 10.2. You can change the name in the **Properties** page. To hide the tab name, use the **Purpose** page.

1. Click **Configuration** > **Layout Designer**.

   The **Layout Designer** appears.
2. To select a layout that you want to update, expand the **Home Page** layout category, and select the **Home** page layout that you want to update.
3. Click **Edit**.

   The **Design** page appears.
4. Drag the components that you want in the **Home** page into the workspace.

   For example, drag the **Task Inbox** and the **Dashboard Reports** components into the workspace.
5. If you want to add a component to the **Home** page layout, drag the component into the workspace.
6. If you want to remove a component from the **Home** page layout, click the **Delete** icon of the component.
7. If you want to update the layout properties, purpose, or template, click **Back** to navigate to the page and make the changes.
8. Click **Save**.

   The changes are saved to the temporary workspace.
9. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Updating Additional Page Layouts

You can update additional page layouts for Data Director in the Layout Designer. For example, you can update the additional page layout to display more components.

1. Click **Configuration** > **Layout Designer**.

   The Layout Designer appears.
2. To update a layout, expand the additional page layout category and select the additional page layout that you want to update.
3. Click **Edit**.

   The **Design** page appears.

4. Drag the components that you want to include into the workspace.
   - If you want to add a component to the additional page layout, drag the component into the workspace.
   - If you want to remove a component from the additional page layout, click the **Delete** icon of the component.
   - If you want to update the layout properties, purpose or template, click **Back** to navigate to the page and make the changes.
5. Click **Save**.

   The changes save to the temporary workspace.
6. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Updating Record View Layouts

You can update a record view layout in the Layout Designer. For example, you can update a record view layout to display additional components or remove a component that you do not require.

Troubleshooting Tip: After you update a layout, the layout tab name is **Untitled Layout** in Data Director. The issue occurs with layouts created in a version prior to 10.2. You can change the name in the **Properties** page. To hide the tab name, use the **Purpose** page.

1. Click **Configuration** > **Layout Designer**.

   The **Layout Designer** appears.
2. To select a layout that you want to update, expand the layout category related to the record view layout, and select the record view layout.
3. Click **Edit**.

   The **Design** page appears.
4. If the layout contains a Record Details component and you want to add or remove fields, configure the component.
   a. Click the **Configure** icon of the data component in the workspace.

      The **Record Details Component Designer** appears.
   b. If you want to use a vertical menu to access the business entity details, enable **Navigation pills**.
   c. If you want to add specific business entity attributes to the record view layout, from the **Attributes** section, drag the business entity attributes to the **Attributes layout** section.
   d. If you want to remove specific business entity attributes from the record view layout, click the **Delete** icons of the business entity attributes in the **Attributes layout** section.
   e. If you want to add child business entities to the record view layout, from the **Child nodes** section, drag the child business entities to the **Child nodes layout** section.
   f. If you want to remove specific child entities from the record view layout, click the **Delete** icons of the child entities in the **Child nodes layout** section.

g. If you want selected fields of the child entities to appear in the record view, click the **Configure** icon next to the child entity.

   The **Record Details Component Designer** appears where you can design the child entity.

h. Click **Done**.

   The **Design** page appears.

5. If you want to add a component to the record view layout, drag the component into the workspace.
6. If you want to remove a component from the record view layout, click the **Delete** icon of the component.
7. If you want to update the layout properties, purpose, or template, click **Back** to navigate to the page and make the changes.
8. Click **Save**.

   The changes are saved to the temporary workspace.
9. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.

   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Configuring Task Manager

The Task Manager displays task notifications and details of the associated records for which the review process generates the task notifications. You can select the fields that you want to display in the task panel of the Task Manager.

1. Click **Configuration** > **General Settings**, and then select **Task Manager Configuration**.
2. Perform one of the following tasks:
   - To configure the Task Manager for the first time, click **Create**.

     Note: You cannot create more than one Task Manager for an Operational Reference Store.
   - To edit the Task Manager configuration, select **TaskManagerConfiguration**.

   The properties panel displays **Basic** and **Advanced** tabs. To configure the Task Manager in the user interface, use the **Basic** tab. To configure the Task Manager with the XML code, use the **Advanced** tab. See "XML Code to Configure the Task Manager" on page 212.
3. On the **Basic** tab, in the **Display Fields** field, click **Edit**.

   The **Table Settings** dialog box appears.
4. To add or remove a field, select the field and drag it to the required section.
5. To set the order for a field, move the field up or down.
6. Click **OK**.
7. Click **Apply**.
8. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.

b. Review the changes or publish without a review.

- To publish without a review, click **Publish**.
- To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

XML Code to Configure the Task Manager

On the **Advanced** tab of the Task Manager configuration page, use the XML code to configure the Task Manager.

The following table lists the elements that you can use in the XML code to configure the Task Manager:

Element	Description
displayFields	Name of the field that you want to display in the task panel of the Task Manager. Use one of the following fields: - TASK_TITLE. A short description of the review task. - TYPE. Name of a people activity within the process. - PRIORITY. Priority of the task. - STATUS_DUEDATE_FILTER. Status of the task and the date the task is due. - OWNER. Name of the signed-in user. - TASK_ID. Instance ID of the review process that generated the task. - SUBMITTED_CREATOR_FILTER. Name of the user who created the task, and the date on which the user created the task.
excludeTaskTypes	Task type that you do not want to view in the task panel of the Task Manager.

The following sample XML code includes all the fields and excludes the draft and merge tasks in the task panel of the Task Manager:

```
<taskManager
    xmlns="http://www.informatica.com/mdm/TaskManager">
    <displayFields>TASK_TITLE</displayFields>
    <displayFields>TYPE</displayFields>
    <displayFields>PRIORITY</displayFields>
    <displayFields>STATUS_DUEDATE_FILTER</displayFields>
    <displayFields>OWNER</displayFields>
    <displayFields>TASK_ID</displayFields>
    <displayFields>SUBMITTED_CREATOR_FILTER</displayFields>
    <excludeTaskTypes>AVOSBeDraftState</excludeTaskTypes>
    <excludeTaskTypes>AVOSBeMerge</excludeTaskTypes>
</taskManager>
```

Hiding the Default Record Views

After you design custom record views, hide the default record views to reduce confusion for your end users.

Note: The default record views cannot be individually hidden.

1. Click **Configuration** > **Application Editor**.
2. Select the IDD application you want to configure.

   The IDD application details appear in the tree view panel and in the properties panel.
3. In the properties panel, clear the **Display Default Records Views** check box.
4. Click **Apply**.

The names of the default record views do not appear as tabs in opened records in Data Director.

Related Topics:

- "Default Record Views" on page 153

CHAPTER 17

Localizing Data Director

This chapter includes the following topics:

- Localizing Data Director Overview, 214
- Localizing Metadata, 214
- Localizing Task Actions, Types, and Messages, 215
- Localizing Lookup Tables, 216
- Localizing Labels and Error Messages, 218

Localizing Data Director Overview

You can display all elements of the Data Director user interface in another language. All elements of the user interface are localized in the selected language. The localized elements are menus, buttons, navigation links, labels, error messages, and metadata.

Data Director users can select their preferred language when they log in to your Data Director application. The selected language becomes the default language for that user.

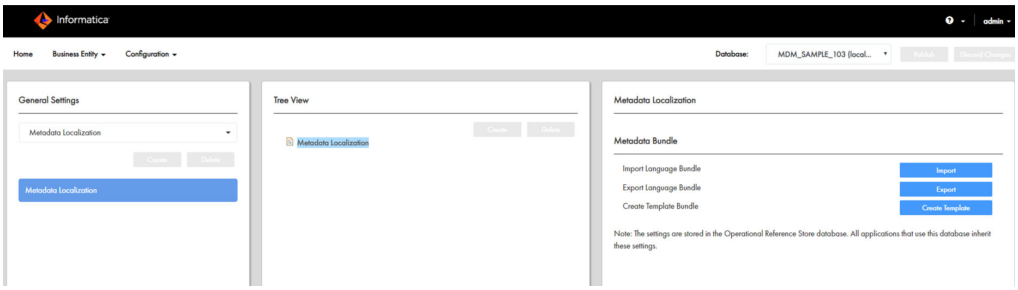
You can select a language to localize Data Director from a list of ten languages. The MDM administrator can add new languages not available in the list to your Data Director application.

Localizing Metadata

You can localize all metadata in Data Director. Metadata provides additional information about data. For example, metadata for a Person business entity includes first name, last name, birth date, and tax ID.

1. Click **Configuration** > **General Settings**, and then select **Metadata Localization**.
2. Click **Metadata Localization**.

The tree view panel and the metadata localization panel display the localization options.



3. In the **Metadata Localization** panel, under **Metadata Bundle**, click **Create Template**.

   The `besMetadata.zip` file downloads.

   Note: If you created a template bundle, click **Export** to download the `besMetadata.zip` file, edit the file, and click **Import** to import the language metadata bundle.

4. Save the `besMetadata.zip` file to a location on your local drive.
5. Extract the `besMetadata.zip` file to your local drive.

   The `besMetadata.properties` file appears.

6. Save a copy of the properties file with language code and country code for the language you want to use. For example, to create the Canadian French version, save the file as `besMetadata_fr_CA.properties`.
7. Open the localized file in a text editor. For each property, add the translation.
   For example, the following property shows a French translation for first name:

   ```
   Person.firstName=prénom
   ```

8. Compress the `besMetadata.properties` file and all the localized property files, such as `besMetadata_fr_CA.properties`, into a ZIP file named `besMetadata.zip`.
9. Import the localized .zip file. In the **Metadata Localization** panel, under **Metadata Bundle**, click **Import**.

   The language metadata bundle is imported.

Localizing Task Actions, Types, and Messages

Task actions, task types, and task action messages are part of the metadata. Define the task metadata in the `besMetadata.properties` file.

1. Click **Configuration** > **General Settings**, and then select **Metadata Localization**.
2. Click **Metadata Localization**.
3. In the **Metadata Localization** panel, under **Metadata Bundle**, click **Export**. The `besMetadata.zip` template file downloads.
4. Unzip the file, and open the `besMetadata.properties` file in an editor of your choice.
5. Add the task action and task action message properties. Edit the task type properties.

Use the following syntax for the task properties:

Property Type	Syntax	French Localization Example
Task action	`taskaction.<Task action>=<Localized task action>`	`taskaction.Cancel_Task=Annuler la tâche` **Note:** Replace white spaces in task action names with an underscore. In the example, the white space in `Cancel Task` is replaced with an underscore, `Cancel_Task`.
Task type	`tasktype.<Task type>=<Localized task type>`	`tasktype.AVOSBeMerge=Fusionner` **Note:** To know which task name to add, you must be aware of the predefined MDM workflows that are deployed to the ActiveVOS server as processes.
Task action message	`taskactionmessage.<Task action configuration>.<Task action>=<Localized message>`	`taskactionmessage.TypicalConfig.Escalate=Vous êtes sur le point de transmettre la tâche à votre responsable.`

6. Save the properties file with the appropriate language code and country code suffix.

   For example, to localize metadata in Canadian French, the language code is `fr` and the country code is `CA`. Save the file as `besMetadata_fr_CA.properties`.

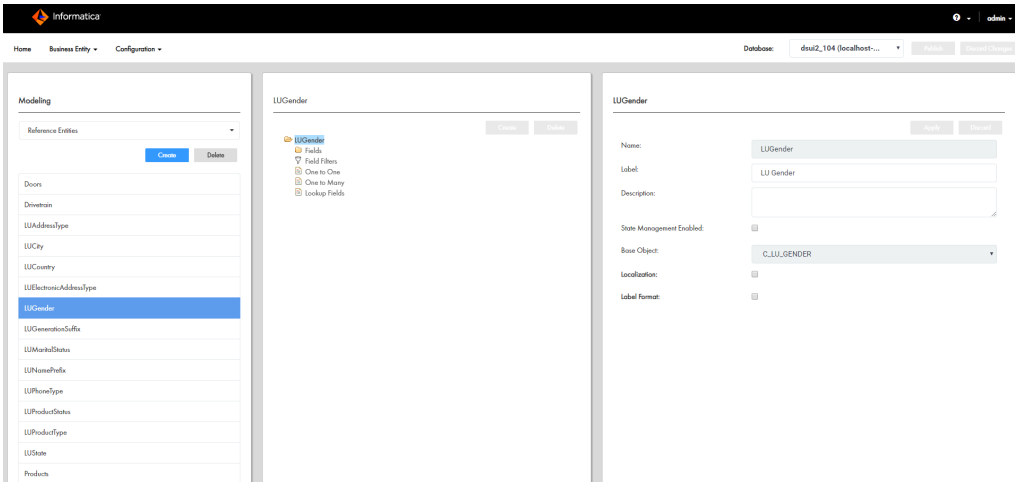
7. Compress the `besMetadata.properties` file and all the localized property files, such as besMetadata_fr_CA.properties, into a ZIP file named besMetadata.zip.
8. In the **Metadata Localization** panel, under **Metadata Bundle**, click **Import**. The `besMetadata.zip` template file is imported.

Localizing Lookup Tables

All lookup tables support localization. To localize lookup tables, you must map the lookup table to the corresponding lookup localization table. For example, to localize a list of gender codes, map the LookupGender lookup table to the LookupGender lookup localization table.

1. Click **Business Entity** > **Modeling**, and then select **Reference Entities**.
2. Select a reference entity lookup to localize.

For example, select **LUGender.**



3. In the properties panel, select the **Localization** checkbox.

   Additional database properties and values appear based on your database table setup. The following table contains example column names from the C_LU_GENDER_LCL base object tables:

Properties	Values
Base Object	C_LU_GENDER_LCL
Key Column	GENDER_CODE
Country Column	COUNTRY_CODE
Language Column	LANGUAGE_CODE
Value Column	LOCALIZED_STRING

   Note: When you localize lookup tables, do not configure the same table column as key and value columns. The lookup key value column displays the localized value.

4. Click **Apply** to localize the LUGender entity.
5. Click **Publish.**
6. Review the changes and click **Confirm.**

Localizing Labels and Error Messages

Labels and error messages are the static text that make up the Data Director user interface. The text cannot be localized through the Provisioning Tool interface. You can localize labels and error messages by editing the Data Director localization bundles.

Important: Informatica provides ant scripts to help you modify the localization bundles. Before you begin, copy the `LocalizationScript` directory that you received to the `resourceKit` directory.

1. Navigate to the following directory:

   `<MDM Hub installation directory>/hub/server`
2. Copy the following files:
   - entity360view-ear.ear
   - provisioning-ear.ear
   - uiwebapp-ear.ear
3. Navigate to the following directory:

   `<MDM Hub installation directory>/resourceKit/LocalizationScript`
4. Paste the copied files, and verify that the directory contains the following files and directory:
   - entity360view-ear.ear
   - provisioning-ear.ear
   - uiwebapp-ear.ear
   - build.xml file
   - library directory
   - ant.sh
   - ant.bat
5. Run the following command from a command prompt:

   ```
   ant generate_all
   ```

   The command generates a `i18n` directory and the localization bundle files.
6. Open a localization bundle file in the command line or in an editor of your choice.
7. If required, edit the bundle file and save it in the following format `bundle_<language name>.properties`. For example, if you update the French localization file, save the file as `bundle_fr.properties`.
8. Generate an updated bundle file by running the following command:

   ```
   ant generate_all
   ```
9. Update the .ear files by running the following command:

   ```
   ant update_all
   ```
10. Copy the updated .ear files to the `<MDM Hub installation directory>/hub/server`.
11. Deploy the updated .ear files to the application server by running the `PostInstallSetup` script.

APPENDIX A

Provisioning Tool Frequently Asked Questions

This appendix includes the following topic:

- Provisioning Tool Frequently Asked Questions, 219

Provisioning Tool Frequently Asked Questions

The following answers to frequently asked questions offer a good introduction to the Provisioning tool:

What does the Provisioning tool do?

You can use the Provisioning tool to define business entity models, tasks, and transformations, and design the user interface for Data Director.

Where should the Provisioning tool be installed?

The Provisioning tool must be running on the same application server as the Hub Server.

When I publish the configuration files to the MDM Hub, is the configuration validated?

Yes. The Repository Manager validates the configuration and reports any errors.

Can I use special characters such as '_' in name fields?

No. You cannot use special characters in names in the Provisioning tool. If you plan to generate a business entity schema from a subject area configuration, you must first remove the special characters from any names in the subject area configuration.

What happens when more than one person configures business entities for the same database at the same time?

If the business entity configuration in the database has changes that are not included in your configuration, the Provisioning tool notifies you. You can choose to publish your configuration and overwrite all configuration that is in the MDM Hub.

Caution: Because it is possible that one person can inadvertently overwrite the changes made by another person, Informatica recommends people co-ordinate with one another before using the Provisioning tool.

What happens when changes are made to the schema configuration in the MDM Hub while you are updating configurations?

As you work on the configuration files, you save your changes to a temporary workspace when you click Apply. The changes are not applied until you publish your changes. If multiple users change the business entity configuration for an ORS simultaneously, the MDM Hub is updated with the most recently published configuration.

Note: After you click Apply, you cannot edit your changes. Discard the changes, or publish the changes to edit them.

What is the maximum number of levels that a business entity model can have?

Unlike subject areas, which have a maximum depth of three, you are not limited in the number of levels the business entity model can have.

How do I create and configure base objects?

Use the Schema tool in the Hub Console to create and configure base objects.

How do I define constraints between base objects?

Use the Relationship tool in the Hub Console to define constraints between base objects.

How do I track creation of a business entity and changes made to the business entity?

When a change occurs, an entry is made in the C_REPAR_CO_CS_CONFIG table. Review the history of the COMPOSITE_OBJECTS entry in the C_REPOS_CO_CS_CONFIG table.

INDEX

A

accessing
  sso 16
analyzer
    character filter
      token filter 89
      tokenizer 89
architecture
  about 13

B

business entities
  configuring state management 26
  setting delete strategy 26
  setting identifying strategy 26
business entity
  about 14
  configuring for applications 19
  copying 34
  creating 24
  defining 34
business entity model
  about 28
business entity structure
  node types 30
business entity view
  copying 56
  defining 56
Business Entity view
  configuring 190
  designing 190

C

chart
  component 162
  creating the component 182
component
  configuration example 171, 173, 174
components
  chart 162
  cross-reference records 160
  dashboard reports 156
  external resources 173
  hierarchy manager 161
  history 160
  matched records 167
  matching records 159
  matching records (advanced) 158
  record details 157
  related records 168
  related tasks 169
components *(continued)*
  similar records 169
  source systems 167
  task inbox 155
configure
  Task Manager 211
configuring a lookup 48
configuring many-to-many relationships
  prerequisites 66
configuring relationships
  overview 63
constraints
  about 29
cross-reference records
  components 160
Cross-reference Records view
  configuring 196
  designing 196
custom data type
  adding 40, 48
custom data types
  about 39, 47

D

DaaS
  integration 132
  transformations 130
DaaS component
  creating 140
DaaS integration
  overview 129
DaaS Provider
  component 132
DaaS provider)
  overview 131
dashboard reports
  component 156
data
  security and filters 114
Data Director
  application 17
Data Director application
  about 17
date format
  configuring 20
datetime format
  configuring 20
default record views
  Business Entity view 153
  Cross-reference Records view 153
  hiding the default record views 212
  Hierarchy view 153
  History view 153
  Matching Records view 153

default record views *(continued)*
Timeline view 153
deleting
DaaS provider 141
extensions 141
SOAP service 141
WSDL file 141
dependent
configuring a lookup 49
Dependent Lookups 43
drilldown chart
linking chart to 183

E

Elasticsearch
index settings 89
enable
sso 15
external calls
configuring 143, 144
overview 142
external resources
component 173

F

field filters
adding 120
rules and user roles 119, 122
field properties
about 35, 44
fields
adding 38
filters
data 114
field, rules 119, 122
foreign keys
about 29

H

hierarchy manager
component 161
history
component 160
History view
configuring 197
designing 197
Home page
about 147
components 154, 172
configuring 187, 209
designing 187
updating 209

I

IDD application
creating 18, 175
index settings
analyzer 89
Informatica Data Director labels
about 40
Informatica Data Director user interface
about 145
components 154, 161, 172
example 146
Home page 147
layout 145, 146
prerequisites 146
record view 148
Informatica Data Director user interface layout
about 146

L

labels
error messages 218
localizing 218
layout
components 154, 161, 172
Home Page 147
record view 148
Layout
Informatica Data Director user interface 146
layout components
custom 172
primary 154
secondary 161
localize
labels and error messages 218
logging in
Provisioning tool 14, 15

M

many
adding a node 32
matched records
component 167
matching records
component 159
matching records (advanced)
component 158
Matching Records view
configuring 194
designing 194
merge task
configuring 201
designing 201
Multiple users
users 16

N

node labels
about 40
configuring 41
node types
about 30

O

one
adding a node 32, 33
overview
DaaS integration 129

P

preface 10
prerequisites
  about 13
  defining the business entity structure 29
  establishing a root node 23
  sso 15
primary keys
  about 29
privilege, role
  business entity 117
  business entity view 118
  fields in a business entity 122
  nodes in a business entity 124
Provisioning Tool
  about 12

Q

query
  configuring 20
  linking chart to 185
query results
  configuring 20

R

record details
  component 157
record view
  about 148
  components 154, 161, 172
  configuring 210
  updating 210
record view layout
  designing 141
record view layout type
  merge task 148, 152
  record view 148, 149
  review task 148, 149
  unmerge task 148, 153
reference entity
  about 14
  creating 25
reference entity fields
  adding 45
related records
  component 168
  creating the component 179
related tasks
  component 169
relationships
  foreign keys 29
review task
  configuring 199
  designing 199
role privilege
  business entity 117
  business entity view 118
  fields in a business entity 122
  nodes in a business entity 124
role privileges 114
root node
  about 14
  overview 22
root node properties
  business entity 23

S

SAM
  privileges 114
search
  configuring 20, 87
  linking chart to 186
  with Elasticsearch 87
search results
  configuring 20
security
  business entity 115
  field filters, defining 120
  filters, defining 46
Security Access Manager
  privileges 114
security filters
  node filters, adding 123
signing in
  Provisioning tool 14, 15
similar records
  component 169
  configuration example 171, 173, 174
  creating the component 180
  with queries 171
  with search 171
SOAP service
  overview 131
  WSDL 130
source systems
  component 167
state management
  business entities 26

T

task inbox
  component 155
  creating the component 176
task manager
  linking chart to 186
Task Manager 211
tasks
  changing the approver 113
token filter
  built-in 91
tokenizer
  built-in 91
transformations
  business entity to XML 130
  DaaS 130
  types 130
  XML to business entity 131
transforming
  business entity to XML 133
  XML to business entity 135

U

unmerge task
  configuring 203
  designing 203

W

WSDL
DaaS provider 130
WSDL *(continued)*
SOAP service 130